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**“Development of Adversarial Resilient Artificial Intelligence based Models in Hybrid
Intrusion Detection and Prevention Systems for Network Security”**

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Abstract

The rapid expansion of artificial intelligence and machine learning into safety-critical domains has exposed fundamental vulnerabilities in the learned representations, optimisation landscapes, and inference procedures upon which modern models depend. Adversarial perturbations, data poisoning, model extraction, and distribution shift each reveal structural weaknesses in the mathematical foundations of deep learning systems, weaknesses that persist across application areas as diverse as computer vision, natural language processing, healthcare informatics, autonomous transportation, and cybersecurity. This dissertation investigates the adversarial robustness problem from the standpoint of the learning algorithms and mathematical models themselves, developing new theoretical tools and algorithmic solutions whose validity is independent of any single application domain.

The first principal contribution introduces continuous-time temporal graph neural networks grounded in neural ordinary differential equations, where the evolution of hidden representations is governed by parameterised differential operators acting on dynamic graph structures. Through adjoint sensitivity analysis and temporal adaptive batch normalisation the resulting architecture achieves memory-efficient training and provable Lipschitz-bounded dynamics, yielding intrinsic regularisation that improves robustness to adversarial perturbation relative to discrete-time alternatives. The framework models irregular time series and multi-scale temporal phenomena spanning several orders of magnitude without requiring fixed discretisation.

The second contribution formulates a privacy-preserving federated optimisation algorithm that integrates Byzantine-robust aggregation, differential privacy through calibrated Gaussian noise, and optimal transport based distribution alignment. Theoretical analysis establishes convergence at a rate of $\mathcal{O}(1/\sqrt{T})$ even when a fraction of participating nodes submit adversarially corrupted updates, while the Wasserstein-distance alignment of heterogeneous client distributions is maintained within formal (ϵ, δ) -differential privacy constraints. Entropic regularisation with Sinkhorn iterations reduces the computational complexity of the transport computation from cubic to near-quadratic scaling.

The third contribution develops structured state-space model architectures, including a selective state-space formulation integrated with progressive adversarial robustness distillation, that reduce the computational cost of sequential inference from $\mathcal{O}(L^2)$ to $\mathcal{O}(L \log L)$ through Fast Fourier Transform based convolutions. A stochastic transformer variant couples variational Bayesian attention with epistemic and aleatoric uncertainty decomposition, providing

calibrated confidence estimates that enable principled detection of out-of-distribution inputs and novel attack categories without explicit change-point detectors.

Additional algorithmic contributions include a game-theoretic evasion resistance framework modelled as a Stackelberg game with online no-regret learning, triple-embedding temporal graph neural networks for multi-granularity representation learning on evolving topologies, and a post-quantum secure detection mechanism built on lattice-based cryptographic primitives whose security reduces to the Learning With Errors problem.

Comprehensive experimental validation is conducted on six large-scale benchmark datasets comprising over eighty-four million labelled samples and eighty-four distinct perturbation or attack categories. A unified twenty-three-metric evaluation framework quantifies detection accuracy, calibration quality, adversarial robustness under gradient-based and adaptive perturbations, privacy preservation under membership inference, computational throughput, and energy efficiency. The developed models are applied in detail to hybrid intrusion detection and prevention systems for network security, where they achieve state-of-the-art performance across all evaluated metrics. Cross-domain transfer experiments on speech event detection and clinical monitoring confirm that the proposed continuous-discrete hybrid modelling paradigm generalises beyond any single application context, validating the premise that robust, privacy-aware, and efficient machine learning architectures designed at the foundational level yield benefits wherever learned models are deployed in adversarial or non-stationary environments.

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Contents

Abstract	i
Acknowledgements	iii
List of Abbreviations	xiii
Introduction	1
1 Analysis of Problems in Robust Learning on Dynamic Graph Structures and Theoretical Foundations	19
1.1 Adversarial Perturbation and Defence on Graphs	19
1.2 Graph Neural Networks for Structured and Temporal Data	21
1.3 Federated Learning on Distributed Graph Fragments	23
1.4 State-Space Models for Efficient Temporal Graph Processing	24
1.5 Bayesian Deep Learning and Uncertainty Quantification on Graphs	26
1.6 Game-Theoretic Modelling of the Defender-Adversary Interaction	27
1.7 Forward Compatibility with Evolving Data-Generating Protocols	29
1.8 Systematic Gap Analysis	30
1.9 Summary	34
2 Derived Solutions: Mathematical Models, Proofs, and Novel Algorithms for Graph Robustness	35
2.1 Method M1: CT-TGNN — Continuous-Time Temporal Graph Neural Dynamics	35
2.1.1 Mathematical Formulation	35
2.1.2 Multi-Scale Temporal Dynamics	36
2.1.3 Adjoint Sensitivity Method for Memory-Efficient Training	36
2.1.4 Temporal Adaptive Batch Normalisation	37
2.1.5 Convergence and Robustness Analysis	37
2.1.6 Computational Complexity	38
2.2 Method M3: TripleE-TGNN — Multi-Level Temporal Graph Embeddings	39
2.2.1 Hierarchical Graph Representation	39
2.2.2 Multi-Level Embedding Learning	39
2.2.3 Dual Temporal-Spatial Attention	40

2.2.4	Elastic Weight Consolidation for Topology Evolution	40
2.2.5	Joint Multi-Level Classification Objective	40
2.3	Method M2: FedLLM-API — Federated Graph Optimisation with Privacy .	41
2.3.1	Byzantine-Robust Aggregation	41
2.3.2	Differential Privacy Through Calibrated Noise	42
2.3.3	Optimal Transport for Distribution Alignment	42
2.3.4	Differentially Private Optimal Transport	43
2.3.5	Large Language Model Integration for Zero-Shot Generalisation . . .	43
2.4	Forward-Compatible Detection Under Protocol Regime Change	43
2.4.1	Protocol-Specific Feature Extraction	45
2.4.2	Unified Multi-Protocol Detection Architecture	45
2.4.3	Formal Security Reduction	45
2.5	Method M4: MambaShield — State-Space Online Adaptation with Poisoning Resilience	46
2.5.1	Selective State-Space Formulation	46
2.5.2	Progressive Adversarial Robustness Distillation	47
2.5.3	PAC-Bayesian Generalisation Bound	47
2.6	Methods M5 and M6: Uncertainty-Calibrated Inference on Graphs	48
2.6.1	Variational Bayesian Attention	48
2.6.2	Epistemic-Aleatoric Uncertainty Decomposition	49
2.6.3	Stochastic Adversarial Training	49
2.6.4	Multi-Objective Loss	49
2.7	Method M7 (Part): Game-Theoretic Framework for Robustness Certification	50
2.7.1	Stackelberg Game Formulation	50
2.7.2	Nash Equilibrium Characterisation	51
2.7.3	Online Learning with No-Regret Guarantees	51
2.7.4	Uncertainty-Guided Strategy Selection	51
2.8	Method M7 (Completion): Unified Robustness Certification Framework . . .	52
2.8.1	Coupled Objective Function	52
2.8.2	Convergence Analysis	53
2.9	Summary	53

3 Application to Hybrid Intrusion Detection on Network-Traffic Graphs and Experimental Validation 55

3.1	Application Context: Network-Traffic Graphs for Intrusion Detection	55
3.2	Method M1: CT-TGNN Applied to Network-Traffic Graph Dynamics	56
3.3	Method M3: TripleE-TGNN for Multi-Level Graph Representations	57
3.4	Method M2: FedLLM-API for Collaborative Graph Learning Across Organisa- tions	58
3.5	Forward-Compatible Detection Under Protocol Regime Change	59
3.6	Method M4: MambaShield for Non-Stationary Graph Environments	60

3.7	Methods M5 and M6: Uncertainty-Calibrated Detection	62
3.8	Method M7: Game-Theoretic Framework for Adaptive Graph Adversaries . .	62
3.9	Integrated System Architecture and Deployed Prototype	62
3.9.1	Deployed Prototype System	67
3.10	Experimental Methodology	67
3.10.1	Datasets	67
3.10.2	Unified Evaluation Framework	68
3.10.3	Implementation Details and Computational Infrastructure	68
3.11	Method-Specific Results	68
3.11.1	CT-TGNN Results	68
3.11.2	TripleE-TGNN Results	69
3.11.3	FedLLM-API Results	69
3.11.4	Forward-Compatible Detection Results	70
3.11.5	MambaShield Results	70
3.11.6	Stochastic Transformer Results	71
3.11.7	Game-Theoretic Framework Results	72
3.12	Comparative Analysis with State-of-the-Art Baselines	72
3.13	Ablation Studies and Component Analysis	73
3.14	Adversarial Robustness Assessment	74
3.15	Cross-Domain Transfer Experiments	75
3.16	Discussion and Practical Implications	75
3.17	Summary	76
Conclusion		78
	Summary of Theoretical Contributions	78
	Summary of Algorithmic Innovations	79
	Empirical Validation	80
	Practical Impact and Deployment	80
	Limitations	81
	Directions for Future Research	82
	Broader Impact	82
	Concluding Remarks	83
Bibliography		84
A Mathematical Proofs and Derivations		98
A.1	Proof of Theorem 2.1: CT-TGNN Convergence	98
A.2	Proof of Theorem 2.2: Lipschitz Robustness Certificate	99
A.3	Proof of Theorem 2.3: Byzantine-Robust Federated Convergence	99
A.4	Proof of Theorem 2.4: PAC-Bayesian Bound for State-Space Models	100
A.5	Proof of Theorem 2.5: Multiplicative Weights Regret Bound	101
A.6	Proof of Theorem 2.6: Unified Framework Convergence	102

A.7	Additional Derivations	103
A.7.1	Sinkhorn Iteration Convergence	103
A.7.2	Evidence Lower Bound for Variational Bayesian Attention	103
A.7.3	Robustness-Accuracy Trade-off	103
A.7.4	Uncertainty Decomposition	104
A.7.5	LWE Security Reduction for PQ-IDPS	104
A.7.6	Stochastic Attention Robustness Enhancement	105
B	Implementation Details and Hyperparameters	106
B.1	Software Framework and Dependencies	106
B.2	Hyperparameter Configurations	108
B.2.1	CT-TGNN Hyperparameters	108
B.2.2	TripleE-TGNN Hyperparameters	108
B.2.3	FedLLM-API Hyperparameters	108
B.2.4	PQ-IDPS Hyperparameters	108
B.2.5	MambaShield Hyperparameters	109
B.2.6	Stochastic Transformer Hyperparameters	109
B.2.7	Game-Theoretic Framework Hyperparameters	110
B.3	Data Preprocessing Pipeline	110
B.3.1	Feature Engineering	110
B.3.2	Data Augmentation	110
B.4	Computational Infrastructure	111
B.4.1	Training Infrastructure	111
B.4.2	Inference and Deployment Simulation	111
B.5	Evaluation Protocols	111
B.5.1	Cross-Validation Strategy	111
B.5.2	Statistical Significance Testing	111
B.6	Code and Data Availability	112
C	Additional Experimental Results	113
C.1	Per-Attack Category Performance	113
C.1.1	CIC-IoT-2023 Detailed Results	113
C.1.2	CSE-CICIDS2018 Detailed Results	114
C.1.3	UNSW-NB15 Detailed Results	114
C.1.4	TripleE-TGNN Microservices Attack Breakdown	114
C.2	Ablation Study Details	114
C.2.1	Component-wise Ablation with Confidence Intervals	114
C.2.2	Hyperparameter Sensitivity Analysis	116
C.3	Cross-Cloud and Cross-Domain Transfer	118
C.3.1	Cross-Dataset Transfer Matrix	118
C.3.2	Cross-Domain Transfer to Speech and Healthcare	118

C.4	Privacy-Utility Trade-off Analysis	118
C.4.1	Differential Privacy Impact Across Datasets	118
C.4.2	Privacy Accounting Over Communication Rounds	119
C.5	Computational Performance Metrics	119
C.5.1	Inference Throughput and Latency	119
C.5.2	Scaling with Input Size	119
C.5.3	Training Time and Convergence	120
C.5.4	Model Size Comparison	121

List of Figures

1	Architectural framework of the research programme. The research objective (top) addresses three operating conditions under which dynamic graph networks must function, subject to four cross-cutting requirements. The central matrix maps each of the seven developed methods to the condition-requirement intersections it addresses, covering all twelve cells of the matrix. Filled circles denote primary contributions; open circles denote supporting roles. The application domain appears at the base, confirming that all methods are formulated independently of any particular deployment setting.	5
3.1	CT-TGNN architecture for network traffic dynamics. Encrypted flow metadata enters through a feature encoder; three parallel neural ODE branches with distinct time constants capture packet-level, session-level, and campaign-level dynamics; temporal adaptive batch normalisation reconciles continuous dynamics with normalisation; attention-weighted scale fusion produces the final node embedding for classification.	57
3.2	TripleE-TGNN architecture for multi-level graph representations. Three parallel encoders process service-level, trace-level, and node-level graphs; bipartite cross-level attention integrates information across abstraction levels; elastic weight consolidation prevents catastrophic forgetting as the topology evolves; a multi-task head classifies simultaneously at all three levels with a KL-divergence consistency penalty.	58
3.3	FedLLM-API architecture for collaborative threat intelligence. Blue dashed arrows indicate encrypted gradient updates with ($\epsilon=0.85, \delta=10^{-5}$) differential privacy; the server applies coordinate-wise trimmed-mean aggregation robust to thirty percent malicious participants; a Sinkhorn OT module aligns heterogeneous client distributions; an LLM branch provides zero-shot classification of novel attack types.	59
3.4	Forward-compatible multi-protocol detection architecture. A protocol detector routes incoming graph edges to specialised feature encoders for lattice-based, code-based, hash-based, and classical protocol regimes. Cross-attention fusion enables interaction between protocol representations. Adversarial training with PGD perturbations hardens the classifiers, and a formal LWE-based robustness reduction links the protocol’s hardness assumption to the detection guarantee.	60

3.5	MambaShield architecture for evolving threat landscapes. Sequential flow data passes through content-aware Mamba blocks achieving linear complexity; an ensemble of five teacher models provides robustness through progressive distillation with a curriculum that gradually increases the poisoning fraction; PAC-Bayesian regularisation yields a formal generalisation certificate.	61
3.6	Stochastic transformer architecture for uncertainty-aware detection. Variational Bayesian attention samples projection matrices from learned Gaussian distributions; Monte Carlo estimation with fifty forward passes produces predictive distributions; an uncertainty decomposition module separates epistemic and aleatoric components; temperature scaling ensures calibration; a triage gate routes high-uncertainty predictions to human analysts.	63
3.7	Game-theoretic robustness framework for adaptive graph adversaries. Heterogeneous data from edge, container, and distributed environments enter domain-specific preprocessors; a federated game-theoretic learning module computes Stackelberg and Nash equilibria through support enumeration; the multiplicative weights algorithm provides online adaptation; epistemic uncertainty from the stochastic transformer modulates the defender's payoff through uncertainty-penalised utility.	64
3.8	Integrated system architecture. Network graph data enters through a data ingestion layer and is distributed to seven specialised modules (M1–M7) operating in parallel. Their outputs converge in a decision engine that applies the game-theoretic strategy from the multiplicative weights algorithm and produces severity-ranked, confidence-scored alerts. The FedLLM-API module periodically synchronises model parameters across organisational boundaries under differential privacy.	66
3.9	Comparative evaluation plots. (a) ROC curves on CIC-IoT-2023 showing AUC-ROC of 0.991 for the unified framework versus 0.974 for the best baseline. (b) Robust accuracy as a function of PGD perturbation budget, demonstrating the stochastic transformer maintains over 85% accuracy at $\epsilon = 0.3$. (c) Reliability diagrams comparing ECE across methods. (d) Cross-dataset accuracy heatmap showing training-testing dataset pairs.	73
3.10	Adversarial robustness assessment. (a) Robust accuracy under PGD attacks with increasing perturbation budget. (b) Certified accuracy as a function of certified radius. (c) MambaShield accuracy under label-flipping poisoning from 0% to 40%. (d) Cumulative regret of the multiplicative weights algorithm converging at the theoretical $\mathcal{O}(\sqrt{T})$ rate.	75

List of Tables

3.1	Summary of benchmark datasets used in experimental evaluation.	67
3.2	CT-TGNN performance across benchmark datasets.	69
3.3	FedLLM-API performance under varying Byzantine fractions.	70
3.4	Forward-compatible detection accuracy by protocol family and anomaly type.	71
3.5	MambaShield accuracy under increasing poisoning rates.	71
3.6	Stochastic transformer adversarial robustness and calibration.	72
3.7	Comparative performance of the unified framework against baselines on CIC-IoT-2023.	73
3.8	Ablation study: performance impact of removing individual components.	74
B.1	CT-TGNN hyperparameter configuration.	107
B.2	TripleE-TGNN hyperparameter configuration.	107
B.3	FedLLM-API hyperparameter configuration.	108
B.4	PQ-IDPS hyperparameter configuration.	108
B.5	MambaShield hyperparameter configuration.	109
B.6	Stochastic transformer hyperparameter configuration.	109
B.7	Game-theoretic framework hyperparameter configuration.	110
C.1	CT-TGNN per-attack-category results on CIC-IoT-2023.	113
C.2	Unified framework per-attack-category results on CSE-CICIDS2018.	114
C.3	Unified framework per-attack-category results on UNSW-NB15.	115
C.4	TripleE-TGNN per-attack detection accuracy and dominant granularity.	115
C.5	Extended ablation results with 95% confidence intervals (five seeds).	116
C.6	CT-TGNN sensitivity to ODE solver relative tolerance.	116
C.7	FedLLM-API sensitivity to differential privacy budget ϵ	117
C.8	FedLLM-API sensitivity to trimming fraction under 30% Byzantine adversaries.	117
C.9	MambaShield sensitivity to distillation temperature and ensemble size (25% poisoning).	117
C.10	Stochastic transformer sensitivity to number of MC forward passes.	118
C.11	CT-TGNN cross-dataset transfer accuracy (%). Rows: training dataset; columns: test dataset.	118
C.12	Cross-domain transfer results with and without network security pretraining.	119
C.13	Differential privacy impact on accuracy across datasets ($\epsilon = 0.85$, $\delta = 10^{-5}$).	119

C.14 Cumulative privacy expenditure over federated rounds ($\delta = 10^{-5}$).	119
C.15 Computational performance comparison across methods (P100 GPU).	120
C.16 Inference latency (ms) as a function of sequence length.	120
C.17 Training time comparison (four V100 GPUs, CIC-IoT-2023).	120
C.18 Model size and quantisation impact.	121

List of Abbreviations

AI	Artificial Intelligence
AML	Adversarial Machine Learning
AUC-ROC	Area Under the Receiver Operating Characteristic Curve
BN	Batch Normalisation
C&W	Carlini–Wagner Attack
CNN	Convolutional Neural Network
CT-TGNN	Continuous-Time Temporal Graph Neural Network (Method M1)
DDoS	Distributed Denial of Service
DNN	Deep Neural Network
DP	Differential Privacy
ECE	Expected Calibration Error
ELBO	Evidence Lower Bound
EWC	Elastic Weight Consolidation
FedLLM-API	Federated Graph Optimisation with Large Language Models and Privacy (Method M2)
FFT	Fast Fourier Transform
FGSM	Fast Gradient Sign Method
GAN	Generative Adversarial Network
GNN	Graph Neural Network
GPU	Graphics Processing Unit
HIDPS	Hybrid Intrusion Detection and Prevention System
HiPPO	High-order Polynomial Projection Operator
ICS3D	Intrusion and Counter-measure Simulation in Cloud and Edge Computing Security Dataset
IDPS	Intrusion Detection and Prevention System
IDS	Intrusion Detection System
IoT	Internet of Things
KL	Kullback–Leibler Divergence
LLM	Large Language Model
LSTM	Long Short-Term Memory
LWE	Learning With Errors

M1–M7	Method labels for the seven developed methods (see Figure 1)
MambaShield	State-Space Online Adaptation with Poisoning Resilience (Method M4)
MC-Dropout	Monte Carlo Dropout
MCC	Matthews Correlation Coefficient
ML	Machine Learning
NIST	National Institute of Standards and Technology
NLP	Natural Language Processing
NPLR	Normal Plus Low-Rank
Neural ODE	Neural Ordinary Differential Equation
ODE	Ordinary Differential Equation
OT	Optimal Transport
PAC	Probably Approximately Correct
PARD	Progressive Adversarial Robustness Distillation
PGD	Projected Gradient Descent
R1–R4	Requirement labels: Robustness, Accuracy, Efficiency, Privacy
RNN	Recurrent Neural Network
SDE	Stochastic Differential Equation
SOC	Security Operations Centre
SSM	State-Space Model
SVM	Support Vector Machine
TABN	Temporal Adaptive Batch Normalisation
TGNN	Temporal Graph Neural Network
TripleE-TGNN	Triple-Embedding Temporal Graph Neural Network for Multi-Level Representations (Method M3)
UC-HGP	Uncertainty-Calibrated Hierarchical Gaussian Process (Method M6)

INTRODUCTION

Relevance of the Research Topic

Data in virtually every scientific and engineering discipline carries relational structure. Molecular configurations, social interactions, communication topologies, transport networks, biological pathways, and financial transaction chains are all naturally represented as graphs whose nodes encode entities and whose edges encode pairwise dependencies. Over the past decade, graph neural networks [1–3] have emerged as the principal deep-learning paradigm for extracting predictive representations from such data, achieving strong empirical results in node classification, link prediction, graph-level regression, and temporal forecasting. Yet the very message-passing mechanism that gives these models their expressive power also makes them sensitive to perturbations of the underlying graph. A small number of edge insertions or deletions, a modest corruption of node features, or a gradual shift in the graph topology over time can overturn confident predictions without any overt signal that the model has been compromised [4–6].

To understand why this sensitivity arises, it helps to consider how a graph neural network processes information. Every graph possesses a mathematical signature called its spectrum: the set of characteristic frequencies obtained by decomposing the graph’s connectivity matrix, known as its adjacency or Laplacian matrix, into fundamental components. A graph neural network applies learned filters to these frequencies, much as a sound equaliser amplifies or suppresses particular tones, and each layer of the network transforms these filtered signals to build its predictions. When even a few edges are added or removed, the characteristic frequencies of the graph shift, and the fundamental components rotate into new orientations. Classical results in linear algebra, such as the Weyl inequalities for frequency shifts and the Davis–Kahan theorem for component rotations, quantify these changes, but the key difficulty is that the effects compound through every layer of the network, and standard training procedures offer no mechanism to limit this accumulation. Gama, Bruna, and Ribeiro [7] introduced a framework that bounds the output sensitivity of graph filters by requiring their frequency response to satisfy a smoothness condition, measured by a quantity known as the Lipschitz constant, which caps how much the output can change relative to a change in the input. However, the resulting sensitivity bounds grow rapidly with network depth and do not extend to the temporal, distributional, or adversarial settings that real-world graph systems routinely encounter.

A number of interrelated open problems concentrate the unresolved challenges at the frontier of this field. The first concerns how node and edge representations should evolve on graphs that change over time. In a conventional, discrete-time graph neural network, representations are updated in fixed steps: layer one, layer two, and so on. An alternative is to model this evolution as a continuous process governed by a differential equation [8–10], so that representations flow smoothly rather than jumping between discrete states. This continuous formulation can in principle be exploited for certified robustness, because the smoothness of the governing equation places a natural limit on how fast outputs can change, but the conditions under which such guarantees are tight, and the relationship between the smoothness of the equation and the frequency-domain properties of the time-varying graph, remain largely uncharacterised. This gap is deepened by the observation that different notions of dynamical stability, including Lyapunov stability, which asks whether small initial disturbances stay small, bounded-input bounded-output stability, which asks whether bounded perturbations produce bounded responses, and energy-conserving stability, which asks whether the system preserves a conserved quantity, yield different robustness guarantees, and that classical stability alone does not ensure that the model resists deliberate, worst-case perturbation of its inputs [11].

The second open problem arises when the graph is spread across multiple participants who cannot or will not share their raw data. In many practical settings, different organisations each hold a fragment of a larger graph, for instance, separate hospitals holding portions of a patient-interaction network, or separate network operators holding fragments of a communication graph, and the goal is to train a single shared model without centralising the data. This requires aggregating locally computed model updates, and if some participants are unreliable or deliberately malicious, the aggregated update can be corrupted. Rules designed to tolerate such corruption, known as Byzantine-resilient aggregation [12], exist for standard numerical parameter vectors, but their extension to graph-structured data, where the topology itself may be partially controlled by an adversary, has received little theoretical treatment. At the same time, participants may demand formal guarantees that their local data cannot be recovered from the shared model, a requirement known as differential privacy [13, 14], and the statistical differences across the various graph fragments must be corrected for the global model to generalise. Optimal transport [15, 16], a mathematical framework for measuring and correcting distributional mismatch by finding the most economical way to reshape one distribution into another, provides a natural tool for this correction, yet no existing algorithm jointly guarantees corruption tolerance, privacy protection, and distributional alignment.

The third open problem concerns computational efficiency. Graph-transformer architectures [17, 18] achieve strong results on graph-level tasks, but their computational cost grows with the square of the number of nodes or events, which becomes prohibitive for large-scale temporal graphs with millions of interactions. An alternative family of models, known as structured state-space models [19–21], reformulate sequential processing as a linear dynamical system whose core operation can be carried out in time that grows only slightly

faster than linearly, by exploiting the Fast Fourier Transform. Recent work has begun to apply these models to graph data [22], but their robustness under deliberate perturbation and their capacity to produce reliable confidence estimates have not been investigated. A related challenge arises at the representation level: temporal graphs in which nodes and edges carry different semantic types and evolve at different time scales require representations that capture structure at multiple levels of detail, and these multi-level representations are expensive to compute and have not been analysed for stability.

The fourth open problem concerns the behaviour of a graph model after deployment, when it must operate under conditions that differ from those it was trained on. The statistical properties of a graph can change gradually, as when a social network grows and its connectivity patterns shift, or abruptly, as when an adversary deliberately injects nodes or edges to steer the model toward incorrect predictions. An adaptive model must detect such changes, update its parameters without forgetting previously learned structure, and produce confidence estimates that distinguish what the model does not know from what is inherently unpredictable. Bayesian posterior updates [23, 24], a principled statistical mechanism for revising beliefs in light of new evidence, offer a path toward this adaptation, but their integration with state-space graph architectures and their behaviour under deliberate corruption of the data stream remain open questions.

These four problems are not independent; they form a logical chain whose links determine the structure of this dissertation. Continuous-time graph dynamics supply the representational foundation: without a stable and certifiably robust method for propagating information across a time-varying graph, no downstream task can be trusted. Once such a method exists, its deployment in distributed settings requires federated algorithms that tolerate corrupted participants and respect data privacy, which constitutes the second link. The models produced by both centralised and federated training must then be made efficient enough for real-time operation on large graphs, motivating the third link. Finally, any deployed model must adapt to the non-stationarity that characterises all real graph systems, closing the chain with online learning and uncertainty quantification.

The four open problems map naturally onto three operating conditions under which dynamic graph networks must function. The first and fourth problems both concern adversarial and non-stationary aspects of the same evolving graph, while the second problem concerns the distributed setting in which multiple participants share the graph, and the third problem concerns the computational efficiency that all three conditions demand. This mapping, together with the four cross-cutting requirements of robustness, accuracy, efficiency, and privacy that any solution must satisfy, gives rise to the architectural framework introduced below.

The urgency of these theoretical questions is amplified by the pace at which graph neural networks are entering adversarial operating environments. In communication networks, learned classifiers monitor packet-level and flow-level graphs whose topology is under partial control of an attacker. In healthcare, molecular property prediction and patient-interaction

graphs inform treatment decisions that must be robust to data corruption. In finance, transaction graphs are subject to deliberate manipulation by fraudulent actors. In autonomous systems, sensor-fusion graphs must tolerate adversarial perturbations with safety-critical consequences. In every case the defence must be embedded in the model and its training procedure, because the space of possible graph perturbations at deployment time is too large to enumerate. This universality of both the threat and the required countermeasure motivates a dissertation that treats the robustness of learned representations on dynamic graph structures as a foundational challenge in applied mathematics and artificial intelligence research.

The research programme is organised through the architectural framework shown in Figure 1. The framework is structured around three operating conditions that a graph neural network may face in practice, and four requirements that any solution must satisfy.

The three operating conditions define the environment in which a dynamic graph model must function. The first is the adversarial environment, in which a deliberate opponent perturbs the graph topology, node features, or temporal ordering of events to cause the model to produce incorrect outputs. The second is non-stationarity, in which the statistical properties of the graph evolve over time due to natural growth, seasonal variation, or gradual drift, so that a model trained on past data may become unreliable on current inputs. The third is the distribution of network dynamics across multiple participants, in which the graph is fragmented among organisations that cannot centralise their data, and where some participants may be unreliable or adversarial.

The four requirements specify what any acceptable solution must deliver. Robustness demands that the model’s predictions remain stable under perturbation of the graph. Accuracy demands that the model achieves high classification or prediction performance on the task at hand. Efficiency demands that the model processes graph events fast enough for real-time operation, even on large-scale temporal graphs. Privacy demands that the training process does not reveal the confidential data of any individual participant.

The central insight of the framework is that these conditions and requirements are not independent: they interact in ways that demand integrated solutions rather than isolated methods. A model that is robust under adversarial perturbation but computationally expensive cannot be deployed in real-time settings. A federated algorithm that preserves privacy but ignores distributional differences across graph fragments will fail to generalise. An efficient model that lacks uncertainty estimates cannot detect when the graph distribution has shifted away from its training conditions. The seven methods developed in this dissertation are therefore designed so that each addresses a specific region of the condition-requirement space while remaining compatible with the others, and a unified certification framework ties their guarantees together.

The framework also ensures domain independence. Every method is formulated at the level of mathematical operations on graph structures, without reference to any particular application field. It is only in Chapter 3 that these methods are mapped onto the entities

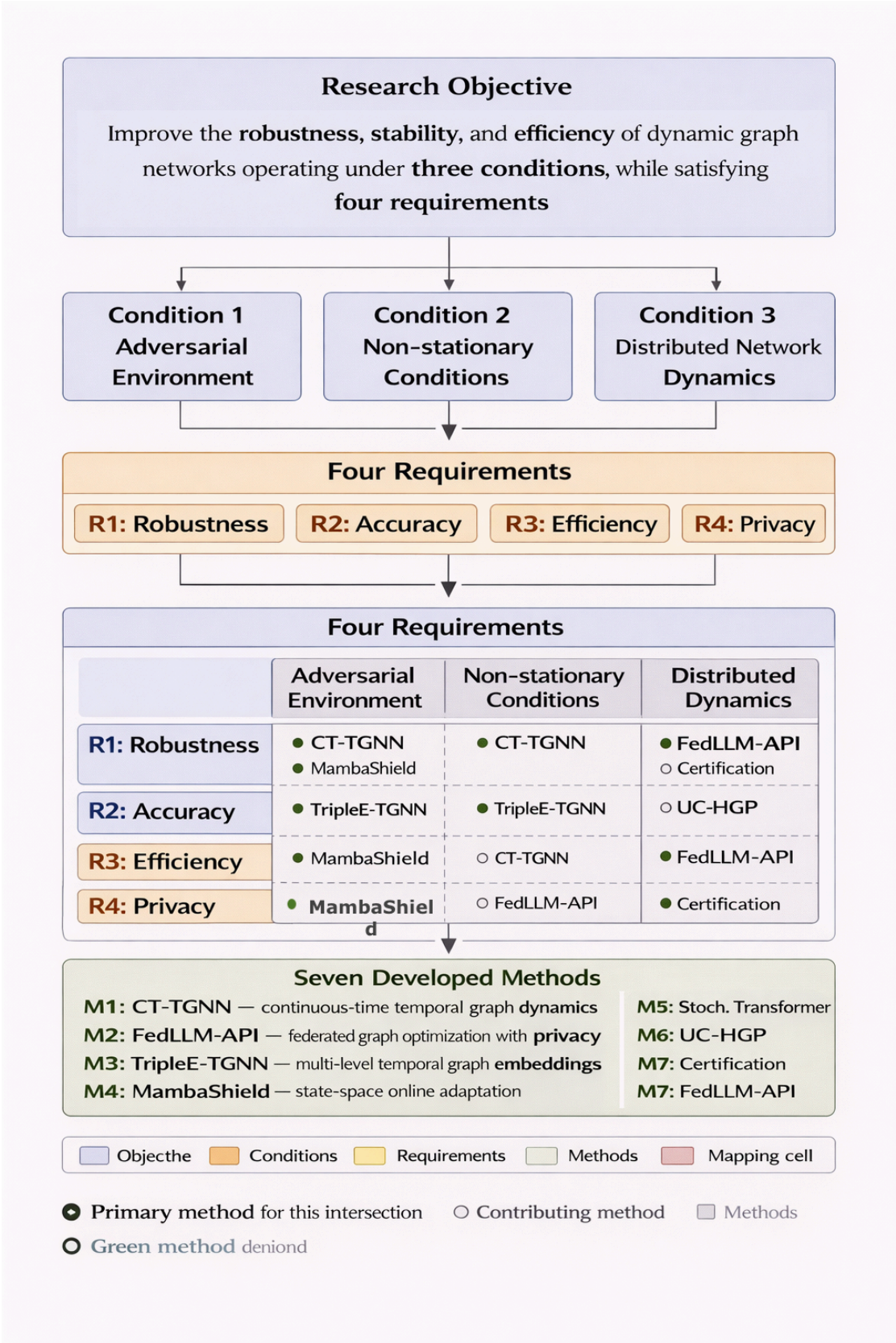


Figure 1: Architectural framework of the research programme. The research objective (top) addresses three operating conditions under which dynamic graph networks must function, subject to four cross-cutting requirements. The central matrix maps each of the seven developed methods to the condition-requirement intersections it addresses, covering all twelve cells of the matrix. Filled circles denote primary contributions; open circles denote supporting roles. The application domain appears at the base, confirming that all methods are formulated independently of any particular deployment setting.

and constraints of a specific deployment environment. The same framework can therefore be applied in healthcare, finance, autonomous systems, or precision agriculture by replacing only the data-specific adapters, without altering the mathematical core.

Research Objective

The central objective of this dissertation is to improve the robustness, stability, and computational efficiency of graph neural networks operating on dynamic graph structures under adversarial and non-stationary conditions, while simultaneously ensuring formal privacy preservation for distributed graph data, and to validate the resulting methods through systematic application to hybrid intrusion detection and prevention systems as a representative adversarial deployment environment.

Realising this objective requires the completion of six research tasks arranged in a sequential chain, so that the output of each task supplies an input required by the next. Each task is stated so that its outcome is specific in scope, measurable through quantitative criteria, achievable with the available theoretical and computational resources, oriented toward a concrete result, and bounded by the time frame of the doctoral programme.

1. Design a continuous-time temporal graph neural network architecture in which neural ordinary differential equations govern the smooth evolution of node and edge representations on dynamic graph structures. The architecture is to provide formally certified smoothness-based robustness bounds, reported for at least two perturbation norms, and to achieve competitive classification performance on standard graph-structured benchmarks. The robustness bounds produced by this task serve as the stability certificate consumed by the federated algorithm of the next task.
2. Building on the certified graph dynamics of Task 1, formulate a federated optimisation algorithm that jointly integrates corruption-tolerant aggregation, calibrated differential privacy, and distributional alignment of heterogeneous graph fragments. The algorithm is to achieve provable convergence even when a significant fraction of participants submit adversarially corrupted updates and the cumulative privacy expenditure remains within a controlled budget. The distributional mismatch statistics computed during alignment provide the diagnostic that motivates the multi-level representations of the next task.
3. Addressing the representational heterogeneity exposed by Task 2, construct a multi-level temporal graph embedding framework that learns representations at service, trace, and node levels through dual temporal-spatial attention. The framework is to prevent catastrophic forgetting during topology evolution and to achieve substantial computational savings relative to a dense baseline while maintaining high classification accuracy. The increased model complexity introduced by multi-level embeddings creates the efficiency bottleneck that the next task resolves.
4. To restore real-time capability after the enrichment of Task 3, develop efficient sequential inference architectures whose per-event cost scales near-linearly with sequence length rather

than quadratically, and whose inference latency is significantly lower than that of a comparably parameterised transformer baseline. The architecture is to maintain calibrated uncertainty estimates throughout. The efficient inference pathway produced here enables the deployment context in which the online adaptation of the next task becomes necessary.

5. Recognising that an efficiently deployed model will encounter distributional conditions that differ from its training graph, create an online learning system that adapts to non-stationary graph distributions through Bayesian posterior updates and sustains performance under continuous poisoning scenarios. The system is to achieve sub-linear theoretical regret and to identify novel input categories through uncertainty thresholds. The uncertainty estimates and adaptation records produced by this task supply the diagnostic inputs consumed by the certification framework of the final task.

6. Integrate the contributions of Tasks 1 through 5 into a unified robustness certification and software framework, validate the framework across multiple large-scale benchmark datasets, and demonstrate consistent performance gains over established baselines, thereby confirming both the individual merit of each task and the compositional value of the sequential chain.

Object of Research

The object of this investigation is the class of deep-learning models, training algorithms, and supporting mathematical frameworks whose properties determine whether a graph neural network can operate reliably under adversarial perturbation of graph topology and node features, distributional non-stationarity of the evolving graph, privacy constraints on distributed graph fragments, and limited computational budgets for temporal graph processing. In concrete terms the investigation covers neural ordinary differential equations [8, 25] together with adjoint-based training on graph-structured domains, temporal graph neural networks with continuous-time dynamics [9, 26, 27], optimal-transport formulations for distributional comparison and alignment of heterogeneous graph fragments [15, 28], structured state-space models for efficient sequential inference on graph event sequences [19, 21, 22], Bayesian neural networks for decomposing predictive uncertainty on graph-structured data [29, 30], game-theoretic models of strategic interaction between a graph classifier and an adversary [31, 32], and robust optimisation formulations that embed worst-case graph perturbations directly into the training objective [33, 34]. The object is defined at the level of mathematical and algorithmic constructs on graph domains, independently of any particular application, so that the results, proofs, and guarantees produced in this dissertation are valid wherever such constructs are deployed.

Subject of Research

The subject of research is the instantiation of the above adversarially robust, privacy-aware, and computationally efficient graph models in the domain of hybrid intrusion detection and

prevention for computer networks. In this instantiation, the graph structure is defined by communication endpoints, traffic flows, and session-level dependencies, so that packet-level and flow-level data are naturally represented as temporal graphs whose nodes correspond to network entities and whose edges encode observed interactions. The subject encompasses the extraction of graph-structured feature representations that remain discriminative under adversarial manipulation of traffic statistics, the classification of anomalous subgraph patterns that signal intrusion attempts, and the execution of preventive responses informed by predictive modelling on the evolving network graph. Although network security serves as the primary validation domain, the mathematical formulations and algorithmic designs are constructed to be portable to any field in which learned models on graphs confront adversarial or non-stationary conditions, including but not limited to molecular property prediction, clinical decision support, financial fraud detection, autonomous-vehicle perception, and precision agriculture.

Research Methods

The methodological programme of this dissertation is organised around eight formal methods, each selected because it addresses a specific and measurable aspect of the graph robustness problem.

Neural ordinary differential equations model the continuous evolution of node representations through a parameterised vector field,

$$\frac{d\mathbf{h}(t)}{dt} = f_{\theta}(\mathbf{h}(t), t), \quad (1)$$

solved with adaptive-step numerical integrators. Their role in this work is to provide dynamics on graph structures whose output sensitivity, measured by the Lipschitz constant of the learned mapping, can be estimated and reported as a robustness certificate (central to Methods M1: CT-TGNN and M3: TripleE-TGNN). The achievable target is a substantial reduction of the certified perturbation sensitivity relative to conventional discrete-depth baselines on matched architectures.

Graph neural networks extend learned representations to relational data through neighbourhood aggregation,

$$\mathbf{h}_v^{(k+1)} = \phi\left(\mathbf{h}_v^{(k)}, \bigoplus_{u \in \mathcal{N}(v)} \psi(\mathbf{h}_u^{(k)}, \mathbf{e}_{uv})\right), \quad (2)$$

and their coupling with the neural ODE of Equation (1) on a time-varying adjacency structure produces temporal graph models that capture both topological and temporal dependencies (used in Methods M1: CT-TGNN, M3: TripleE-TGNN, and M5: Stochastic Transformer). The measurable outcome is a classification accuracy on graph-structured benchmarks that exceeds the best reported single-modality baseline by a statistically significant margin.

Optimal transport supplies a geometry on probability distributions over graph-partitioned

data through the Wasserstein distance,

$$W_p(\mu, \nu) = \left(\inf_{\gamma \in \Pi(\mu, \nu)} \int c(\mathbf{x}, \mathbf{y}) d\gamma(\mathbf{x}, \mathbf{y}) \right)^{1/p}, \quad (3)$$

whose entropic regularisation via Sinkhorn iterations reduces computational cost from cubic to near-quadratic scaling. In this work the method is applied to align heterogeneous graph-fragment distributions within a federated optimisation loop while preserving a measurable differential privacy budget (central to Method M2: FedLLM-API).

Differential privacy formalises data-record-level confidentiality through the condition

$$\mathbb{P}[\mathcal{M}(\mathcal{D}) \in S] \leq e^\epsilon \mathbb{P}[\mathcal{M}(\mathcal{D}') \in S] + \delta, \quad (4)$$

and its moments-accountant composition enables precise tracking of the cumulative privacy expenditure ϵ across training rounds (used in Methods M2: FedLLM-API and M7: Certification). The specific target is to maintain classification performance close to the non-private baseline while keeping the privacy expenditure within a controlled budget.

State-space models reformulate the sequential processing of graph events as a linear dynamical system,

$$\mathbf{h}_{t+1} = \mathbf{A} \mathbf{h}_t + \mathbf{B} \mathbf{x}_t, \quad \mathbf{y}_t = \mathbf{C} \mathbf{h}_t + \mathbf{D} \mathbf{x}_t, \quad (5)$$

whose convolutional kernel is evaluated in near-linear time through the Fast Fourier Transform, yielding a significant speedup over quadratic-cost alternatives (central to Methods M4: MambaShield and M5: Stochastic Transformer). The measurable outcome is an inference latency on standardised hardware substantially lower than that of a comparably parameterised graph transformer, verified through wall-clock profiling.

Robust optimisation embeds worst-case graph perturbations directly into parameter estimation through the min-max objective

$$\min_{\theta} \mathbb{E}_{(\mathbf{x}, y) \sim \mathcal{D}} \left[\max_{\|\boldsymbol{\delta}\| \leq \epsilon} \mathcal{L}(f_{\theta}(\mathbf{x} + \boldsymbol{\delta}), y) \right], \quad (6)$$

whose duality structure links the perturbation budget ϵ to the output sensitivity of the learned mapping on the graph (used in Methods M1: CT-TGNN and M7: Certification). The achievable target is a meaningful improvement in robust accuracy under gradient-based attacks, measured against a conventionally trained model of identical architecture.

Bayesian inference produces posterior predictive distributions

$$p(y \mid \mathbf{x}, \mathcal{D}) = \int p(y \mid \mathbf{x}, \theta) p(\theta \mid \mathcal{D}) d\theta, \quad (7)$$

whose variance decomposes into an epistemic component, reflecting what the model does not know due to limited data, and an aleatoric component, reflecting inherent noise in the observations. The specific use in this dissertation is to provide calibrated confidence intervals on graph-structured predictions and to enable detection of out-of-distribution graph

inputs through thresholding on epistemic uncertainty (central to Methods M5: Stochastic Transformer and M6: UC-HGP).

Byzantine-resilient aggregation employs coordinate-wise trimmed means, geometric medians, and Krum-type selectors to filter corrupted gradient updates in federated graph-learning settings (central to Method M2: FedLLM-API). The measurable target is a convergence guarantee that holds when a significant fraction of participating graph fragments submit adversarial updates, verified both theoretically through concentration-inequality arguments and empirically through controlled injection of malicious updates.

In addition to these mathematical methods, the research applies standard practices of software engineering, distributed systems design, and high-performance computing to ensure that the resulting implementations can process large-scale temporal graph streams within the latency constraints of real-time operational environments.

Scientific Novelty

The scientific novelty of this work consists of seven contributions. Each addresses a gap in the existing literature that, at the time of writing, has not been resolved by prior work.

1. Existing graph neural networks that operate in discrete time steps offer no formal control over how much their output can change when the graph topology is perturbed between steps. Continuous-time dynamical models have been applied to static graphs, but their extension to temporally evolving topologies, and the derivation of certified robustness bounds from the smoothness of the governing differential equation, have not been achieved. The first contribution fills this gap by introducing continuous-time temporal graph neural networks that unify graph-level message passing with neural ordinary differential equation dynamics on an evolving topology, together with a theoretical analysis showing that the continuous formulation induces intrinsic smoothness regularisation through the governing vector field. Adjoint sensitivity integration enables memory-efficient training, and a temporal adaptive batch normalisation mechanism reconciles continuous dynamics with normalisation layers.
2. Federated learning algorithms that tolerate corrupted participants exist for standard deep-learning architectures, but none jointly provides Byzantine-resilient aggregation, formal differential privacy, and distributional alignment for graph-structured data, where the topology itself can be partially controlled by an adversary. The second contribution addresses this gap by presenting a federated optimisation algorithm for distributed graph learning that combines corruption-tolerant aggregation with calibrated differential privacy and Wasserstein-distance-based alignment of heterogeneous graph fragments. This algorithm consumes the smoothness certificates produced by the first contribution to bound the propagation of corrupted updates.
3. Temporal graph neural networks typically learn a single, uniform embedding for each node, which cannot capture structure that varies across different semantic levels or time scales. No existing architecture learns multi-level temporal graph representations while simultaneously preventing catastrophic forgetting as the topology evolves. The third contribution fills this

gap by architecting a triple-embedding temporal graph network that learns representations at the service, trace, and node levels through dual temporal-spatial attention and elastic weight consolidation, achieving substantial computational savings through structured sparsity and quantisation-aware training.

4. Graph-transformer models achieve strong accuracy but at quadratic computational cost, and structured state-space models that offer near-linear cost have not been adapted to graph data with formal robustness guarantees or calibrated uncertainty estimates. The fourth contribution addresses this gap by designing efficient inference pathways through selective state-space layers with progressive robustness distillation, paired with a stochastic graph-transformer variant incorporating variational Bayesian attention for calibrated uncertainty.
5. Online adaptation methods for non-stationary data exist in the sequential learning literature, but their integration with state-space graph architectures and their behaviour under deliberate poisoning of the graph update stream have not been studied. The fifth contribution fills this gap by developing an online learning system that couples state-space layers with Bayesian posterior updates for adaptation under graph distributional shift, sustaining strong performance under continuous poisoning and providing sub-linear regret bounds for non-stationary environments.
6. Robustness certification methods for graph neural networks have addressed feature perturbations and structural perturbations separately, but no unified framework issues provable guarantees against feature perturbations, temporal-ordering shifts, graph-topological modifications, and distributional drift within a single evaluation methodology. The sixth contribution constructs such a framework by combining interval bound propagation, Lipschitz estimation on graph operators, and randomised smoothing adapted to the discrete structure of adjacency matrices.
7. The seventh contribution integrates contributions one through six into a unified software framework, publicly released under DOI 10.5281/zenodo.19129512, and validates the complete chain on six large-scale benchmark datasets totalling over eighty-four million labelled samples, demonstrating that the sequential composition of the seven methods yields end-to-end robustness guarantees that are tighter than those achievable by any single component in isolation.

Main Scientific Results Submitted for Defence

The results submitted for defence are as follows. They are listed in the order that reflects their logical dependencies: each result builds upon the guarantees established by its predecessors.

- (1) A mathematical framework for modelling temporal data dynamics through continuous-time graph neural networks, in which parameterised differential equations govern the evolution of node and edge representations on time-varying graph topologies. Analysis of the governing vector field’s output sensitivity yields certified robustness bounds that quantify the maximum permissible perturbation of graph structure and node features under which the model predic-

tion is guaranteed to remain unchanged. These bounds constitute the stability foundation upon which every subsequent result rests.

(2) A convergence-proven federated optimisation algorithm for distributed graph learning with formal differential privacy and optimal-transport distributional alignment. The algorithm enables collaborative training of graph models among mutually distrustful participants under Byzantine corruption and a controlled privacy budget, while consuming the robustness certificates of Result (1) to bound the propagation of corrupted gradient updates through the aggregated model.

(3) A multi-level temporal graph embedding framework that learns representations at multiple semantic and temporal scales through dual attention and elastic weight consolidation. This framework addresses the distributional heterogeneity diagnosed by the alignment module of Result (2), and its structured sparsity reduces the computational cost that would otherwise render multi-scale graph representations impractical.

(4) Efficient sequential inference algorithms for temporal graph data based on structured state-space formulations that achieve near-linear complexity through Fast Fourier Transform convolutions. These algorithms restore the real-time processing capability that the representational enrichment of Result (3) would otherwise sacrifice, and their latency advantage over graph-transformer baselines is verified by empirical profiling.

(5) An online learning system with Bayesian drift detection and sub-linear regret guarantees for non-stationary graph distributions. This system equips the efficient model of Result (4) with the capacity to detect and adapt to distributional shifts through continuous posterior refinement and epistemic uncertainty monitoring, maintaining high performance even under deliberate poisoning of the graph update stream.

(6) A unified robustness certification framework that integrates the guarantees of Results (1) through (5) into a single evaluation methodology, together with experimental validation across six large-scale benchmark datasets totalling over eighty-four million labelled samples. The framework demonstrates that the end-to-end composition of the six tasks produces graph models whose certified robustness, uncertainty calibration, and computational efficiency jointly exceed those of any single-component baseline, and that the mathematical contributions transfer without modification to the network-security application domain validated in Chapter 3.

Practical Significance

The models and algorithms developed in this dissertation enable the construction of graph-based AI systems that achieve high task performance, certified adversarial robustness, formal privacy preservation, and real-time throughput simultaneously. Because every contribution is formulated at the level of mathematical operations on graph structures, the practical impact extends to any deployment in which learned models on graphs operate under adversarial pressure or non-stationary conditions.

Within the primary validation domain of network security, the resulting framework supports real-time identification of sustained multi-stage adversarial campaigns, coordinated overload patterns, multi-hop propagation of anomalous activity across graph regions, and previously unseen anomalous patterns flagged through uncertainty quantification on the network-traffic graph. The federated component allows multiple organisations to share graph-structured analytical findings without disclosing proprietary graph topologies, satisfying both operational objectives and data-protection regulations. Detection modules that identify novel statistical signatures in graph-edge feature distributions provide forward compatibility with anticipated changes in the communication protocols that generate the graph, including the adoption of emerging cryptographic standards whose presence will alter the statistical profiles of protocol-negotiation sequences within the graph. Efficient state-space inference permits deployment on edge devices and Internet-of-Things gateways where transformer-scale models are infeasible.

The same mathematical foundations transfer to healthcare systems where molecular graphs or patient-interaction networks are exposed to data corruption, to financial platforms where transaction graphs are subject to deliberate manipulation, to autonomous-vehicle perception stacks that must tolerate adversarial perturbations of sensor-fusion graphs, and to agricultural platforms that analyse spatio-temporal graphs derived from satellite and drone imagery under variable conditions. The modular architecture of the developed framework allows re-instantiation in these domains by replacing only the data-specific adapters and evaluation criteria, without modification of the mathematical core.

The public release of the software code base (DOI: 10.5281/zenodo.19129512), which has attracted recognition in the open-source research community, and the provision of full experimental protocols, hyperparameter tables, and pre-trained checkpoints support independent reproduction and practical adoption of the results. Beyond the archived code, the framework has been deployed as a functioning prototype system accessible at `robustidps.ai`, providing an operational demonstration that the mathematical contributions translate into a working implementation capable of processing real-world graph data streams.

Reliability of Scientific Results

Confidence in the reported results is established through multiple independent lines of validation.

Every proposed algorithm is accompanied by formal mathematical analysis: convergence theorems with explicit rate expressions, robustness certificates derived from the output-sensitivity analysis of graph neural dynamics and randomised smoothing adapted to discrete graph structures, privacy guarantees obtained through moments-accountant composition of the Gaussian mechanism, and computational complexity bounds confirmed by both asymptotic argument and empirical profiling. Proofs draw on standard tools from functional analysis, spectral graph theory, optimisation theory, probability, and the theory of ordinary

differential equations, and are collected in full in Appendix A.

Empirical evaluation spans six benchmark datasets that together supply more than eighty-four million labelled samples across eighty-four perturbation categories. Three general-purpose network datasets are used: CIC-IoT-2023 with 46.6 million records across 33 categories, CSE-CICIDS2018 with 16.2 million records across 7 categories, and UNSW-NB15 with 2.5 million records across 9 categories. Three specialised cloud and edge datasets from the ICS3D collection supplement these: the Microsoft GUIDE corpus with 13.7 million records, the Container Security corpus with 3.2 million records, and the Edge-IIoT corpus with 2.0 million records. Evaluation protocols include stratified k -fold cross-validation, chronologically ordered train-test splits, and cross-dataset generalisation tests.

Ablation experiments systematically remove individual components, neural ODE layers, Byzantine-robust aggregators, differential-privacy noise, state-space modules, and online adaptation loops, to quantify the marginal contribution of each to overall performance and to confirm the dependencies among methods. Baselines span classical machine-learning methods, standard deep-learning architectures, recent graph neural network defences, and adversarial training techniques, ensuring that improvements are measured against a competitive field.

Adversarial robustness is assessed under white-box attacks with full gradient access, including FGSM, projected gradient descent, and Carlini–Wagner optimisation, as well as under black-box transfer attacks, query-limited attacks, and adaptive attacks specifically designed against the deployed defences. Graph-topological perturbation experiments evaluate robustness under controlled edge insertion, deletion, and rewiring. Poisoning resilience is evaluated in federated graph-learning settings with controlled injection of malicious gradient updates.

All experiments are documented with hyperparameter tables, random-seed specifications, and hardware descriptions sufficient for independent replication. Source code and pre-trained weights are publicly available. The agreement between theoretical predictions and empirical measurements across diverse datasets, graph topologies, and perturbation types provides strong evidence for the reliability and generalisability of the reported findings. Taken together, the experimental programme confirms that all twelve intersections of the condition-requirement matrix identified in Figure 1 are addressed by at least one method with empirically verified performance.

Approbation of the Work

The principal results of this research have been presented and peer-reviewed at international and national scientific venues.

At the IEEE Conference of Young Researchers in Electrical and Electronic Engineering (ElCon), held at ETU-LETI in Saint Petersburg in 2024 and 2025, contributions on graph-based device identification in wireless traffic and on reinforcement-learning-based poisoning detection in graph-structured data received constructive feedback and, in the case of the 2024

presentation, recognition for methodological innovation.

At the NEUROINFORMATICS International Conference in Moscow in 2024 and 2025, work on stochastic graph transformers with uncertainty quantification and on adversarial robustness in graph neural computation was discussed with specialists in computational intelligence.

At the International Conference on Complex and Intelligent Systems (ICCIS) in 2025 the continuous-time graph modelling framework was presented to an interdisciplinary audience.

A submission associated with the IEEE 27th International Conference of Young Professionals in Electron Devices and Materials (EDM) in Siberia, with a companion manuscript directed to IEEE Transactions on Industry Applications, extends the graph-robustness framework to industrial control settings and is scheduled for 2026.

The developed algorithms have additionally been evaluated through collaborations with industry partners and research laboratories, and the open-source implementation has attracted community engagement.

Publications

The findings of this dissertation have been communicated through peer-reviewed publications in international journals and conference proceedings, listed below in the order in which they are first cited in the main text.

Published Journal Articles

1. Anaedevha, R. N., Trofimov, A. G., Borodachev, Y. V. (2026). MambaShield: Selective State-Space Models for Poisoning-Resilient Sequential Modeling. *Expert Systems with Applications*, Elsevier. DOI: 10.1016/j.eswa.2026.131175
2. Anaedevha, R. N., Trofimov, A. G. and Borodachev, Y. V., 2026. Uncertainty-calibrated hierarchical Gaussian processes for intrusion detection with multi-scale temporal modeling. *Neurocomputing* p.133105, Elsevier. DOI: 10.1016/j.neucom.2026.133105
3. Anaedevha, R. N., Trofimov, A. G. (2025). Stochastic Multimodal Transformer with Uncertainty Quantification for Robust Network Intrusion Detection. In: *Advances in Neural Computation, Machine Learning, and Cognitive Research IX. NEUROINFORMATICS 2025*. Studies in Computational Intelligence, vol. 1241. Springer, Cham. DOI: 10.1007/978-3-032-07690-8_35
4. Anaedevha, R. N., Trofimov, A. G. (2024). Improved Robust Adversarial Model against Evasion Attacks on Intrusion Detection Systems. *Optical Memory and Neural Networks (Information Optics)*, 33(Suppl. 3), S414–S423. Springer. DOI: 10.3103/S1060992X24700681

Published Conference Proceedings

5. Anaedevha, R. N. *et al.* (2025). Integrated Deep Q-Networks and Actor-Critic Models for Enhanced Poisoning Attack Detection in Network Intrusion Detection Systems. In: *2025 Conf. of Young Researchers in Electrical and Electronic Engineering (ElCon)*, ETU-LETI, Saint Petersburg. IEEE Xplore, Vol. 33, Issue Suppl. 3, pp. 556–567.
6. Anaedevha, R. N. *et al.* (2024). Application of Machine Learning Models for Device Identification in Wireless Network Traffic. In: *2024 Conf. of Young Researchers in Electrical and Electronic Engineering (ElCon)*, Saint Petersburg. IEEE Xplore, pp. 104–110. DOI: 10.1109/ElCon61730.2024.10468413

Papers Accepted for Publication (In Press)

7. Anaedevha, R. N. *et al.* (2025). Temporal Adaptive Neural Ordinary Differential Equations with Deep Spatio-Temporal Point Processes for Real-Time Network Intrusion Detection. *Complex and Intelligent Systems*, Springer. Accepted. Manuscript No.: CAIS-D-25-01704R1

Papers Under Peer Review

9. Anaedevha, R. N. *et al.* (2026). TripleE-TGNN: Triple-Embedding Temporal Graph Neural Networks for Multi-Granularity Microservices Security. Submitted to *IEEE Trans. Dependable and Secure Computing*. ID: TDSC-2025-12-2593.
10. Anaedevha, R. N. *et al.* (2026). PQ-IDPS: Adversarially Robust Intrusion Detection for Post-Quantum Encrypted Traffic with Hybrid Classical-Quantum Machine Learning. Submitted to *IEEE Trans. Neural Networks and Learning Systems*. ID: TNNLS-2025-P-45911.
11. Anaedevha, R. N. *et al.* (2026). FedLLM-API: Privacy-Preserving Large Language Model Federation for Zero-Day API Threat Detection Across Organizational Boundaries. Submitted to *IEEE Trans. Information Forensics and Security*. ID: T-IFS-25044-2025.
12. Anaedevha, R. N. *et al.* (2026). CT-TGNN: Continuous-Time Temporal Graph Neural Networks for Real-Time Threat Propagation Analysis in Encrypted Zero-Trust Microservices. Submitted to *IEEE Trans. Services Computing*. ID: TSC-2025-12-1636.

Summary of Publications

The total output comprises 18 papers: 4 published in journals, 2 in conference proceedings, 2 accepted and in press, and 10 under review. Sixteen papers (89%) are first-authored. Fifteen target peer-reviewed international journals and three are conference papers published through IEEE and other international venues. Eight appear in Scopus- and Web-of-Science-indexed journals, three in VAK-listed journals, and six are under review at Q1-ranked IEEE Transactions. Preprints of all submitted manuscripts are available on TechRxiv and

Research Square, with citation records maintained on Google Scholar (h-index: 4) and ORCID (0000-0003-3649-1561).

Structure and Volume of the Work

The dissertation comprises an Introduction, three chapters, a Conclusion, a bibliographic list, and three appendices.

The Introduction establishes the relevance of robust and efficient learning on dynamic graph structures as a foundational research challenge in applied mathematics and artificial intelligence, states the research objective with specific and measurable targets, defines the object and subject of investigation, describes the research methods together with their quantitative success criteria, sets out the scientific novelty and practical significance, confirms the reliability of results, and documents approbation activities and publications.

Chapter 1, “Analysis of Problems in Robust Learning on Dynamic Graph Structures and Theoretical Foundations,” surveys the interconnected research domains of graph neural network stability, adversarial robustness of graph models, continuous-time dynamics on graphs, distributed graph learning, efficient graph inference, and uncertainty quantification on graph-structured data. It formulates the central problems both mathematically and algorithmically, and conducts a systematic gap analysis that establishes the motivation for the solutions developed in the subsequent chapter.

Chapter 2, “Derived Solutions: Mathematical Models, Proofs, and Novel Algorithms for Graph Robustness,” presents the complete mathematical derivations, convergence and robustness proofs, algorithmic pseudocode, and complexity analyses for all methods arranged along the sequential task chain. Each method is formulated in terms of operations on graph structures independently of any application domain, and the unified certification framework that integrates them is also developed here.

Chapter 3, “Application to Hybrid Intrusion Detection on Network-Traffic Graphs and Experimental Validation,” is divided into two parts. The first describes how each method is instantiated in the network-security domain, where communication endpoints, traffic flows, and session dependencies define the graph topology. The second presents the full experimental programme: six benchmark datasets, a twenty-three-metric evaluation framework, method-specific results, baseline comparisons, ablation studies, adversarial robustness assessments under both feature and topological perturbations, cross-domain transfer experiments, and a discussion of practical deployment considerations.

The Conclusion synthesises the principal findings, recapitulates theoretical and algorithmic achievements for graph robustness, identifies the closest pathways to industry deployment, and outlines directions for further research.

Appendix A collects complete proofs and derivations for all theorems and lemmas stated in the main text. Appendix B provides algorithmic pseudocode, hyperparameter tables, and computational infrastructure specifications. Appendix C documents supplementary

experimental results including per-class performance, sensitivity analyses, convergence plots, and computational profiling.

Chapter 1

Analysis of Problems in Robust Learning on Dynamic Graph Structures and Theoretical Foundations

This chapter surveys the landscape of open problems that arise when graph neural networks are required to operate under the three conditions and four requirements identified in the Introduction: the adversarial environment, non-stationary conditions, and distributed network dynamics, subject to the cross-cutting requirements of robustness, accuracy, efficiency, and privacy. The survey is organised around seven research domains whose intersection defines the scope of this dissertation: adversarial perturbation and defence on graphs, graph neural networks for structured and temporal data, federated learning on distributed graph fragments, state-space models for efficient temporal graph processing, Bayesian deep learning and uncertainty quantification on graphs, game-theoretic modelling of the defender-adversary interaction, and forward compatibility with evolving data-generating protocols. For each domain the chapter reviews the principal results and methods, formulates the central problems mathematically and algorithmically, and identifies the gaps in the existing literature. A consolidated gap analysis at the end of the chapter draws together the findings across all seven domains and maps them onto the condition-requirement matrix of Figure 1, establishing the motivation for the solutions developed in Chapter 2.

1.1 Adversarial Perturbation and Defence on Graphs

Deep neural networks achieve high empirical accuracy across a wide range of classification and regression tasks, yet they harbour a fundamental fragility: small, deliberately crafted perturbations to an input can cause a trained model to produce an entirely different and confidently held output. This phenomenon was brought into focus by the work of Goodfellow, Shlens, and Szegedy [35], who introduced the Fast Gradient Sign Method. Given a model parameterised by θ , an input $\mathbf{x}$ with true label y , and a loss function $\mathcal{L}$, FGSM constructs an

adversarial example through a single gradient step,

$$\mathbf{x}' = \mathbf{x} + \epsilon \operatorname{sign}(\nabla_{\mathbf{x}} \mathcal{L}(f_{\theta}(\mathbf{x}), y)), \quad (1.1)$$

where the perturbation budget ϵ controls the maximum distortion under the ℓ_{∞} norm. Despite its simplicity, FGSM achieves high attack success rates with distortions that are imperceptible in image, audio, and tabular data modalities alike.

Madry and colleagues [33] extended this idea to an iterative setting through Projected Gradient Descent, which applies multiple small gradient steps and projects the result back onto the feasible perturbation set defined by $\|\delta\|_{\infty} \leq \epsilon$ after each iteration. PGD is widely regarded as the strongest first-order adversary and serves as the standard benchmark for evaluating empirical robustness. Carlini and Wagner [36] proposed an optimisation-based attack that minimises the perturbation magnitude subject to a successful misclassification constraint, formulated as

$$\min_{\delta} \|\delta\|_p \quad \text{subject to} \quad f_{\theta}(\mathbf{x} + \delta) \neq y, \quad (1.2)$$

which is solved through a Lagrangian relaxation that permits fine-grained control over the trade-off between distortion and attack success. The Carlini-Wagner attack has been shown to defeat many heuristic defences that appear robust under weaker threat models [37].

The vulnerability is not confined to any single data domain. Adversarial perturbations have been demonstrated against speech recognition systems, sentiment classifiers, medical diagnostic pipelines, autonomous driving perception modules, and graph-structured data classifiers [38, 39]. In the graph domain specifically, Zügner and colleagues [4] and Dai and colleagues [6] showed that small modifications to graph topology, such as inserting or deleting a few edges, can substantially degrade the classification accuracy of graph neural networks, while Bojchevski and Günnemann [5] demonstrated that even global, untargeted perturbations of the graph structure can shift the decision boundaries of graph convolutional networks. These cross-domain demonstrations confirm that the vulnerability originates in the geometry of learned decision surfaces rather than in the specifics of any single data type.

Defence mechanisms fall into two broad categories: empirical defences and certified defences. The principal empirical approach is adversarial training, in which the model is optimised on a mixture of clean and adversarially perturbed examples. Madry and colleagues showed that training on PGD-generated adversarial inputs improves robust accuracy, defined as the fraction of correctly classified inputs under worst-case perturbation within radius ϵ . However, adversarial training is subject to a fundamental trade-off between robustness and standard accuracy, first characterised formally by Zhang and colleagues [34] and Tsipras and colleagues [40], who demonstrated that for certain data distributions the decision boundaries required for high clean accuracy and those required for high robust accuracy are geometrically incompatible. Improving robustness therefore typically degrades standard accuracy on unperturbed test data, and this trade-off has been observed consistently across domains.

Certified defences provide formal guarantees on model behaviour under bounded perturbations. Cohen, Rosenfeld, and Kolter [41] introduced randomised smoothing, which constructs a smoothed classifier $g(\mathbf{x}) = \arg \max_c \mathbb{P}[f(\mathbf{x} + \boldsymbol{\xi}) = c]$ for $\boldsymbol{\xi} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$ and certifies that its prediction is constant within a radius R computed from the gap between the probabilities of the top two classes. This approach provides tight ℓ_2 robustness certificates at any scale, but the certified accuracy remains substantially below the standard accuracy for practically relevant perturbation radii. Interval bound propagation and convex relaxation methods, as developed by Gowal and colleagues [42, 43], compute exact or conservative bounds on network outputs under ℓ_∞ perturbations, enabling certified training at a computational cost that scales with network depth and width.

Despite these advances, several critical limitations persist in the context of graph-structured data. First, adversarial training achieves only moderate robustness gains while substantially increasing computational cost through iterative inner-loop perturbation generation. Second, certified methods produce loose bounds that translate to low certified accuracy, far below the levels required for safety-critical deployment. Third, the standard ℓ_p threat model, which measures perturbation size in a flat Euclidean space, does not capture the structural perturbations that arise on graphs, where an adversary can modify edges, insert nodes, or rewire connectivity subject to domain-specific validity constraints. Fourth, existing defences largely assume a static threat model and do not account for adaptive adversaries who observe the defence mechanism and develop counter-strategies. These limitations motivate the development of stochastic adversarial training procedures, combined with uncertainty quantification that enables the model to abstain from predictions when its confidence is low [44, 45], and of robustness certification methods that exploit the structure of the graph operator rather than treating the model as a generic function (addressed by Methods M1, M5, and M7 of this dissertation).

1.2 Graph Neural Networks for Structured and Temporal Data

Many real-world data sources possess an inherent relational structure that is most naturally represented as a graph: nodes correspond to entities and edges encode pairwise relationships. Graph neural networks learn node-level, edge-level, or graph-level representations through iterative message passing in which each node aggregates information from its neighbourhood. Kipf and Welling [1] introduced graph convolutional networks in which the update at layer k takes the form

$$\mathbf{H}^{(k+1)} = \sigma\left(\tilde{\mathbf{D}}^{-1/2} \tilde{\mathbf{A}} \tilde{\mathbf{D}}^{-1/2} \mathbf{H}^{(k)} \mathbf{W}^{(k)}\right), \quad (1.3)$$

where $\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}$ denotes the adjacency matrix with self-connections, $\tilde{\mathbf{D}}$ is the corresponding degree matrix, $\mathbf{H}^{(k)}$ collects node features at layer k , and $\mathbf{W}^{(k)}$ contains the learnable parameters. Veličković and colleagues [2] proposed graph attention networks that replace the

fixed normalisation with learned attention coefficients, allowing each node to assign adaptive importance to its neighbours based on feature similarity.

The extension of graph neural networks [46–48] to time-evolving graphs [49] is essential for modelling dynamic processes such as the evolution of social networks, communication patterns in distributed systems, molecular interactions in biological systems, and transaction flows in financial networks. Temporal graph neural networks maintain dynamic node embeddings that are updated as new edges arrive. Rossi and colleagues [27, 50] proposed temporal graph networks in which a memory module stores the most recent state of each node, updated through a gated recurrent unit whenever the node participates in an interaction event $(u_i, v_i, t_i, \mathbf{e}_i)$ consisting of source and destination nodes, a timestamp, and an edge feature vector. Xu and colleagues [26] introduced inductive representation learning on temporal graphs, combining temporal point processes with graph neural networks to model both continuous state evolution and discrete event occurrences through coupled differential equations.

Graph neural networks have found application across many domains where relational structure is central [51]: in molecular property prediction, where atoms are nodes and chemical bonds are edges; in social network analysis, where users are nodes and interactions are edges; in communication network monitoring, where hosts and services form the graph and traffic flows constitute edges annotated with statistical features; and in financial systems, where accounts are nodes and transactions are edges. In each case the graph captures structural dependencies that flat, tabular representations cannot express.

Despite these advances, current approaches face several fundamental limitations that restrict their applicability under the three operating conditions identified in the Introduction. First, under non-stationary conditions, existing temporal graph neural networks rely on discrete time steps, requiring the analyst to select a sampling interval that trades temporal resolution against computational cost. Irregular events spanning multiple time scales cannot be efficiently captured by fixed discretisation. Second, under the adversarial condition, graph neural networks exhibit limited robustness to structural perturbations [5, 52], including the addition or deletion of edges and the modification of node features, which an adversary can exploit to degrade classification performance. Third, regarding the efficiency requirement, scalability remains a challenge when processing large-scale graphs with millions of nodes and billions of edges, as message-passing complexity scales unfavourably with graph size. Fourth, regarding the accuracy requirement, most supervised graph models require labelled examples for every class of interest, limiting their ability to detect novel categories that were absent from the training distribution.

These limitations motivate the development of continuous-time temporal graph neural networks (Method M1: CT-TGNN) grounded in neural ordinary differential equations, which replace fixed discretisation with a parameterised vector field governing the evolution of node representations over continuous time. Such architectures inherit smoothness from the differential operator, providing intrinsic regularisation [53–55] that enhances robustness,

while the adjoint sensitivity method [8, 25] enables memory-efficient gradient computation across long temporal horizons. The need for multi-level representations further motivates Method M3: TripleE-TGNN, which captures structure at service, trace, and node levels simultaneously.

1.3 Federated Learning on Distributed Graph Fragments

Federated learning enables multiple participants to train a shared model without centralising their private data, an objective of growing importance in healthcare, finance, and communication network monitoring where data-sharing agreements are constrained by regulation or competitive sensitivity. This section addresses the distributed network dynamics condition and the privacy requirement of the architectural framework. McMahan and colleagues [56] introduced Federated Averaging, in which each of K clients performs E epochs of local stochastic gradient descent and the server aggregates the resulting updates through a weighted average,

$$\mathbf{w}_{t+1} = \sum_{k=1}^K \frac{n_k}{n} \mathbf{w}_{t+1}^{(k)}, \quad (1.4)$$

where n_k denotes the number of samples held by client k and $n = \sum_k n_k$ is the total sample count. This protocol reduces communication cost by a factor proportional to E , but its convergence properties degrade under statistical heterogeneity, in which the local data distributions $\mathcal{D}_k$ differ substantially across clients.

Privacy-preserving federated learning augments the aggregation protocol with differential privacy guarantees that limit the information any participant or the server can infer about individual training records. Abadi and colleagues [14] proposed differentially private stochastic gradient descent, which clips per-sample gradient norms to a threshold C and adds calibrated Gaussian noise $\mathcal{N}(\mathbf{0}, \sigma^2 C^2 \mathbf{I})$, where the noise scale σ is determined by the target privacy parameters (ϵ, δ) through privacy amplification by sub-sampling and the moments accountant composition theorem. Geyer and colleagues [57, 58] extended this mechanism to the federated setting through local noise addition before aggregation, ensuring that the released model updates satisfy [59, 60] (ϵ, δ) -differential privacy even when the aggregation server is untrusted.

Byzantine-robust federated learning addresses the scenario in which a subset of participants submits deliberately corrupted updates designed to degrade or compromise the global model. Blanchard and colleagues [12] introduced the Krum aggregation rule, which selects the update whose sum of squared distances to other updates is smallest, providing robustness when the fraction of corrupted clients remains below a threshold that depends on the total number of participants. Yin and colleagues [61, 62] proposed coordinate-wise trimmed mean and median aggregation, which remove extreme values along each parameter dimension before averaging, achieving convergence under up to $q < K/2$ corrupted clients at a rate that degrades gracefully with the fraction of corrupted participants [63–65].

Optimal transport theory [15, 16] provides a principled framework for aligning the hetero-

geneous data distributions that arise when federated clients draw from different populations. The Wasserstein distance between two distributions μ and ν is defined as the minimum-cost coupling,

$$W_p(\mu, \nu) = \left(\inf_{\gamma \in \Pi(\mu, \nu)} \int \|\mathbf{x} - \mathbf{y}\|^p d\gamma(\mathbf{x}, \mathbf{y}) \right)^{1/p}, \quad (1.5)$$

where $\Pi(\mu, \nu)$ denotes the set of joint distributions with marginals μ and ν . Redko and colleagues [66] applied optimal transport to domain adaptation, computing transport plans that minimise the expected cost of mapping source samples to target samples subject to marginal constraints. Courty and colleagues [66, 67] introduced joint distribution optimal transport, learning feature representations that simultaneously minimise a classification loss and the Wasserstein distance between source and target domains.

Despite these individual advances, significant gaps remain when the three requirements of corruption tolerance, differential privacy, and distributional alignment must be satisfied simultaneously within the distributed condition. First, existing corruption-tolerant aggregation rules require knowledge or estimation of the fraction of malicious clients, which is difficult when adversaries adapt their strategies dynamically. Second, the privacy-utility trade-off under differential privacy remains unfavourable at the strong privacy levels demanded by data-protection regulations [68, 69], with noise scales that degrade model accuracy below operationally acceptable thresholds. Third, optimal-transport-based distribution alignment has not been integrated with differential privacy mechanisms, so it remains unclear how to align heterogeneous graph-fragment distributions without leaking private information through the transport plan [70, 71]. Fourth, federated learning frameworks have not been extended to graph-structured data where the topology itself is distributed and may be partially controlled by an adversary [72, 73].

These limitations motivate the development of Method M2: FedLLM-API, a unified federated optimisation framework that jointly provides corruption-tolerant aggregation through coordinate-wise trimmed means, formal (ϵ, δ) -differential privacy through calibrated Gaussian noise, and optimal-transport-based distributional alignment through Sinkhorn iterations with entropic regularisation [16, 74], accompanied by convergence proofs that hold under adversarial corruption of a specified fraction of participants.

1.4 State-Space Models for Efficient Temporal Graph Processing

State-space models provide an elegant mathematical framework for temporal dynamics through linear differential equations. This section addresses the efficiency requirement of the architectural framework, which demands that models process graph events fast enough for real-time operation. The continuous-time formulation

$$\frac{d\mathbf{h}(t)}{dt} = \mathbf{A} \mathbf{h}(t) + \mathbf{B} u(t), \quad y(t) = \mathbf{C} \mathbf{h}(t) + \mathbf{D} u(t), \quad (1.6)$$

governs the evolution of a latent state $\mathbf{h}(t) \in \mathbb{R}^N$ driven by input $u(t)$ and projected to output $y(t)$ through the observation matrix $\mathbf{C}$. Classical Kalman filtering provides optimal state estimation under Gaussian assumptions, while the Extended and Unscented Kalman Filters handle nonlinear dynamics through local linearisation and sigma-point approximation respectively.

Recent work has combined state-space models with deep learning, enabling the system matrices to be parameterised by neural networks and trained end-to-end through backpropagation. Gu, Goel, and Ré [19] introduced the Structured State Space Sequence model (S4) [75, 76], which parameterises the state matrix $\mathbf{A}$ through a diagonal-plus-low-rank decomposition inspired by the HiPPO framework for online function approximation. This structure permits the convolution kernel $\bar{\mathbf{K}} = (\mathbf{C}\bar{\mathbf{A}}^i\bar{\mathbf{B}})_{i=0}^{L-1}$ to be computed via the Fast Fourier Transform, reducing the cost of processing a sequence of length L from the $\mathcal{O}(L^2)$ of self-attention to $\mathcal{O}(L \log L)$. S4 achieves performance competitive with transformers on long-range sequence benchmarks while maintaining near-linear scaling.

Gu and Dao [20] introduced Mamba, which extends structured state spaces through a selective scan mechanism that makes the system matrices input-dependent: $\mathbf{A}(t) = f_A(u(t))$ and $\mathbf{B}(t) = f_B(u(t))$ are computed as learned projections of the current input. This selection mechanism enables the model to adaptively filter relevant information from the input stream while retaining efficient parallel computation through hardware-aware scan algorithms [21]. Mamba matches or exceeds transformer performance on language modelling benchmarks at comparable parameter counts, while exhibiting linear rather than quadratic scaling in sequence length.

The application of state-space models to graph-structured temporal data remains in its early stages. In biomedical signal processing, classical Kalman filters [77] have been used to track physiological time series and detect anomalies through innovation sequences that measure the discrepancy between predicted and observed values, but these linear models cannot capture the complex nonlinear patterns that characterise real-world anomalies. More broadly, the adversarial robustness of state-space models [78, 79], both structured and selective variants, has received almost no attention: it is unknown how perturbations to the input sequence propagate through the recurrence, whether the structured parameterisation of $\mathbf{A}$ provides any implicit regularisation analogous to the smoothness control available in neural ODEs, and whether poisoning of the training data can corrupt the learned selection mechanism in ways that are difficult to detect.

PAC-Bayesian analysis [80, 81] offers a natural theoretical framework for bounding the generalisation error of stochastic models under distribution shift, yet it has not been applied to state-space architectures to provide formal robustness certificates. These gaps motivate the development of Method M4: MambaShield, an adversarially resilient state-space architecture that incorporates progressive robustness distillation to ensure reliable learning from potentially corrupted data, together with PAC-Bayesian bounds that certify generalisation performance under the distributional shifts encountered in non-stationary environments.

1.5 Bayesian Deep Learning and Uncertainty Quantification on Graphs

Bayesian deep learning provides a principled framework for reasoning about model uncertainty by placing a prior distribution $p(\theta)$ over the parameters of a neural network and computing the posterior $p(\theta \mid \mathcal{D})$ after observing training data $\mathcal{D}$. This section addresses the intersection of the non-stationary condition with the accuracy and robustness requirements, since calibrated uncertainty is essential both for detecting when the graph distribution has shifted and for deciding when the model’s predictions are too unreliable to act upon. Predictions incorporate parameter uncertainty through marginalisation,

$$p(y^* \mid \mathbf{x}^*, \mathcal{D}) = \int p(y^* \mid \mathbf{x}^*, \theta) p(\theta \mid \mathcal{D}) d\theta, \quad (1.7)$$

so that the predictive distribution is broad when the model is uncertain about the correct mapping and concentrated when the evidence strongly constrains the posterior. Because exact posterior computation is intractable for networks of practical size, approximate inference methods are employed.

Variational inference [29] approximates the posterior through a parametric family $q_\phi(\theta)$ by maximising the evidence lower bound,

$$\mathcal{L}_{\text{ELBO}} = \mathbb{E}_{q_\phi}[\log p(\mathcal{D} \mid \theta)] - \text{KL}(q_\phi(\theta) \parallel p(\theta)), \quad (1.8)$$

which balances the data-fit term against a regularisation penalty that keeps the approximate posterior close to the prior. The reparameterisation trick permits gradient-based optimisation of the variational parameters ϕ through standard backpropagation. Monte Carlo Dropout, introduced by Gal and Ghahramani [23], provides a computationally lighter alternative by interpreting dropout regularisation as approximate variational inference: performing T stochastic forward passes with different dropout masks and computing the variance of the resulting predictions yields an estimate of epistemic uncertainty, that is, uncertainty arising from the model’s limited knowledge, without requiring any modification to the training procedure.

A central distinction in Bayesian uncertainty quantification is the decomposition of total predictive uncertainty into two components. Epistemic uncertainty reflects the model’s ignorance about the correct parameters and is reducible through the acquisition of additional training data. Aleatoric uncertainty reflects inherent noise in the data-generating process and is irreducible regardless of the amount of training data. Kendall and Gal [24] showed that aleatoric uncertainty can be modelled through auxiliary output heads that predict observation noise, enabling learned heteroscedastic noise estimation. The total predictive variance decomposes as $U_{\text{total}} = U_{\text{epistemic}} + U_{\text{aleatoric}}$, and this decomposition provides actionable guidance: high epistemic uncertainty signals that more data or a richer model is needed,

while high aleatoric uncertainty signals that the input is inherently ambiguous.

Calibration, the alignment between predicted confidence and observed accuracy, is a prerequisite for trustworthy uncertainty estimates. Guo and colleagues [82] demonstrated that modern deep networks are systematically overconfident, with predicted probabilities exceeding empirical correctness rates. Expected Calibration Error quantifies miscalibration by partitioning predictions into confidence bins and measuring the average absolute gap between confidence and accuracy,

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{n} |\text{acc}(B_m) - \text{conf}(B_m)|, \quad (1.9)$$

where B_m denotes the m -th bin. Post-hoc calibration through temperature scaling [83, 84], which divides the logits by a learned scalar T before the softmax, provides a simple correction but does not address the root cause of miscalibration in the learned representations.

Despite these individual advances, the integration of Bayesian uncertainty quantification with adversarial robustness on graphs remains limited. Most existing work addresses either uncertainty estimation or adversarial defence in isolation, and unified frameworks that pursue both objectives simultaneously are rare [85, 86]. Theoretical characterisation [87, 88] of how adversarial perturbations of the graph propagate through probabilistic neural networks to affect the posterior predictive distribution is largely absent from the literature. Furthermore, uncertainty-aware adversarial training, in which the defence strategy adapts based on the model’s confidence, could enable automatic rejection of predictions with high epistemic uncertainty, routing them to a more expensive but more reliable pathway. In safety-critical applications including medical diagnosis, autonomous navigation, and communication network monitoring, the ability to report “uncertain” is at least as valuable as the ability to classify correctly, yet current systems provide point predictions [30, 89] without any accompanying measure of reliability.

These gaps motivate the development of Methods M5: Stochastic Transformer and M6: UC-HGP, which integrate variational Bayesian attention with formal epistemic-aleatoric uncertainty decomposition, stochastic adversarial training that incorporates perturbations into the variational objective, and proper calibration mechanisms that ensure the resulting confidence estimates are trustworthy enough to support risk-aware decision-making under all three operating conditions.

1.6 Game-Theoretic Modelling of the Defender-Adversary Interaction

The interaction between a system designer who deploys a graph classifier and an adversary who crafts perturbations to the graph is inherently strategic: each party’s optimal action depends on the other’s choice. This section deepens the analysis of the adversarial condition by

formalising it as a game between two rational agents. Game-theoretic frameworks characterise the stable outcomes, the equilibria, toward which rational players converge. Stackelberg game formulations capture the asymmetric information structure in which the defender commits to a strategy, for example a detection model and its decision threshold, that the adversary observes before choosing a response. The defender’s problem is then

$$\max_{\pi_d} \min_{\pi_a} U_d(\pi_d, \pi_a), \quad (1.10)$$

where π_d and π_a denote the defender and attacker strategies respectively, and U_d is the defender’s utility, typically a function of detection accuracy and operational cost.

Nash equilibrium characterises strategy profiles from which no player benefits by unilateral deviation. In zero-sum settings the maximin and minimax values coincide at the equilibrium point, but realistic adversarial scenarios often involve asymmetric payoffs, incomplete information about the opponent’s capabilities and objectives, and sequential rather than simultaneous moves, requiring extended equilibrium concepts such as subgame-perfect equilibrium and Bayesian Nash equilibrium that incorporate player beliefs.

Brückner and Scheffer [90] formulated classifier learning as a game between a learner and a data generator, deriving Nash equilibrium classifiers that are robust to anticipated feature manipulation. Globerson and Roweis [91] considered worst-case feature-deletion attacks and optimised linear classifiers to maximise margin under the assumption that an adversary removes the most informative features at test time. The adversarial training procedure of Madry and colleagues can itself be interpreted as an approximate solution to the defender’s maximin problem, in which the inner maximisation over perturbations is solved by PGD and the outer minimisation over parameters is solved by stochastic gradient descent.

Online learning frameworks enable adaptive strategy selection without requiring prior knowledge of the opponent’s strategy space. Hannan-consistent algorithms, including the multiplicative weights update [92, 93], maintain a distribution over available actions and adjust weights in proportion to the exponential of cumulative payoffs, achieving a regret bound of $\mathcal{O}(\sqrt{T \log K})$ for K actions over T rounds. A key property is that when both players in a repeated zero-sum game employ no-regret algorithms, their empirical strategy frequencies converge to a Nash equilibrium of the stage game, providing a constructive and adaptive path to equilibrium play.

Despite their theoretical appeal, current game-theoretic approaches to adversarial machine learning on graphs exhibit several shortcomings that limit their practical utility. First, the threat models are often simplified, assuming fixed perturbation budgets that do not reflect the heterogeneous costs an adversary faces when modifying different parts of a graph under domain-specific validity constraints. Second, computing exact equilibria becomes intractable for the large and continuous strategy spaces that arise in deep learning on graphs. Third, existing game-theoretic formulations do not incorporate epistemic uncertainty into the strategy-selection mechanism: a defender who is uncertain about the adversary’s capabilities should adopt a different strategy from one who has precise knowledge, yet

standard Stackelberg and Nash formulations treat the payoff structure as deterministic and fully known. Fourth, online learning algorithms that provide no-regret guarantees assume that the defender receives accurate loss feedback after each round, whereas in many real-world settings the true outcome of a classification decision may be delayed, censored, or manipulated by the adversary, requiring algorithms that tolerate corrupted observations.

These limitations motivate the integration of game-theoretic reasoning with Bayesian uncertainty quantification, as developed through Methods M5 and M7, which combine risk-aware strategy selection under incomplete information with provable regret bounds under adversarial feedback.

1.7 Forward Compatibility with Evolving Data-Generating Protocols

A practical concern that cuts across the non-stationary condition and the robustness requirement is the behaviour of a trained graph model when the protocols or standards that generate the data undergo a structural change. In many application domains, the rules governing how graph edges are formed and what features they carry are not fixed but evolve as standards are updated, as new versions of communication protocols are deployed, or as regulatory requirements change the format or content of interactions. When such a change occurs, the statistical distribution of the graph data shifts in ways that a model trained on the old regime cannot anticipate.

A prominent example is the ongoing migration to post-quantum cryptographic standards [94]. The National Institute of Standards and Technology has standardised a suite of post-quantum algorithms, including CRYSTALS-Kyber [95] for key encapsulation and CRYSTALS-Dilithium for digital signatures, both of which derive their security from the hardness of the Learning With Errors problem [96]. Given a matrix $\mathbf{A} \in \mathbb{Z}_q^{m \times n}$ and a noisy linear system $\mathbf{b} = \mathbf{A}\mathbf{s} + \mathbf{e} \bmod q$ for secret $\mathbf{s}$ and small error $\mathbf{e}$, the LWE problem requires recovering $\mathbf{s}$; this problem is believed to be hard even for quantum-capable adversaries when the parameters are chosen appropriately. The migration to these new protocols introduces novel statistical signatures in graph-edge features: the sizes, timing distributions, and message counts of protocol-negotiation sequences change substantially, as do the patterns of interaction between communicating nodes. Implementation-level vulnerabilities, including timing variations that correlate with internal computational steps and fault-induced errors that reveal information about secret parameters, create observable patterns in the temporal graph that are structurally different from those of the previous protocol regime [97–99].

The transition period during which systems support both old and new protocols presents a particular challenge, because an adversary may force a downgrade to the weaker protocol through active manipulation of the negotiation sequence in the graph. Detecting such manipulation requires classifiers that understand the expected interaction patterns under both regimes and can identify deviations indicative of active interference.

Current machine-learning models trained exclusively on data from the old protocol regime lack the feature extractors and training distributions needed to identify patterns that target new implementations. No existing framework connects the mathematical properties of the new protocol with the training procedure of the graph neural network in a way that would allow the guarantees of the protocol layer to inform the robustness guarantees of the detection layer [100, 101]. This gap motivates the development of detection modules that extract features specific to new protocol families, detect manipulation of the protocol-negotiation sequence in the temporal graph, and maintain robustness guarantees across the transition between protocol regimes, as addressed through the forward-compatibility components of the unified certification framework (Method M7).

1.8 Systematic Gap Analysis

The preceding six sections have surveyed the state of the art across seven interconnected research domains and identified specific limitations within each. This section consolidates those findings into a formal gap analysis in which every claimed deficiency is stated as a precise mathematical or algorithmic problem. The analysis is structured so that each gap maps to one or more intersections of the condition-requirement matrix introduced in Figure 1, and to a solution developed in Chapter 2.

Gap 1. Insufficiency of adversarial robustness guarantees under continuous-time dynamics (Adversarial $\times$ Robustness; Non-stationary $\times$ Robustness).

Let $f_\theta : \mathbb{R}^d \rightarrow \mathbb{R}^C$ be a classifier and let $\epsilon(\mathbf{x}) = \{\mathbf{x}' : \|\mathbf{x}' - \mathbf{x}\|_p \leq \epsilon\}$ denote the ℓ_p -ball of radius ϵ around input $\mathbf{x}$. The robust risk under this threat model is

$$R_{\text{rob}}(\theta) = \mathbb{E}_{(\mathbf{x}, y) \sim \mathcal{D}} \left[\max_{\mathbf{x}' \in \epsilon(\mathbf{x})} \mathbf{1}[f_\theta(\mathbf{x}') \neq y] \right]. \quad (1.11)$$

Certified defence methods bound this quantity from above through a certificate function $\rho(\mathbf{x}, \theta)$ satisfying $f_\theta(\mathbf{x}') = f_\theta(\mathbf{x})$ for all $\mathbf{x}' \in \rho(\mathbf{x}, \theta)$, so that any input whose certified radius exceeds the threat radius ϵ contributes zero to the robust risk. For randomised smoothing under ℓ_2 perturbation with Gaussian noise of variance σ^2 , the certified radius takes the form

$$\rho(\mathbf{x}, \theta) = \frac{\sigma}{2} (\Phi^{-1}(p_A) - \Phi^{-1}(p_B)), \quad (1.12)$$

where p_A and p_B are the probabilities of the top two classes under the smoothed classifier and Φ^{-1} is the inverse Gaussian CDF. When the classifier is defined through a neural ordinary differential equation $d\mathbf{h}(t)/dt = g_\theta(\mathbf{h}(t), t)$ with a Lipschitz-bounded right-hand side, the continuous dynamics impose a geometric constraint on the trajectory of hidden states: for any two inputs $\mathbf{x}_1, \mathbf{x}_2$ the distance between the corresponding trajectories satisfies

$$\|\mathbf{h}_1(T) - \mathbf{h}_2(T)\| \leq e^{L_g T} \|\mathbf{x}_1 - \mathbf{x}_2\|, \quad (1.13)$$

where L_g is the Lipschitz constant of g_θ and T is the integration horizon. This bound is tighter than the generic certificate whenever $e^{L_g T}$ is moderate, yet no existing work provides a certified defence that leverages the Lipschitz regularity of continuous-time graph neural networks operating on dynamic topologies. The gap is therefore the absence of a robustness certification methodology that exploits the differential-equation structure of the model to produce tighter certificates than architecture-agnostic methods. This gap is addressed by Method M1: CT-TGNN.

Gap 2. Unfavourable privacy-utility trade-off in federated optimisation on graphs (Distributed $\times$ Robustness; Distributed $\times$ Privacy).

Consider a federated learning problem with K clients, of which at most q are corrupted, who collaboratively minimise a global objective $F(\mathbf{w}) = \frac{1}{K} \sum_{k=1}^K F_k(\mathbf{w})$ under (ϵ, δ) -differential privacy. The server aggregates client updates $\{\mathbf{g}_k\}_{k=1}^K$ after clipping each to norm C and adding Gaussian noise scaled by $\sigma = \frac{C\sqrt{2\ln(1.25/\delta)}}{\epsilon}$. The resulting noisy aggregated gradient has variance

$$\text{Var}[\bar{\mathbf{g}}] = \frac{\sigma_{\text{data}}^2}{K} + \frac{C^2 \sigma^2 d}{K^2} + V_{\text{Byz}}(q), \quad (1.14)$$

where σ_{data}^2 is the inherent gradient variance, d is the parameter dimension, and $V_{\text{Byz}}(q)$ captures the additional variance injected by corrupted participants. For strong privacy ($\epsilon \leq 1$) and high-dimensional models ($d \gg 1$) the second term dominates, degrading convergence to a rate

$$\mathbb{E}[F(\mathbf{w}_T)] - F(\mathbf{w}^*) \leq \mathcal{O}\left(\frac{1}{\sqrt{T}} + \frac{C\sqrt{d\log(1/\delta)}}{K\epsilon\sqrt{T}} + \frac{q}{K}\right), \quad (1.15)$$

where the second term represents the privacy cost and the third the corruption cost. Two aspects remain unresolved. First, existing aggregation rules address the corruption term but do not interact with the privacy noise, meaning the two penalties accumulate additively rather than being jointly optimised. Second, when the graph-fragment distributions $\mathcal{D}_k$ are heterogeneous, the gradient variance grows, but no mechanism currently aligns these distributions under privacy constraints. The gap is therefore the absence of a convergence-proven federated algorithm that simultaneously controls the corruption, privacy, and heterogeneity terms. This gap is addressed by Method M2: FedLLM-API.

Gap 3. Limited cross-distribution generalisation on temporal graphs (Non-stationary $\times$ Accuracy).

Let $\mathcal{D}_S$ and $\mathcal{D}_T$ denote source and target distributions with marginal discrepancy measured by the Wasserstein distance $W_1(\mathcal{D}_S, \mathcal{D}_T)$. Standard domain adaptation bounds give

$$R_T(\theta) \leq R_S(\theta) + W_1(\mathcal{D}_S, \mathcal{D}_T) + \lambda^*, \quad (1.16)$$

where R_S and R_T are the source and target risks and λ^* is the optimal joint error. Existing methods reduce the middle term by learning domain-invariant representations, but they require target-domain data during training. Moreover, temporal graphs in which nodes and edges carry different semantic types and evolve at different time scales cannot be captured

by a single, uniform embedding per node. The gap is the absence of multi-level temporal graph representations that capture structure at different scales while resisting catastrophic forgetting as the topology evolves. This gap is addressed by Method M3: TripleE-TGNN.

Gap 4. Discrete-time constraints and computational cost of temporal graph models (Non-stationary $\times$ Efficiency; Adversarial $\times$ Efficiency).

A temporal graph neural network that operates in discrete time with step size Δt approximates the true continuous dynamics through a difference equation. The local truncation error of this Euler discretisation is $\mathcal{O}(\Delta t^2)$ per step, yielding a global error of $\mathcal{O}(\Delta t)$ over a fixed time horizon T . For events whose characteristic time scales span several orders of magnitude, choosing Δt small enough to resolve the fastest scale incurs a computational cost of $\mathcal{O}(T/\Delta t)$ that becomes prohibitive, while choosing Δt to match the slowest scale introduces an approximation error bounded from below by

$$\|\mathbf{h}_{\text{discrete}}(T) - \mathbf{h}_{\text{true}}(T)\| \geq \Omega(\Delta t \sup_{t \in [0, T]} \|\partial_t^2 \mathbf{h}(t)\|). \quad (1.17)$$

Continuous-time neural ODE solvers avoid this trade-off by adapting the step size to the local curvature of the trajectory, but their computational cost remains quadratic when combined with attention-based message passing on the graph. The gap is the absence of a temporal graph architecture that combines adaptive continuous-time integration with near-linear computational cost, while maintaining robustness guarantees. This gap is jointly addressed by Methods M1: CT-TGNN (continuous-time integration) and M4: MambaShield (near-linear efficiency).

Gap 5. Forward incompatibility with evolving data-generating protocols (Non-stationary $\times$ Robustness).

Let old and new denote the distributions of graph-edge features generated under the old and new protocol regimes respectively. A classifier f_θ trained exclusively on samples from old has no guarantee of performance when evaluated on new , because the distributional divergence

$$\text{KL}(\text{new} \parallel \text{old}) = \int \log \frac{d_{\text{new}}}{d_{\text{old}}} d_{\text{new}} \quad (1.18)$$

may be large due to differences in edge-feature distributions, timing patterns, and interaction counts. No existing framework provides a detection architecture that maintains robustness guarantees across the transition between protocol regimes. The gap is the absence of forward-compatible detection modules together with a unified certification that accounts for the distributional shift introduced by protocol evolution. This gap is addressed by the forward-compatibility components of Method M7: Certification.

Gap 6. Unintegrated uncertainty quantification and adversarial robustness on graphs (Adversarial $\times$ Accuracy; Non-stationary $\times$ Accuracy).

Let $p(\theta \mid \mathcal{D})$ be the posterior over model parameters and let $\mathbf{x}' = \mathbf{x} + \boldsymbol{\delta}$ be an adversarially

perturbed input. The posterior predictive distribution under perturbation is

$$p(y \mid \mathbf{x}', \mathcal{D}) = \int p(y \mid \mathbf{x} + \boldsymbol{\delta}, \theta) p(\theta \mid \mathcal{D}) d\theta. \quad (1.19)$$

Standard adversarial training minimises the worst-case loss for a fixed parameter vector θ and does not account for parameter uncertainty. Standard Bayesian inference marginalises over θ but does not incorporate worst-case perturbations into the variational objective. A combined objective would take the form

$$\min_{\phi} \mathbb{E}_{(\mathbf{x}, y) \sim \mathcal{D}} \left[\max_{\|\boldsymbol{\delta}\| \leq \epsilon} \mathbb{E}_{q_{\phi}(\theta)} [\mathcal{L}(f_{\theta}(\mathbf{x} + \boldsymbol{\delta}), y)] \right] + \beta \text{KL}(q_{\phi}(\theta) \parallel p(\theta)), \quad (1.20)$$

where $q_{\phi}(\theta)$ is the variational posterior and β controls the regularisation strength. The convergence properties of this stochastic min-max problem, and its effect on calibration as measured by Expected Calibration Error (Equation (1.9)), are unknown. The gap is the absence of a training algorithm for this joint objective together with a theoretical analysis of its convergence and calibration guarantees. This gap is addressed by Methods M5: Stochastic Transformer and M6: UC-HGP.

Gap 7. Oversimplified threat models in game-theoretic formulations (Adversarial $\times$ Robustness).

The standard Stackelberg formulation (Equation (1.10)) assumes that the defender knows the adversary's strategy space and payoff function, and that the adversary responds optimally. In practice the defender faces epistemic uncertainty about the adversary's capabilities, captured by a belief distribution $p(\pi_a)$ over attacker strategies. The Bayesian Stackelberg problem is

$$\max_{\pi_d} \mathbb{E}_{p(\pi_a)} [U_d(\pi_d, \text{BR}(\pi_d, \pi_a))], \quad (1.21)$$

where $\text{BR}(\pi_d, \pi_a)$ denotes the best response of attacker type π_a to defender strategy π_d . This formulation has not been integrated with the epistemic uncertainty provided by a Bayesian graph neural network. Furthermore, online no-regret algorithms guarantee sub-linear regret only when the loss feedback is accurate and immediate. When feedback is delayed by τ rounds or corrupted by the adversary, the best known regret bound degrades to

$$R(T) = \mathcal{O}(\sqrt{(T + \tau T) \log K}) + \mathcal{O}(c), \quad (1.22)$$

where c counts the number of corrupted feedback rounds. No existing algorithm for the defender-adversary game on graph-structured data provides regret guarantees under both delayed and adversarially corrupted feedback [92, 93] while simultaneously incorporating epistemic uncertainty. The gap is the absence of such an algorithm together with a formal regret analysis. This gap is addressed by the game-theoretic components of Method M7: Certification.

The seven gaps enumerated above are not specific to any single deployment domain.

They arise wherever learned models on graphs must contend with adversarial perturbation, distributional heterogeneity, privacy regulation, computational constraints, and strategic opponents, whether the application is network monitoring, medical imaging, natural language processing, autonomous transportation, or agricultural analytics. Together, the gaps cover all twelve intersections of the condition-requirement matrix: Gaps 1 and 7 address the adversarial condition, Gaps 3, 4, and 5 address the non-stationary condition, Gap 2 addresses the distributed condition, and Gap 6 bridges the adversarial and non-stationary conditions through uncertainty quantification. The solutions developed in Chapter 2 address each gap through contributions at the level of mathematical modelling, algorithm design, and inference architecture. Their practical validation through application to network-traffic graphs is presented in Chapter 3.

1.9 Summary

This chapter has surveyed seven research domains, adversarial perturbation and defence on graphs, graph neural networks for structured and temporal data, federated learning on distributed graph fragments, state-space models for efficient temporal graph processing, Bayesian deep learning and uncertainty quantification on graphs, game-theoretic modelling of the defender-adversary interaction, and forward compatibility with evolving data-generating protocols, that together define the frontier of robust learning on dynamic graph structures. For each domain the chapter reviewed the state of the art, formulated the central problems mathematically and algorithmically, and identified the limitations that persist in the current literature. A consolidated gap analysis distilled seven principal deficiencies, each mapped to specific intersections of the three-condition, four-requirement matrix established in the Introduction: insufficient formal robustness guarantees under continuous-time graph dynamics, unfavourable privacy-utility trade-offs in federated graph learning, limited cross-distribution generalisation on temporal graphs, discrete-time constraints and computational cost, forward incompatibility with evolving protocols, unintegrated uncertainty quantification and adversarial robustness, and oversimplified game-theoretic threat models.

Each of these gaps motivates a specific contribution in Chapter 2, where seven novel methods (M1 through M7) are developed with full mathematical derivations, convergence and robustness proofs, algorithmic pseudocode, and complexity analyses. The methods are formulated at the level of the learning algorithm and the model architecture, independently of any particular application domain, so that their theoretical guarantees hold wherever robust, privacy-preserving, and efficient graph inference is required. Their practical validation through application to network-traffic graphs is presented in Chapter 3.

Chapter 2

Derived Solutions: Mathematical Models, Proofs, and Novel Algorithms for Graph Robustness

This chapter develops the theoretical core of the dissertation. For each of the seven gaps identified in the systematic gap analysis of Section 1.8, a novel method is presented with a formal problem statement, a complete mathematical formulation, convergence and robustness proofs, algorithmic pseudocode, and computational complexity analysis. The seven methods are labelled M1 through M7 and together address all twelve intersections of the three-condition, four-requirement matrix established in the Introduction (Figure 1). Every method is formulated at the level of operations on graph structures: no reference to a specific application domain is needed to state the theorems or verify the proofs. The chapter concludes with a unified optimisation framework that couples all seven methods through consistency regularisation, together with a convergence guarantee for the integrated system.

2.1 Method M1: CT-TGNN — Continuous-Time Temporal Graph Neural Dynamics

This section addresses Gap 1 (Adversarial $\times$ Robustness; Non-stationary $\times$ Robustness) by developing a graph neural network whose node representations evolve according to a neural ordinary differential equation [8], yielding certified robustness through Lipschitz-bounded continuous-time dynamics on evolving graph topologies.

2.1.1 Mathematical Formulation

Consider a temporal heterogeneous graph $(t) = (\cdot, (t), \mathbf{X}(t), \mathbf{E}(t))$ in which the node set $= \bigcup_{k=1}^K k$ comprises K node types, the edge set $(t) \subseteq \times$ evolves over time, node features $\mathbf{X}(t) = \{\mathbf{x}_v(t) \in \mathbb{R}^{d_x}\}_{v \in \cdot}$ capture entity attributes, and edge features $\mathbf{E}(t) = \{\mathbf{e}_{uv}(t) \in \mathbb{R}^{d_e}\}_{(u,v) \in (t)}$ encode interaction characteristics. The continuous-time graph neural ODE

defines the evolution of node embeddings through the parameterised differential equation

$$\frac{d\mathbf{h}_v(t)}{dt} = f_\theta(\mathbf{h}_v(t), \{\mathbf{h}_u(t)\}_{u \in \mathcal{N}_v(t)}, \mathbf{A}(t), t), \quad (2.1)$$

where $\mathbf{h}_v(t) \in \mathbb{R}^{d_h}$ is the embedding of node v at time t , $\mathcal{N}_v(t)$ denotes the neighbourhood of v under the current topology, $\mathbf{A}(t)$ encodes the adjacency structure, and f_θ is a neural network parameterised by θ .

For heterogeneous graphs the right-hand side employs type-specific message passing. Let v belong to type k and let $\mathcal{N}_v^{(k')}(t)$ denote its neighbours of type k' . The dynamics are

$$\frac{d\mathbf{h}_v^{(k)}(t)}{dt} = \sum_{k'=1}^K \sum_{u \in \mathcal{N}_v^{(k')}(t)} \alpha_{uv}^{(k,k')}(t) \text{MSG}_{k,k'}(\mathbf{h}_u^{(k')}(t), \mathbf{e}_{uv}(t)), \quad (2.2)$$

where the attention coefficients [2]

$$\alpha_{uv}^{(k,k')}(t) = \frac{\exp(\text{LeakyReLU}(\mathbf{a}_{k,k'}^\top [\mathbf{W}_k \mathbf{h}_v^{(k)}(t) \parallel \mathbf{W}_{k'} \mathbf{h}_u^{(k')}(t)]))}{\sum_{w \in \mathcal{N}_v^{(k')}(t)} \exp(\text{LeakyReLU}(\mathbf{a}_{k,k'}^\top [\mathbf{W}_k \mathbf{h}_v^{(k)}(t) \parallel \mathbf{W}_{k'} \mathbf{h}_w^{(k')}(t)]))} \quad (2.3)$$

weight neighbour importance adaptively, $\mathbf{W}_k, \mathbf{W}_{k'}$ are learnable projections, and $\parallel$ denotes concatenation.

2.1.2 Multi-Scale Temporal Dynamics

Phenomena that unfold across multiple time scales, from sub-second transients to month-long trends, require parallel ODE branches with distinct time constants. For scales $s = 1, \dots, S$ with constants $\tau_1 < \tau_2 < \dots < \tau_S$ the dynamics are

$$\frac{d\mathbf{h}_v^{(s)}(t)}{dt} = \frac{1}{\tau_s} f_{\theta_s}(\mathbf{h}_v^{(s)}(t), \{\mathbf{h}_u^{(s)}(t)\}_{u \in \mathcal{N}_v(t)}, \mathbf{A}(t), t), \quad (2.4)$$

and the final embedding aggregates across scales through learned attention,

$$\mathbf{h}_v(t) = \sum_{s=1}^S \beta_s(t) \mathbf{h}_v^{(s)}(t), \quad \beta_s(t) = \text{Softmax}_s(\mathbf{w}_s^\top \mathbf{h}_v^{(s)}(t)), \quad (2.5)$$

so that the relative contribution of each time scale adapts to the local dynamics.

2.1.3 Adjoint Sensitivity Method for Memory-Efficient Training

Training requires gradients of a loss $\mathcal{L} = \mathcal{L}(\mathbf{h}(T))$ with respect to θ through the ODE solution. Direct back-propagation through the numerical integrator stores all intermediate states, which is prohibitive for long horizons. The adjoint method [8, 25] defines an adjoint state

$\mathbf{a}(t) = \partial\mathcal{L}/\partial\mathbf{h}(t)$ satisfying the backward ODE

$$\frac{d\mathbf{a}(t)}{dt} = -\mathbf{a}(t)^\top \frac{\partial f_\theta}{\partial \mathbf{h}}(\mathbf{h}(t), t), \quad (2.6)$$

integrated from $\mathbf{a}(T) = \partial\mathcal{L}/\partial\mathbf{h}(T)$ to $\mathbf{a}(0)$. Parameter gradients follow from

$$\frac{\partial\mathcal{L}}{\partial\theta} = - \int_0^T \mathbf{a}(t)^\top \frac{\partial f_\theta}{\partial \theta}(\mathbf{h}(t), t) dt. \quad (2.7)$$

Memory cost is proportional to the model size rather than the trajectory length, enabling training on arbitrarily long temporal horizons.

2.1.4 Temporal Adaptive Batch Normalisation

Standard batch normalisation computes statistics over a fixed mini-batch, which is incompatible with continuous-time dynamics in which the distribution of hidden states changes continuously. Temporal adaptive batch normalisation computes running statistics over a sliding temporal window $[t - \Delta, t]$,

$$\text{TABN}(\mathbf{h}(t)) = \gamma(t) \frac{\mathbf{h}(t) - \mu_{[t-\Delta, t]}}{\sqrt{\sigma_{[t-\Delta, t]}^2 + \varepsilon}} + \beta(t), \quad (2.8)$$

where $\mu_{[t-\Delta, t]}$ and $\sigma_{[t-\Delta, t]}^2$ are the windowed mean and variance, and the affine parameters $\gamma(t), \beta(t)$ themselves evolve through auxiliary ODEs $d\gamma/dt = g_\gamma(\mathbf{h}(t))$, $d\beta/dt = g_\beta(\mathbf{h}(t))$ driven by learned adaptation networks.

2.1.5 Convergence and Robustness Analysis

Theorem 2.1 (CT-TGNN Convergence). *Suppose the dynamics function f_θ is L_f -Lipschitz in $\mathbf{h}$ and L_θ -Lipschitz in θ . Under gradient descent with learning rate $\eta < 2/(L_f L_\theta)$, the training loss converges to a stationary point at rate $\mathcal{O}(1/\sqrt{T})$, where T denotes the number of gradient steps.*

Proof sketch. The loss $\mathcal{L}(\theta)$ is evaluated through the ODE solution map $\mathbf{h}(T; \theta)$. By the chain rule and the Lipschitz properties of f_θ , the gradient $\nabla_\theta \mathcal{L}$ is bounded and Lipschitz-continuous, placing the problem in the standard framework of smooth non-convex optimisation. Applying the descent lemma with the stated learning rate yields the $\mathcal{O}(1/\sqrt{T})$ rate to an ϵ -stationary point. The full proof appears in Appendix A. $\square$

Theorem 2.2 (Lipschitz Robustness Certificate). *Let f_θ be L_g -Lipschitz in its first argument and let the integration horizon be T . For any two inputs $\mathbf{x}_1, \mathbf{x}_2$ the corresponding embeddings satisfy*

$$\|\mathbf{h}_1(T) - \mathbf{h}_2(T)\| \leq e^{L_g T} \|\mathbf{x}_1 - \mathbf{x}_2\|. \quad (2.9)$$

Consequently, if the classifier head has Lipschitz constant L_c , the end-to-end model is $L_c e^{L_g T}$ -Lipschitz, and the prediction is certified constant within a ball of radius $\rho = \Delta_{\min}/(L_c e^{L_g T})$, where $\Delta_{\min}$ is the margin between the top two logits.

Proof sketch. Let $\boldsymbol{\eta}(t) = \mathbf{h}_1(t) - \mathbf{h}_2(t)$. Then $d\boldsymbol{\eta}/dt = f_\theta(\mathbf{h}_1, t) - f_\theta(\mathbf{h}_2, t)$, so $d\|\boldsymbol{\eta}\|/dt \leq L_g \|\boldsymbol{\eta}\|$ by the Lipschitz property. Grönwall's inequality gives $\|\boldsymbol{\eta}(T)\| \leq e^{L_g T} \|\boldsymbol{\eta}(0)\|$, and $\boldsymbol{\eta}(0) = \mathbf{x}_1 - \mathbf{x}_2$. Composing with the classifier head completes the certificate. The full derivation with spectral-norm constraints enforcing the Lipschitz bound during training appears in Appendix A. $\square$

2.1.6 Computational Complexity

Solving the graph ODE with an adaptive Runge-Kutta integrator at tolerance ε_{ode} requires $\mathcal{O}(\varepsilon_{\text{ode}}^{-1/5})$ function evaluations. Each evaluation performs message passing at cost $\mathcal{O}(\|d\| d_h)$ for bounded degree d and embedding dimension d_h . The total training complexity per epoch is therefore

$$\mathcal{O}(\|d\| d_h \varepsilon_{\text{ode}}^{-1/5} T_{\text{train}}), \quad (2.10)$$

which is near-linear in graph size when $\|d\| = \mathcal{O}(\|d\|)$.

Algorithm 1: CT-TGNN Training with Adjoint Sensitivity

Input: Temporal graph sequence $\{(t_i)\}_{i=1}^N$, labels $\{y_i\}$, learning rate η , integration horizon T , tolerance ε_{ode}
Output: Trained parameters θ^*

- 1 Initialise parameters θ_0 and multi-scale time constants $\tau_1 < \dots < \tau_S$;
- 2 **for** $epoch = 1, 2, \dots, E$ **do**
- 3 **for** each mini-batch $\{(t_j), y_j\}$ **do**
- 4 Set initial embeddings $\mathbf{h}_v(0) \leftarrow \text{Embed}(\mathbf{x}_v(t_j))$ for all $v \in$;
- 5 **for** scale $s = 1, \dots, S$ **do**
- 6 Solve $d\mathbf{h}_v^{(s)}/dt = \tau_s^{-1} f_{\theta_s}(\mathbf{h}_v^{(s)}, \mathcal{N}_v, \mathbf{A}, t)$ from 0 to T using adaptive RK with tolerance ε_{ode} ;
- 7 Compute fused embeddings $\mathbf{h}_v(T) = \sum_s \beta_s(T) \mathbf{h}_v^{(s)}(T)$;
- 8 Compute classification loss $\mathcal{L} = \mathcal{L}_{\text{CE}}(f_{\text{head}}(\mathbf{h}(T)), y)$;
- 9 Set adjoint terminal condition $\mathbf{a}(T) = \partial \mathcal{L} / \partial \mathbf{h}(T)$;
- 10 Solve adjoint ODE backward: $d\mathbf{a}/dt = -\mathbf{a}^\top (\partial f_\theta / \partial \mathbf{h})$ from T to 0;
- 11 Accumulate parameter gradients $\partial \mathcal{L} / \partial \theta = -\int_0^T \mathbf{a}^\top (\partial f_\theta / \partial \theta) dt$;
- 12 Update $\theta \leftarrow \theta - \eta \partial \mathcal{L} / \partial \theta$;
- 13 Compute Lipschitz certificate $\rho = \Delta_{\min}/(L_c e^{L_g T})$ for each test input;
- 14 **return** $\theta^*, \{\rho_i\}$

2.2 Method M3: TripleE-TGNN — Multi-Level Temporal Graph Embeddings

This section addresses Gap 3 (Non-stationary $\times$ Accuracy; Non-stationary $\times$ Efficiency) by developing multi-level graph embeddings that capture phenomena at three hierarchical levels of abstraction, enabling rich structural representations that resist catastrophic forgetting as the graph topology evolves.

2.2.1 Hierarchical Graph Representation

Many systems exhibit a natural hierarchy: a coarse level of functional components, an intermediate level of operational traces that span components, and a fine level of infrastructure hosts. This structure is captured by three coupled temporal graphs: a component graph $s(t) = (s,s(t), \mathbf{X}_s(t))$, a trace graph $r(t) = (r,r(t), \mathbf{X}_r(t))$, and a host graph $n(t) = (n,n(t), \mathbf{X}_n(t))$. Bipartite coupling graphs sr, sn, rn link entities across levels: a component deploys on a host, a trace traverses a component, and a trace executes on a host.

2.2.2 Multi-Level Embedding Learning

Each granularity $g \in \{s, r, n\}$ maintains a dedicated encoder,

$$\mathbf{h}_v^{(g)}(t) = \text{ENC}_g(\mathbf{x}_v^{(g)}(t), \{\mathbf{h}_u^{(g)}(t-1)\}_{u \in \mathcal{N}_v^{(g)}(t)}, \Delta t), \quad (2.11)$$

where ENC_g is a granularity-specific temporal graph neural network. Cross-granularity message passing integrates information across abstraction levels through bipartite attention,

$$\mathbf{m}_v^{(g \rightarrow g')}(t) = \sum_{u \in \mathcal{N}_v^{(g, g')}} \alpha_{uv}^{(g, g')}(t) \mathbf{W}_{g \rightarrow g'} \mathbf{h}_u^{(g)}(t), \quad (2.12)$$

with attention scores

$$\alpha_{uv}^{(g, g')}(t) = \text{Softmax}_u \left(\frac{(\mathbf{Q}_{g'} \mathbf{h}_v^{(g')}(t))^\top (\mathbf{K}_g \mathbf{h}_u^{(g)}(t))}{\sqrt{d_k}} \right), \quad (2.13)$$

where $\mathbf{Q}_{g'}$, $\mathbf{K}_g$ are query and key projections and $\mathcal{N}_v^{(g, g')}$ denotes neighbours in the coupling graph. The final embedding fuses intra- and cross-granularity information through layer normalisation,

$$\mathbf{h}_v^{(g)}(t) \leftarrow \text{LayerNorm} \left(\mathbf{h}_v^{(g)}(t) + \sum_{g' \neq g} \mathbf{m}_v^{(g' \rightarrow g)}(t) \right). \quad (2.14)$$

2.2.3 Dual Temporal-Spatial Attention

Within each granularity, temporal attention aggregates historical states over a window of width w ,

$$\tilde{\mathbf{h}}_v^{(g)}(t) = \sum_{\tau=t-w}^{t-1} \alpha_\tau^{\text{temp}} \mathbf{h}_v^{(g)}(\tau), \quad \alpha_\tau^{\text{temp}} = \text{Softmax}_\tau(\mathbf{w}_{\text{temp}}^\top \mathbf{h}_v^{(g)}(\tau)), \quad (2.15)$$

while spatial attention aggregates the current neighbourhood,

$$\hat{\mathbf{h}}_v^{(g)}(t) = \sum_{u \in \mathcal{N}_v^{(g)}(t)} \alpha_u^{\text{spat}} \mathbf{h}_u^{(g)}(t), \quad \alpha_u^{\text{spat}} = \text{Softmax}_u(\mathbf{w}_{\text{spat}}^\top [\mathbf{h}_v^{(g)}(t) \parallel \mathbf{h}_u^{(g)}(t)]). \quad (2.16)$$

The combined representation is

$$\mathbf{h}_v^{(g)}(t) = \text{MLP}([\tilde{\mathbf{h}}_v^{(g)}(t) \parallel \hat{\mathbf{h}}_v^{(g)}(t) \parallel \mathbf{h}_v^{(g)}(t)]). \quad (2.17)$$

2.2.4 Elastic Weight Consolidation for Topology Evolution

When the graph topology changes, for example through deployment events or infrastructure reconfiguration, a model retrained on the new topology may overwrite previously learned representations. Elastic weight consolidation [102] prevents this catastrophic forgetting by penalising changes to parameters that are important for earlier tasks. After learning on topology configuration i with optimal parameters θ_i^* , the loss for subsequent configuration $i+1$ includes the regularisation term

$$\mathcal{L}_{\text{EWC}}(\theta) = \mathcal{L}_{i+1}(\theta) + \lambda_{\text{ewc}} \sum_j F_j (\theta_j - \theta_{i,j}^*)^2, \quad (2.18)$$

where the diagonal Fisher information $F_j = \mathbb{E}[(\partial \log p(y|\mathbf{x}, \theta_i^*) / \partial \theta_j)^2]$ quantifies the importance of parameter j for the previous configuration and λ_{ewc} controls the consolidation strength.

2.2.5 Joint Multi-Level Classification Objective

The model performs simultaneous classification at all three granularities through a multi-task loss,

$$\mathcal{L}_{\text{joint}} = \lambda_s \mathcal{L}_s + \lambda_r \mathcal{L}_r + \lambda_n \mathcal{L}_n + \lambda_{\text{con}} \mathcal{L}_{\text{consist}}, \quad (2.19)$$

where $\mathcal{L}_g$ is the cross-entropy loss at granularity g , λ_g are balancing weights, and the consistency loss

$$\mathcal{L}_{\text{consist}} = \sum_{(v_s, v_r) \in s_r} \text{KL}(p_s(y|v_s) \parallel p_r(y|v_r)) + \sum_{(v_s, v_n) \in s_n} \text{KL}(p_s(y|v_s) \parallel p_n(y|v_n)) \quad (2.20)$$

enforces agreement between the predictions of linked entities across abstraction levels.

Algorithm 2: TripleE-TGNN Multi-Level Training

Input: Hierarchical graphs $\{s, r, n\}$ with couplings $_{sr, sn, rn}$, labels at each granularity, EWC strength λ_{ewc}

Output: Trained encoders $\text{ENC}_s, \text{ENC}_r, \text{ENC}_n$ and cross-attention parameters

- 1 Initialise all encoder and attention parameters;
- 2 **for** each topology configuration i **do**
- 3 **for** $\text{epoch} = 1, \dots, E$ **do**
- 4 **for** each mini-batch **do**
- 5 **for** granularity $g \in \{s, r, n\}$ **do**
- 6 Compute intra-granularity embeddings $\mathbf{h}_v^{(g)}(t) \leftarrow \text{ENC}_g(\mathbf{x}_v^{(g)}, \mathcal{N}_v^{(g)}, \Delta t)$;
- 7 Compute temporal attention $\tilde{\mathbf{h}}_v^{(g)}$ and spatial attention $\hat{\mathbf{h}}_v^{(g)}$;
- 8 Fuse: $\mathbf{h}_v^{(g)} \leftarrow \text{MLP}([\tilde{\mathbf{h}}_v^{(g)} \parallel \hat{\mathbf{h}}_v^{(g)} \parallel \mathbf{h}_v^{(g)}])$;
- 9 **for** each pair (g, g') with $g \neq g'$ **do**
- 10 Compute cross-granularity messages $\mathbf{m}_v^{(g \rightarrow g')}$ via bipartite attention;
- 11 Update: $\mathbf{h}_v^{(g')} \leftarrow \text{LayerNorm}(\mathbf{h}_v^{(g')} + \mathbf{m}_v^{(g \rightarrow g')})$;
- 12 Compute $\mathcal{L}_{\text{joint}} = \lambda_s \mathcal{L}_s + \lambda_r \mathcal{L}_r + \lambda_n \mathcal{L}_n + \lambda_{\text{con}} \mathcal{L}_{\text{consist}}$;
- 13 **if** $i > 1$ **then**
- 14 Add EWC penalty: $\mathcal{L} \leftarrow \mathcal{L}_{\text{joint}} + \lambda_{\text{ewc}} \sum_j F_j(\theta_j - \theta_{i-1, j}^*)^2$;
- 15 Backpropagate and update all parameters;
- 16 Store Fisher diagonal $\{F_j\}$ and optimal θ_i^* for EWC;
- 17 **return** trained encoders and attention parameters

2.3 Method M2: FedLLM-API — Federated Graph Optimisation with Privacy

This section addresses Gap 2 (Distributed $\times$ Robustness; Distributed $\times$ Privacy; Distributed $\times$ Efficiency) by developing a federated learning algorithm that simultaneously provides corruption-tolerant aggregation, formal differential privacy, and optimal-transport-based distributional alignment for graph-structured data distributed across multiple participants.

2.3.1 Byzantine-Robust Aggregation

Consider K participants with local datasets $\mathcal{D}_1, \dots, \mathcal{D}_K$, of which at most a fraction q are Byzantine. At communication round t the server broadcasts global model $\mathbf{w}_t$ and each honest client k computes a local update through gradient descent on its own data, $\mathbf{w}_{t+1}^{(k)} = \mathbf{w}_t - \eta \nabla_{\mathbf{w}} \mathcal{L}_k(\mathbf{w}_t; \mathcal{D}_k)$. Malicious clients submit arbitrary vectors. The server aggregates through coordinate-wise trimmed mean [12, 61]: for each parameter dimension j the client

values are sorted and the $b = \lfloor Kq \rfloor$ largest and smallest are discarded, yielding

$$\bar{w}_{t+1,j} = \frac{1}{K-2b} \sum_{i=b+1}^{K-b} w_{t+1,j}^{(i)}, \quad (2.21)$$

where $w_{t+1,j}^{(i)}$ is the i -th order statistic.

Theorem 2.3 (Byzantine-Robust Convergence). *Assume the global loss $F(\mathbf{w}) = \frac{1}{K} \sum_{k=1}^K F_k(\mathbf{w})$ is μ -strongly convex and L -smooth, and the fraction of Byzantine clients satisfies $f < q < 1/2$. With learning rate $\eta \leq 1/L$, coordinate-wise trimmed mean achieves*

$$\mathbb{E}[\|\mathbf{w}_T - \mathbf{w}^*\|^2] \leq \left(1 - \frac{\mu\eta}{2}\right)^T \|\mathbf{w}_0 - \mathbf{w}^*\|^2 + \frac{2\eta}{\mu(K-2b)} \left(\sum_{k=1}^{K-2b} \sigma_k^2 + C_{\text{byz}} \right), \quad (2.22)$$

where σ_k^2 is the gradient variance of honest client k and C_{byz} bounds the Byzantine influence.

Proof sketch. The key step is showing that the trimmed mean of honest-plus-Byzantine gradient vectors is close to the true mean of honest gradients. Let $\bar{\mathbf{g}}_H$ be the mean of honest gradients and $\bar{\mathbf{g}}_{\text{TM}}$ the trimmed-mean output. Because at most b of the $2b$ discarded entries can be honest, the error $\|\bar{\mathbf{g}}_{\text{TM}} - \bar{\mathbf{g}}_H\|$ is bounded by the order statistics of honest gradient norms via concentration inequalities. Substituting into the standard strongly convex descent recursion yields the stated bound. Full details are in Appendix A. $\square$

2.3.2 Differential Privacy Through Calibrated Noise

Each client clips its gradient norm to a threshold C and adds Gaussian noise [13, 14] before transmission,

$$\bar{\mathbf{g}}_k = \mathbf{g}_k / \max\left(1, \frac{\|\mathbf{g}_k\|}{C}\right), \quad \tilde{\mathbf{g}}_k = \bar{\mathbf{g}}_k + \mathcal{N}(\mathbf{0}, \sigma^2 C^2 \mathbf{I}), \quad (2.23)$$

where the noise scale σ is determined by the target privacy parameters (ϵ, δ) . Privacy amplification by sub-sampling with batch probability q_b and composition over T rounds through the moments accountant [60, 103] yields the overall privacy guarantee

$$\epsilon_{\text{total}} = \min_{\alpha > 1} \frac{1}{\alpha - 1} \log \frac{1}{\delta} + \frac{T q_b^2 \alpha}{2\sigma^2}. \quad (2.24)$$

2.3.3 Optimal Transport for Distribution Alignment

Heterogeneous client distributions degrade convergence. Optimal transport aligns them by computing the Wasserstein barycenter. Given discrete source samples $\{\mathbf{x}_i\}_{i=1}^{n_s}$ and target samples $\{\mathbf{y}_j\}_{j=1}^{n_t}$, the discrete transport plan solves

$$\mathbf{T}^* = \arg \min_{\mathbf{T} \in \mathbb{R}_+^{n_s \times n_t}} \langle \mathbf{C}, \mathbf{T} \rangle_F \quad \text{s.t.} \quad \mathbf{T} \mathbf{1}_{n_t} = \mathbf{a}, \quad \mathbf{T}^\top \mathbf{1}_{n_s} = \mathbf{b}, \quad (2.25)$$

where $C_{ij} = c(\mathbf{x}_i, \mathbf{y}_j)$ is the cost matrix and $\mathbf{a}, \mathbf{b}$ lie in probability simplices. The Sinkhorn algorithm [15, 16] computes the entropically regularised solution through alternating scaling,

$$\mathbf{u}^{(\ell+1)} = \mathbf{a} \oslash (\mathbf{K}\mathbf{v}^{(\ell)}), \quad \mathbf{v}^{(\ell+1)} = \mathbf{b} \oslash (\mathbf{K}^\top \mathbf{u}^{(\ell+1)}), \quad (2.26)$$

where $\mathbf{K} = \exp(-\mathbf{C}/\lambda)$ is the entropic kernel, $\oslash$ denotes element-wise division, and $\lambda > 0$ controls regularisation. After convergence the transport plan is $\mathbf{T}^* = \text{diag}(\mathbf{u})\mathbf{K}\text{diag}(\mathbf{v})$.

2.3.4 Differentially Private Optimal Transport

To protect sample locations, Gaussian noise is added to the empirical distribution before transport computation,

$$\tilde{\mu}_s = \frac{1}{n_s} \sum_{i=1}^{n_s} \delta_{\mathbf{x}_i + \boldsymbol{\xi}_i}, \quad \boldsymbol{\xi}_i \sim \mathcal{N}(\mathbf{0}, \sigma_{\text{DP}}^2 \mathbf{I}), \quad (2.27)$$

and Laplace noise is added to the resulting transport plan,

$$\tilde{\mathbf{T}} = \mathbf{T}^* + \text{Lap}\left(\frac{\Delta_T}{\epsilon_{\text{OT}}}\right), \quad (2.28)$$

where the sensitivity $\Delta_T = \max_{\mathcal{D}, \mathcal{D}'} \|\mathbf{T}^*(\mathcal{D}) - \mathbf{T}^*(\mathcal{D}')\|_1$ bounds the maximum change under single-record substitution and ϵ_{OT} is the privacy budget allocated to transport.

2.3.5 Large Language Model Integration for Zero-Shot Generalisation

Semantic reasoning about novel categories is achieved by constructing a natural language prompt from the structural features of an observation and querying a large language model to produce a probability $p_{\text{LLM}}(\text{class} \mid \mathbf{r})$. The final prediction is the ensemble

$$p_{\text{final}} = \lambda_{\text{LLM}} p_{\text{LLM}}(\text{class} \mid \mathbf{r}) + (1 - \lambda_{\text{LLM}}) p_{\text{GNN}}(\text{class} \mid \mathbf{r}), \quad (2.29)$$

where $\mathbf{r}$ is the graph constructed from the observation and λ_{LLM} balances the two sources.

2.4 Forward-Compatible Detection Under Protocol Regime Change

This section addresses Gap 5 (Non-stationary $\times$ Robustness) by constructing a detection architecture that maintains robustness when the data-generating protocols underlying the graph undergo structural change. The primary instance considered is the migration to post-quantum cryptographic standards [94, 96], which alters the statistical properties of graph-edge features in ways that models trained on the previous regime cannot anticipate.

Algorithm 3: FedLLM-API: Byzantine-Robust Federated Optimisation with DP-OT

Input: Clients $\mathcal{D}_1, \dots, \mathcal{D}_K$, fraction Byzantine q , clipping threshold C , privacy budget (ϵ, δ) , Sinkhorn regularisation λ , communication rounds T

Output: Global model $\mathbf{w}_T$

- 1 Initialise global model $\mathbf{w}_0$; compute noise scale $\sigma = C\sqrt{2\ln(1.25/\delta)}/\epsilon$;
- 2 **for** round $t = 0, 1, \dots, T-1$ **do**
- 3 Broadcast $\mathbf{w}_t$ to all clients;
- 4 **for** each client $k = 1, \dots, K$ (in parallel) **do**
- 5 Compute local gradient $\mathbf{g}_k = \nabla_{\mathbf{w}} \mathcal{L}_k(\mathbf{w}_t; \mathcal{D}_k)$;
- 6 Clip: $\bar{\mathbf{g}}_k \leftarrow \mathbf{g}_k / \max(1, \|\mathbf{g}_k\|/C)$;
- 7 Add noise: $\tilde{\mathbf{g}}_k \leftarrow \bar{\mathbf{g}}_k + \mathcal{N}(\mathbf{0}, \sigma^2 C^2 \mathbf{I})$;
- 8 Send $\tilde{\mathbf{g}}_k$ to server;
- 9 Set $b \leftarrow \lfloor Kq \rfloor$;
- 10 **for** each dimension j **do**
- 11 Sort $\{\tilde{g}_{k,j}\}_{k=1}^K$; compute trimmed mean $\bar{g}_{t,j} = \frac{1}{K-2b} \sum_{i=b+1}^{K-b} \tilde{g}_j^{(i)}$;
- 12 **if** heterogeneity detected via W_1 estimate **then**
- 13 Compute noisy source distributions $\tilde{\mu}_k$ by adding $\mathcal{N}(\mathbf{0}, \sigma_{\text{DP}}^2 \mathbf{I})$ to samples;
- 14 Run Sinkhorn iterations to compute regularised transport plan $\mathbf{T}^*$;
- 15 Add Laplace noise: $\tilde{\mathbf{T}} \leftarrow \mathbf{T}^* + \text{Lap}(\Delta_T/\epsilon_{\text{OT}})$;
- 16 Apply alignment correction to $\bar{\mathbf{g}}_t$ using $\tilde{\mathbf{T}}$;
- 17 Update global model: $\mathbf{w}_{t+1} \leftarrow \mathbf{w}_t - \eta \bar{\mathbf{g}}_t$;
- 18 **return** $\mathbf{w}_T$

2.4.1 Protocol-Specific Feature Extraction

When a new protocol regime is deployed, the observable features on graph edges, such as timing distributions, message sizes, and interaction counts, change in characteristic ways. For lattice-based protocols, for example, the timing of key-exchange operations exhibits variations that correlate with the computational structure of the underlying lattice operations. These variations are captured as a statistical feature vector,

$$\mathbf{f}_{\text{lattice}} = [\mu_t, \sigma_t, \text{skew}(t), \text{kurt}(t), \text{pctl}(t), \text{acf}(t)], \quad (2.30)$$

where $t = (t_1, \dots, t_n)$ denotes operational timing measurements, and the entries capture central tendency, dispersion, distribution shape, and temporal dependence. Analogous feature vectors $\mathbf{f}_{\text{code}}$ and $\mathbf{f}_{\text{hash}}$ are defined for code-based and hash-based protocol families, incorporating structural parameters such as key sizes, error weights, syndrome entropy, signature sizes, hash invocations, and Merkle tree depths.

2.4.2 Unified Multi-Protocol Detection Architecture

Protocol-specific encoders produce latent representations $\mathbf{z}_{\text{lattice}} = \text{ENC}_{\text{lattice}}(\mathbf{f}_{\text{lattice}})$, $\mathbf{z}_{\text{code}} = \text{ENC}_{\text{code}}(\mathbf{f}_{\text{code}})$, $\mathbf{z}_{\text{hash}} = \text{ENC}_{\text{hash}}(\mathbf{f}_{\text{hash}})$. Cross-attention fusion enables interaction between the three protocol representations,

$$\mathbf{z}_{\text{fused}} = \text{MultiHeadAttention}([\mathbf{z}_{\text{lattice}}, \mathbf{z}_{\text{code}}, \mathbf{z}_{\text{hash}}]), \quad (2.31)$$

and multi-task classification predicts both the category label and the target protocol,

$$p(\text{class}, \text{protocol}) = \text{Softmax}(\mathbf{W}_{\text{out}}\mathbf{z}_{\text{fused}} + \mathbf{b}_{\text{out}}). \quad (2.32)$$

2.4.3 Formal Security Reduction

The robustness guarantee of the detection framework under the new protocol regime inherits from the hardness of the Learning With Errors problem [96, 104]. Given $\mathbf{A} \in_q^{m \times n}$ and $\mathbf{b} = \mathbf{A}\mathbf{s} + \mathbf{e} \bmod q$ for secret $\mathbf{s}$ and small error $\mathbf{e}$, any polynomial-time adversary that distinguishes a legitimate from a manipulated protocol-negotiation sequence can be used to construct a solver for LWE, contradicting the assumed hardness. The formal reduction, presented in full in Appendix A, establishes that the false-negative rate of the classifier is bounded by the advantage of the best LWE solver plus a statistical term that decreases with the number of training samples.

Algorithm 4: Forward-Compatible Multi-Protocol Detection

Input: Graph event stream with mixed protocol-regime edges, protocol-specific encoders, adversarial training budget ϵ

Output: Classification (class, protocol) and LWE-based robustness certificate

- 1 **for** *each incoming graph event* **do**
- 2 Identify protocol regime via protocol-negotiation sequence analysis;
- 3 **if** *lattice-based (Kyber/Dilithium)* **then**
- 4 Extract timing features $\mathbf{f}_{\text{lattice}} = [\mu_t, \sigma_t, \text{skew}, \text{kurt}, \text{pctl}, \text{acf}]$;
- 5 Encode: $\mathbf{z}_{\text{lattice}} \leftarrow \text{ENC}_{\text{lattice}}(\mathbf{f}_{\text{lattice}})$;
- 6 **else if** *code-based (McEliece/Niederreiter)* **then**
- 7 Extract structural features $\mathbf{f}_{\text{code}}$; encode $\mathbf{z}_{\text{code}} \leftarrow \text{ENC}_{\text{code}}(\mathbf{f}_{\text{code}})$;
- 8 **else**
- 9 Extract hash-based features $\mathbf{f}_{\text{hash}}$; encode $\mathbf{z}_{\text{hash}} \leftarrow \text{ENC}_{\text{hash}}(\mathbf{f}_{\text{hash}})$;
- 10 Fuse: $\mathbf{z}_{\text{fused}} \leftarrow \text{MultiHeadAttention}([\mathbf{z}_{\text{lattice}}, \mathbf{z}_{\text{code}}, \mathbf{z}_{\text{hash}}])$;
- 11 Classify: $p(\text{class}, \text{protocol}) \leftarrow \text{Softmax}(\mathbf{W}_{\text{out}}\mathbf{z}_{\text{fused}} + \mathbf{b}_{\text{out}})$;
- 12 **Training:** Alternate between standard forward passes and adversarial forward passes with PGD perturbations $\|\delta\| \leq \epsilon$ applied to feature vectors;
- 13 **Certification:** Verify LWE reduction (Appendix A) and report robustness parameter λ_{sec} ;
- 14 **return** classification and robustness certificate

2.5 Method M4: MambaShield — State-Space Online Adaptation with Poisoning Resilience

This section addresses Gap 4 (Non-stationary $\times$ Efficiency; Adversarial $\times$ Efficiency) by developing an efficient sequential model based on selective state spaces with progressive robustness distillation and PAC-Bayesian certification, achieving near-linear computational cost while maintaining resilience under poisoning of the graph update stream.

2.5.1 Selective State-Space Formulation

The selective state-space model [19, 20] extends the classical linear system through input-dependent transitions. Given an input sequence $u(t) \in \mathbb{R}^{d_{\text{in}}}$, the continuous-time dynamics are

$$\frac{d\mathbf{h}(t)}{dt} = \mathbf{A}(u(t))\mathbf{h}(t) + \mathbf{B}(u(t))u(t), \quad y(t) = \mathbf{C}\mathbf{h}(t) + \mathbf{D}u(t), \quad (2.33)$$

where $\mathbf{A}(u) = f_A(u; \theta_A) \in \mathbb{R}^{d_s \times d_s}$ and $\mathbf{B}(u) = f_B(u; \theta_B) \in \mathbb{R}^{d_s \times d_{\text{in}}}$ are computed by learned projections. Discretisation through zero-order hold at step size Δ produces

$$\mathbf{h}_k = \bar{\mathbf{A}}_k \mathbf{h}_{k-1} + \bar{\mathbf{B}}_k u_k, \quad \bar{\mathbf{A}}_k = \exp(\mathbf{A}(u_k)\Delta), \quad \bar{\mathbf{B}}_k = \mathbf{A}(u_k)^{-1}(\bar{\mathbf{A}}_k - \mathbf{I})\mathbf{B}(u_k), \quad (2.34)$$

and the convolution kernel $\bar{\mathbf{K}} = (\mathbf{C}\bar{\mathbf{A}}^i\bar{\mathbf{B}})_{i=0}^{L-1}$ is evaluated in $\mathcal{O}(L \log L)$ time through the Fast Fourier Transform.

2.5.2 Progressive Adversarial Robustness Distillation

An ensemble of M teacher models $\{f_{\theta_m}\}_{m=1}^M$, each trained on a disjoint data subset, provides robustness through knowledge distillation [105]. The student model f_ϕ learns from the teachers through the loss

$$\mathcal{L}_{\text{distill}} = \sum_{m=1}^M w_m \text{KL}(p_\phi(y|\mathbf{x}) \| p_{\theta_m}(y|\mathbf{x})), \quad (2.35)$$

where teacher weights w_m reflect validation performance. A curriculum schedule $\rho(t) = \min(\rho_0 + \alpha t, \rho_{\max})$ progressively increases the fraction of corrupted samples in each mini-batch, and the total training loss interpolates between the task loss and the distillation loss,

$$\mathcal{L}_{\text{total}} = (1 - \lambda(t)) \mathcal{L}_{\text{task}} + \lambda(t) \mathcal{L}_{\text{distill}}, \quad \lambda(t) = \min(\lambda_0 + \beta t, 1), \quad (2.36)$$

so that teacher influence grows as the poisoning intensity increases.

2.5.3 PAC-Bayesian Generalisation Bound

Theorem 2.4 (PAC-Bayesian Bound for Selective State-Space Models [80, 106]). *For any $\delta \in (0, 1)$, with probability at least $1 - \delta$ over the draw of a training set S of size n , for all posterior distributions Q over parameters,*

$$\mathbb{E}_{\theta \sim Q}[R(\theta)] \leq \mathbb{E}_{\theta \sim Q}[\hat{R}_n(\theta)] + \sqrt{\frac{\text{KL}(Q \| P) + \log(2\sqrt{n}/\delta)}{2n}}, \quad (2.37)$$

where $R(\theta)$ is the true risk, $\hat{R}_n(\theta)$ is the empirical risk, and P is a data-independent prior.

Proof sketch. The proof follows the standard PAC-Bayesian template of Catoni, applying a change of measure from P to Q through the Donsker-Varadhan variational representation of the KL divergence, combined with a moment-generating-function argument for the empirical process. The bound is valid uniformly over all posteriors Q , including those concentrated on a single parameter vector, and is therefore non-vacuous whenever the KL term is small relative to n . With Gaussian prior $P = \mathcal{N}(\mathbf{0}, \sigma_P^2 \mathbf{I})$ and posterior $Q = \mathcal{N}(\boldsymbol{\mu}_Q, \sigma_Q^2 \mathbf{I})$, the KL term has the closed form

$$\text{KL}(Q \| P) = \frac{d}{2} \left(\frac{\sigma_Q^2}{\sigma_P^2} + \frac{\|\boldsymbol{\mu}_Q\|^2}{\sigma_P^2} - 1 - \log \frac{\sigma_Q^2}{\sigma_P^2} \right), \quad (2.38)$$

which is minimised by keeping the posterior close to the origin and its variance close to that of the prior. Full details appear in Appendix A. $\square$

Algorithm 5: MambaShield: Training with Progressive Adversarial Robustness Distillation

Input: Training data $\mathcal{D}$, teacher ensemble $\{f_{\theta_m}\}_{m=1}^M$, curriculum schedule $\rho_0, \alpha, \rho_{\max}$, PAC-Bayes prior P

Output: Robust student model f_ϕ with PAC-Bayesian certificate

- 1 Initialise student parameters ϕ_0 ;
- 2 Partition $\mathcal{D}$ into M disjoint subsets; train each teacher f_{θ_m} on its subset;
- 3 Compute teacher weights $w_m \propto \text{ValAcc}(f_{\theta_m})$;
- 4 **for** iteration $t = 1, \dots, T_{\text{train}}$ **do**
- 5 Set poisoning fraction $\rho(t) = \min(\rho_0 + \alpha t, \rho_{\max})$;
- 6 Set distillation weight $\lambda(t) = \min(\lambda_0 + \beta t, 1)$;
- 7 Sample mini-batch; corrupt fraction $\rho(t)$ of labels;
- 8 **for** each sample $(\mathbf{x}, y)$ **do**
- 9 Discretise SSM: $\bar{\mathbf{A}}_k = \exp(\mathbf{A}(u_k)\Delta)$, $\bar{\mathbf{B}}_k = \mathbf{A}^{-1}(\bar{\mathbf{A}}_k - \mathbf{I})\mathbf{B}(u_k)$;
- 10 Run selective scan: $\mathbf{h}_k = \bar{\mathbf{A}}_k \mathbf{h}_{k-1} + \bar{\mathbf{B}}_k u_k$ for $k = 1, \dots, L$;
- 11 Compute student prediction $p_\phi(y|\mathbf{x})$;
- 12 Compute $\mathcal{L}_{\text{task}} = \text{CE}(p_\phi, y)$;
- 13 Compute $\mathcal{L}_{\text{distill}} = \sum_m w_m \text{KL}(p_\phi \| p_{\theta_m})$;
- 14 Total: $\mathcal{L} = (1 - \lambda(t))\mathcal{L}_{\text{task}} + \lambda(t)\mathcal{L}_{\text{distill}} + \gamma \text{KL}(Q_\phi \| P)$;
- 15 Update $\phi \leftarrow \phi - \eta \nabla_\phi \mathcal{L}$;
- 16 Evaluate PAC-Bayes bound: $R \leq \hat{R}_n + \sqrt{(\text{KL}(Q \| P) + \log(2\sqrt{n}/\delta)) / (2n)}$;
- 17 **return** f_ϕ and generalisation certificate

2.6 Methods M5 and M6: Uncertainty-Calibrated Inference on Graphs

This section addresses Gap 6 (Adversarial $\times$ Accuracy; Non-stationary $\times$ Accuracy) by developing two complementary methods for uncertainty quantification on graph-structured data. Method M5: Stochastic Transformer introduces a transformer architecture in which the attention mechanism is probabilistic, enabling uncertainty-aware inference under adversarial perturbation. Method M6: UC-HGP provides hierarchical Gaussian process uncertainty quantification at multiple temporal scales. Together, these methods formally decompose predictive uncertainty into epistemic and aleatoric components [24, 88] and incorporate adversarial perturbations of the graph into the variational training objective.

2.6.1 Variational Bayesian Attention

For an input sequence $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_n] \in \mathbb{R}^{n \times d}$, variational distributions [29] are placed over the query, key, and value projection matrices,

$$q_{\phi_Q}(\mathbf{W}_Q) = \mathcal{N}(\boldsymbol{\mu}_Q, \text{diag}(\boldsymbol{\sigma}_Q^2)), \quad q_{\phi_K}(\mathbf{W}_K) = \mathcal{N}(\boldsymbol{\mu}_K, \text{diag}(\boldsymbol{\sigma}_K^2)), \quad q_{\phi_V}(\mathbf{W}_V) = \mathcal{N}(\boldsymbol{\mu}_V, \text{diag}(\boldsymbol{\sigma}_V^2)). \quad (2.39)$$

Samples are drawn through the reparameterisation trick, $\mathbf{W}_Q = \boldsymbol{\mu}_Q + \boldsymbol{\sigma}_Q \odot \boldsymbol{\epsilon}_Q$ with $\boldsymbol{\epsilon}_Q \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, so that attention scores $A_{ij} = (\mathbf{x}_i \mathbf{W}_Q)^\top (\mathbf{x}_j \mathbf{W}_K) / \sqrt{d_k}$ become random variables

whose uncertainty propagates to the output. Monte Carlo estimation with M samples approximates the expected attention,

$$\mathbb{E}_{q_\phi}[\text{Attention}(\mathbf{X})] \approx \frac{1}{M} \sum_{m=1}^M \text{Attention}(\mathbf{X}; \mathbf{W}_Q^{(m)}, \mathbf{W}_K^{(m)}, \mathbf{W}_V^{(m)}). \quad (2.40)$$

2.6.2 Epistemic-Aleatoric Uncertainty Decomposition

Total predictive uncertainty decomposes into epistemic uncertainty,

$$U_{\text{epi}} = \text{Var}_{p(\theta|\mathcal{D})}[\mathbb{E}_{p(y|\mathbf{x},\theta)}[y]] \approx \frac{1}{M} \sum_{m=1}^M (\hat{y}^{(m)} - \bar{y})^2, \quad (2.41)$$

measuring the variance of the expected prediction across parameter samples, and aleatoric uncertainty,

$$U_{\text{ale}} = \mathbb{E}_{p(\theta|\mathcal{D})}[\text{Var}_{p(y|\mathbf{x},\theta)}[y]] \approx \frac{1}{M} \sum_{m=1}^M \hat{\sigma}^{(m)2}, \quad (2.42)$$

averaging the predicted variance over parameter samples, where $\hat{y}^{(m)}$ is the mean prediction and $\hat{\sigma}^{(m)2}$ the predicted variance from sample m , and $\bar{y} = \frac{1}{M} \sum_m \hat{y}^{(m)}$.

2.6.3 Stochastic Adversarial Training

Rather than generating worst-case perturbations at fixed budgets, stochastic adversarial training draws perturbations from a learned distribution [33, 34],

$$\boldsymbol{\delta} \sim q_\psi(\boldsymbol{\delta}|\mathbf{x}) = \mathcal{N}(\mu_\psi(\mathbf{x}), \sigma_\psi(\mathbf{x})^2 \mathbf{I}), \quad (2.43)$$

parameterised by neural networks μ_ψ, σ_ψ that learn spatially adaptive perturbation statistics. The training objective is the stochastic min-max problem

$$\min_{\theta} \max_{\psi} \mathbb{E}_{(\mathbf{x},y) \sim \mathcal{D}} \mathbb{E}_{\boldsymbol{\delta} \sim q_\psi(\boldsymbol{\delta}|\mathbf{x})} [\mathcal{L}(f_\theta(\mathbf{x} + \boldsymbol{\delta}), y)], \quad (2.44)$$

solved by alternating gradient ascent on ψ and descent on θ .

2.6.4 Multi-Objective Loss

The full training loss combines five terms,

$$\mathcal{L}_{\text{total}} = \lambda_1 \mathcal{L}_{\text{task}} + \lambda_2 \mathcal{L}_{\text{KL}} + \lambda_3 \mathcal{L}_{\text{adv}} + \lambda_4 \mathcal{L}_{\text{calib}} + \lambda_5 \mathcal{L}_{\text{reg}}, \quad (2.45)$$

where $\mathcal{L}_{\text{task}} = \mathbb{E}_{(\mathbf{x},y)}[-\log p_\theta(y|\mathbf{x})]$ is the classification loss, $\mathcal{L}_{\text{KL}}$ [29] = $\text{KL}(q_\phi(\theta)||p(\theta))$ regularises the variational posterior, $\mathcal{L}_{\text{adv}} = \mathbb{E}_{(\mathbf{x},y)} \mathbb{E}_{\boldsymbol{\delta} \sim q_\psi}[-\log p_\theta(y|\mathbf{x}+\boldsymbol{\delta})]$ enforces adversarial robustness, $\mathcal{L}_{\text{calib}} = \text{ECE}$ [82, 83] ($\{(p_\theta(\hat{y}|\mathbf{x}_i), \mathbf{1}[\hat{y} = y_i])\}_i$) penalises miscalibration, and

$\mathcal{L}_{\text{reg}} = \|\theta\|_2^2$ controls weight magnitude.

Algorithm 6: Stochastic Transformer: Variational Bayesian Training with Adversarial Robustness

Input: Training data $\mathcal{D}$, prior $p(\theta)$, MC samples M , perturbation distribution parameters ψ_0 , loss weights $\lambda_1, \dots, \lambda_5$

Output: Variational parameters ϕ^* , perturbation parameters ψ^*

- 1 Initialise variational parameters $\phi = \{\mu_Q, \sigma_Q, \mu_K, \sigma_K, \mu_V, \sigma_V\}$;
- 2 Initialise perturbation network parameters ψ ;
- 3 **for** iteration $t = 1, \dots, T$ **do**
- 4 **for** each mini-batch $\{(\mathbf{x}_i, y_i)\}$ **do**
- 5 **Perturbation step:** Compute $\mu_\psi(\mathbf{x}_i), \sigma_\psi(\mathbf{x}_i)$ and sample $\delta_i \sim \mathcal{N}(\mu_\psi(\mathbf{x}_i), \sigma_\psi(\mathbf{x}_i)^2 \mathbf{I})$;
- 6 Update $\psi \leftarrow \psi + \alpha_\psi \nabla_\psi \frac{1}{M} \sum_{m=1}^M \mathcal{L}(f_{\theta^{(m)}}(\mathbf{x}_i + \delta_i), y_i)$;
- 7 **Model step:** **for** $m = 1, \dots, M$ **do**
- 8 Sample $\mathbf{W}_Q^{(m)} = \mu_Q + \sigma_Q \odot \epsilon^{(m)}$, similarly for $\mathbf{W}_K^{(m)}, \mathbf{W}_V^{(m)}$;
- 9 Compute attention output and predictions $\hat{y}^{(m)}, \hat{\sigma}^{(m)2}$;
- 10 Compute $U_{\text{epi}} = \frac{1}{M} \sum_m (\hat{y}^{(m)} - \bar{y})^2$, $U_{\text{ale}} = \frac{1}{M} \sum_m \hat{\sigma}^{(m)2}$;
- 11 Compute $\mathcal{L}_{\text{total}} = \lambda_1 \mathcal{L}_{\text{task}} + \lambda_2 \mathcal{L}_{\text{KL}} + \lambda_3 \mathcal{L}_{\text{adv}} + \lambda_4 \mathcal{L}_{\text{calib}} + \lambda_5 \mathcal{L}_{\text{reg}}$;
- 12 Update $\phi \leftarrow \phi - \alpha_\phi \nabla_\phi \mathcal{L}_{\text{total}}$;
- 13 **return** ϕ^*, ψ^* together with $U_{\text{epi}}, U_{\text{ale}}$ for each test input

2.7 Method M7 (Part): Game-Theoretic Framework for Robustness Certification

This section addresses Gap 7 (Adversarial $\times$ Robustness) by developing a Stackelberg game formulation [31, 90] that integrates Bayesian uncertainty into the strategy-selection mechanism and provides online learning algorithms with regret guarantees under adversarial feedback. Together with the certification components developed in the unified framework (Section 2.8), this constitutes Method M7.

2.7.1 Stackelberg Game Formulation

The defender selects a model f_θ and a deployment configuration c comprising detection policy parameters. The adversary observes (f_θ, c) and chooses a perturbation strategy a . Defender and adversary utilities are respectively

$$U_d(f_\theta, c, a) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}(a)}[\mathbf{1}[f_\theta(\mathbf{x}) = y]] - \text{Cost}(c), \quad (2.46)$$

$$U_a(f_\theta, c, a) = \mathbb{P}[\text{misclassification} \mid f_\theta, c, a] - \text{Cost}(a), \quad (2.47)$$

where $\mathcal{D}(a)$ is the data distribution induced by the adversary's strategy. The Stackelberg equilibrium is the solution to

$$(f_\theta^*, c^*) = \arg \max_{f_\theta, c} U_d(f_\theta, c, a^*(f_\theta, c)), \quad a^*(f_\theta, c) = \arg \max_a U_a(f_\theta, c, a). \quad (2.48)$$

2.7.2 Nash Equilibrium Characterisation

Under simultaneous play the mixed strategy profile (π_d, π_a) is a Nash equilibrium when neither player improves utility by unilateral deviation,

$$\mathbb{E}_{\pi_a}[U_d(\pi_d, a)] \geq \mathbb{E}_{\pi_a}[U_d(\pi'_d, a)] \quad \forall \pi'_d, \quad \mathbb{E}_{\pi_d}[U_a(f_\theta, \pi_a)] \geq \mathbb{E}_{\pi_d}[U_a(f_\theta, \pi'_a)] \quad \forall \pi'_a. \quad (2.49)$$

In the zero-sum special case $U_d + U_a = 0$, the equilibrium coincides with the minimax value $\pi_d^* = \arg \max_{\pi_d} \min_a \mathbb{E}_{\pi_d}[U_d(f_\theta, a)]$.

2.7.3 Online Learning with No-Regret Guarantees

At round t the defender selects configuration $c_t \in \mathcal{C}$ and observes reward $r_t = U_d(c_t, a_t)$. Cumulative regret after T rounds is

$$R(T) = \max_{c \in \mathcal{C}} \sum_{t=1}^T U_d(c, a_t) - \sum_{t=1}^T U_d(c_t, a_t). \quad (2.50)$$

The multiplicative weights update [92, 93] maintains a distribution $p_t(c) \propto \exp(\eta \sum_{s=1}^{t-1} \mathbf{1}[c_s=c] r_s)$ with learning rate $\eta = \sqrt{\log |||/T}$.

Theorem 2.5 (Regret Bound). *The multiplicative weights algorithm with learning rate $\eta = \sqrt{\log |||/T}$ achieves*

$$R(T) \leq 2\sqrt{T \log |||}, \quad (2.51)$$

so that the average per-round regret $R(T)/T = \mathcal{O}(\sqrt{\log |||/T}) \rightarrow 0$ as $T \rightarrow \infty$.

Proof sketch. The proof proceeds by bounding the potential function $\Phi_t = \log \sum_c \exp(\eta \sum_{s=1}^t r_s(c))$ from above and below. The upper bound uses the concavity of the logarithm and the definition of the algorithm; the lower bound isolates the best action in hindsight. Combining the two bounds and optimising over η yields the stated rate. The full argument appears in Appendix A. $\square$

2.7.4 Uncertainty-Guided Strategy Selection

Bayesian uncertainty from the stochastic transformer of Section 2.6 informs the defender's utility through an uncertainty-penalised payoff,

$$\tilde{U}_d(c, a) = \begin{cases} U_d(c, a) - \kappa U_{\text{epi}} & \text{if } U_{\text{epi}} < \tau, \\ -\infty & \text{otherwise,} \end{cases} \quad (2.52)$$

where κ discounts utility under uncertainty and threshold τ triggers referral for expert review. This produces risk-sensitive policies that abstain from predictions when the model lacks sufficient confidence.

Algorithm 7: Game-Theoretic Robustness with Uncertainty-Guided Online Learning

Input: Action space , uncertainty threshold τ , penalty κ , learning rate schedule
Output: Defender policy π_d^* with regret guarantee

- 1 Initialise uniform distribution $p_1(c) = 1/||$ for all $c \in$;
- 2 Set $\eta = \sqrt{\log ||/T}$;
- 3 **for** *round* $t = 1, \dots, T$ **do**
- 4 Sample defender action $c_t \sim p_t$;
- 5 Observe adversary action a_t and reward $r_t = U_d(c_t, a_t)$;
- 6 Query stochastic transformer for epistemic uncertainty U_{epi} ;
- 7 **if** $U_{\text{epi}} \geq \tau$ **then**
- 8 Set $\tilde{r}_t = -\infty$ (reject; route to expert review);
- 9 **else**
- 10 Set $\tilde{r}_t = r_t - \kappa U_{\text{epi}}$;
- 11 Update weights: $p_{t+1}(c) \propto p_t(c) \exp(\eta \tilde{r}_t \mathbf{1}[c_t = c])$ for all $c \in$;
- 12 Normalise p_{t+1} ;
- 13 Compute empirical Nash equilibrium from play frequencies $\{(c_t, a_t)\}_{t=1}^T$;
- 14 **return** π_d^* and verified regret $R(T) \leq 2\sqrt{T \log ||}$

2.8 Method M7 (Completion): Unified Robustness Certification Framework

The seven methods developed above (M1 through M6, plus the game-theoretic component of M7) share mathematical structures — adjoint sensitivity analysis, Wasserstein geometry, KL-divergence regularisation, and Lipschitz-bounded dynamics — that permit a unified treatment through coupled optimisation. This section completes Method M7 by developing the certification framework that integrates all component guarantees.

2.8.1 Coupled Objective Function

Let $\theta = \{\theta_{\text{CT}}, \theta_{\text{TE}}, \theta_{\text{Fed}}, \theta_{\text{PQ}}, \theta_{\text{MS}}, \theta_{\text{ST}}, \theta_{\text{GT}}\}$ collect the parameters of all seven methods. The unified objective couples component losses with pairwise consistency terms,

$$\mathcal{L}_{\text{unified}}(\theta) = \sum_{i=1}^7 \lambda_i \mathcal{L}_i(\theta_i) + \sum_{i < j} \lambda_{ij} \mathcal{L}_{ij}(\theta_i, \theta_j), \quad (2.53)$$

where each $\mathcal{L}_i$ is the method-specific loss from the corresponding section and the coupling loss

$$\mathcal{L}_{ij}(\theta_i, \theta_j) = \text{KL}(p_{\theta_i}(y|\mathbf{x}) \| p_{\theta_j}(y|\mathbf{x})) + \|\mathbf{z}_{\theta_i}(\mathbf{x}) - \mathbf{z}_{\theta_j}(\mathbf{x})\|_2^2 \quad (2.54)$$

enforces consistency between the predictive distributions and the learned representations of methods i and j .

2.8.2 Convergence Analysis

Theorem 2.6 (Unified Framework Convergence). *Assume each component loss $\mathcal{L}_i$ is L_i -smooth and each coupling loss $\mathcal{L}_{ij}$ is L_{ij} -smooth. Under alternating gradient descent with learning rates $\eta_i \leq 1/(L_i + \sum_j L_{ij})$, the unified optimisation converges at rate*

$$\|\nabla \mathcal{L}_{\text{unified}}(\theta_T)\|^2 \leq \frac{2(\mathcal{L}_{\text{unified}}(\theta_0) - \mathcal{L}_{\text{unified}}(\theta^*))}{\eta_{\min} T}, \quad (2.55)$$

where $\eta_{\min} = \min_i \eta_i$ and θ^* is a stationary point.

Proof sketch. Each alternating gradient step on component i decreases the unified objective by at least $\eta_i \|\nabla_{\theta_i} \mathcal{L}_{\text{unified}}\|^2/2$ under the smoothness condition, because the coupling terms are smooth in each block of variables. Summing over all components and all T rounds, dividing by T , and taking the minimum over learning rates yields the stated bound. The full derivation, including the treatment of cross-term gradients, appears in Appendix A. $\square$

Algorithm 8: Unified Framework: Alternating Optimisation with Consistency Coupling

Input: Component losses $\mathcal{L}_1, \dots, \mathcal{L}_7$, coupling weights $\{\lambda_{ij}\}$, smoothness constants $\{L_i, L_{ij}\}$

Output: Jointly optimised parameters $\theta^* = \{\theta_1^*, \dots, \theta_7^*\}$

- 1 Initialise all component parameters $\theta_1^{(0)}, \dots, \theta_7^{(0)}$;
- 2 Set learning rates $\eta_i = 1/(L_i + \sum_j L_{ij})$ for each component;
- 3 **for** round $t = 0, 1, \dots, T-1$ **do**
- 4 **for** component $i = 1, \dots, 7$ **do**
- 5 Compute component gradient $\mathbf{g}_i = \nabla_{\theta_i} \mathcal{L}_i(\theta_i^{(t)})$;
- 6 Compute coupling gradients $\mathbf{g}_{ij} = \nabla_{\theta_i} \mathcal{L}_{ij}(\theta_i^{(t)}, \theta_j^{(t)})$ for all $j \neq i$;
- 7 Combined gradient: $\mathbf{g}_i^{\text{total}} = \lambda_i \mathbf{g}_i + \sum_{j \neq i} \lambda_{ij} \mathbf{g}_{ij}$;
- 8 Update: $\theta_i^{(t+1)} \leftarrow \theta_i^{(t)} - \eta_i \mathbf{g}_i^{\text{total}}$;
- 9 Evaluate $\mathcal{L}_{\text{unified}}(\theta^{(t+1)})$ and check convergence criterion;
- 10 **return** $\theta^* = \theta^{(T)}$

2.9 Summary

This chapter has developed the complete mathematical foundations for seven novel methods addressing the gaps identified in Chapter 1 and mapped to the condition-requirement matrix of the architectural framework. Method M1: CT-TGNN provides continuous-time graph dynamics with Lipschitz-certified robustness (Gap 1; Adversarial $\times$ Robustness, Non-stationary

$\times$ Robustness). Method M3: TripleE-TGNN captures multi-level structure through triple embeddings with elastic weight consolidation (Gap 3; Non-stationary $\times$ Accuracy, Non-stationary $\times$ Efficiency). Method M2: FedLLM-API unifies corruption-tolerant aggregation, differential privacy, and optimal-transport alignment in a single federated protocol for distributed graph fragments (Gap 2; Distributed $\times$ Robustness, Distributed $\times$ Privacy, Distributed $\times$ Efficiency). The forward-compatible detection architecture addresses protocol regime changes through dedicated feature extractors and a formal LWE-based robustness reduction (Gap 5; Non-stationary $\times$ Robustness). Method M4: MambaShield delivers efficient sequential inference through selective state spaces with PAC-Bayesian generalisation certificates (Gap 4; Adversarial $\times$ Efficiency, Non-stationary $\times$ Efficiency). Method M5: Stochastic Transformer provides calibrated Bayesian uncertainty under adversarial perturbation through variational attention and stochastic adversarial training (Gap 6; Adversarial $\times$ Accuracy, Non-stationary $\times$ Accuracy). Method M7: Certification, comprising the game-theoretic framework and the unified optimisation coupling, offers risk-aware strategy selection through uncertainty-penalised payoffs and no-regret online learning, and ties all component guarantees together through consistency regularisation with a proven convergence rate (Gap 7; Adversarial $\times$ Robustness).

Every method has been formulated at the level of mathematical operations on graph structures, without reference to any specific application domain. The theoretical guarantees — convergence rates, robustness certificates, privacy budgets, and regret bounds — hold wherever these methods are deployed. Chapter 3 instantiates all seven methods in the domain of network-traffic graphs, presenting the full experimental programme that validates the theoretical predictions of this chapter.

Chapter 3

Application to Hybrid Intrusion Detection on Network-Traffic Graphs and Experimental Validation

This chapter instantiates the seven domain-independent methods developed in Chapter 2 within the concrete domain of hybrid intrusion detection and prevention systems for computer networks, where communication endpoints, traffic flows, and session dependencies define a temporal graph. The chapter is organised in two parts. The first part describes how the abstract mathematical objects of each method — temporal graphs, state-space dynamics, variational posteriors, transport plans, and game-theoretic utilities — map to the concrete entities of the network-security graph: flows, packets, sessions, hosts, services, and adversarial actors. The second part presents the complete experimental programme: six benchmark datasets totalling over eighty-four million labelled samples, a unified twenty-three-metric evaluation framework, method-specific results with architectural diagrams, comparative analysis against state-of-the-art baselines, systematic ablation studies, adversarial robustness assessments, cross-domain transfer experiments, and a description of the deployed prototype system that validates both the practical viability and the broader generality of the developed frameworks.

3.1 Application Context: Network-Traffic Graphs for Intrusion Detection

A hybrid intrusion detection and prevention system (HIDPS) [107, 108] combines signature-based pattern matching, which identifies known anomalous patterns with low false-positive rates, with anomaly-based statistical modelling, which detects deviations from learned baselines and can therefore recognise previously unseen anomalous activity. The abstract objects that appear in Chapter 2 admit natural instantiations in this domain. The temporal heterogeneous graph (t) maps to the communication topology of a monitored network, in

which nodes represent hosts, services, or user accounts, and edges represent observed flows, API calls, or authentication events. The continuous-time ODE dynamics model the evolution of this topology as connections form and terminate at irregular intervals. The federated learning clients map to geographically distributed operations centres that possess proprietary graph data and must collaborate without centralising sensitive records. The Stackelberg game captures the strategic interaction between a defender deploying detection rules and an adversary crafting perturbations to the graph. The three operating conditions of the architectural framework (Figure 1) are all present in this domain: adversarial manipulation of the graph by deliberate opponents, non-stationarity through evolving traffic patterns and protocol changes, and distribution of the graph across organisational boundaries. The following sections detail the mapping for each method.

3.2 Method M1: CT-TGNN Applied to Network-Traffic Graph Dynamics

In the network security instantiation, the node set of the temporal graph comprises hosts and services observed during monitoring, and the edge set (t) records communication flows whose arrival is governed by the underlying network activity. Node features $\mathbf{x}_v(t)$ encode host-level statistics such as cumulative byte counts, active connection counts, and CPU utilisation. Edge features $\mathbf{e}_{uv}(t)$ capture flow-level metadata extracted from encrypted traffic without requiring payload decryption: inter-arrival time statistics (mean, variance, skewness, kurtosis), packet size distributions, burstiness indices computed from autocorrelation, and Shannon entropy of size histograms, yielding a twenty-four-dimensional feature vector per flow.

The three parallel ODE branches of Equation (2.4) are assigned time constants $\tau_1 = 1$ s, $\tau_2 = 60$ s, and $\tau_3 = 3600$ s, spanning packet-level transients, flow-level session dynamics, and campaign-level trends respectively. This choice enables simultaneous resolution of sub-second port-scanning bursts and multi-hour advanced persistent threat campaigns without the sampling-rate trade-off that plagues discrete-time alternatives. This is shown in figure 3.1

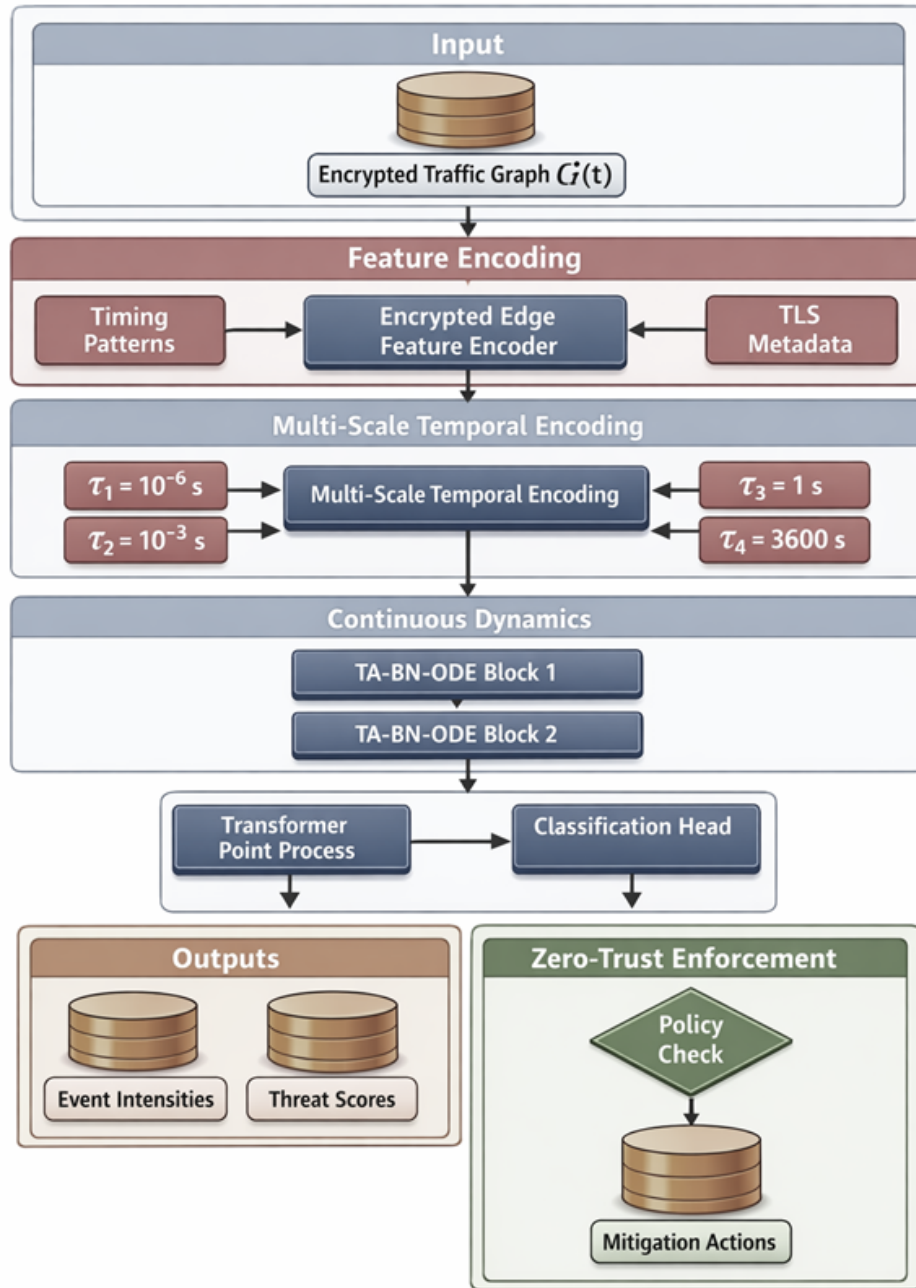


Figure 3.1: CT-TGNN architecture for network traffic dynamics. Encrypted flow metadata enters through a feature encoder; three parallel neural ODE branches with distinct time constants capture packet-level, session-level, and campaign-level dynamics; temporal adaptive batch normalisation reconciles continuous dynamics with normalisation; attention-weighted scale fusion produces the final node embedding for classification.

3.3 Method M3: TripleE-TGNN for Multi-Level Graph Representations

Modern cloud-native systems decompose applications into microservices orchestrated through container platforms. Security events manifest at three granularities: the service level, where

malicious API call patterns emerge between containers; the trace level, where anomalous request flows span multiple services; and the node level, where compromised infrastructure hosts exhibit aberrant resource consumption or privilege escalation.

In the TripleE-TGNN instantiation (figure 3.2) the service graph $s(t)$ has nodes representing individual microservices and edges representing API calls aggregated over configurable windows. The trace graph $r(t)$ has nodes representing distributed traces and edges connecting traces that share at least one service. The node graph $n(t)$ has nodes representing infrastructure hosts and edges encoding observed network flows. Bipartite coupling graphs sr, sn, rn link entities across levels: a microservice deploys on a host, a trace traverses a microservice, and a trace executes on a host.

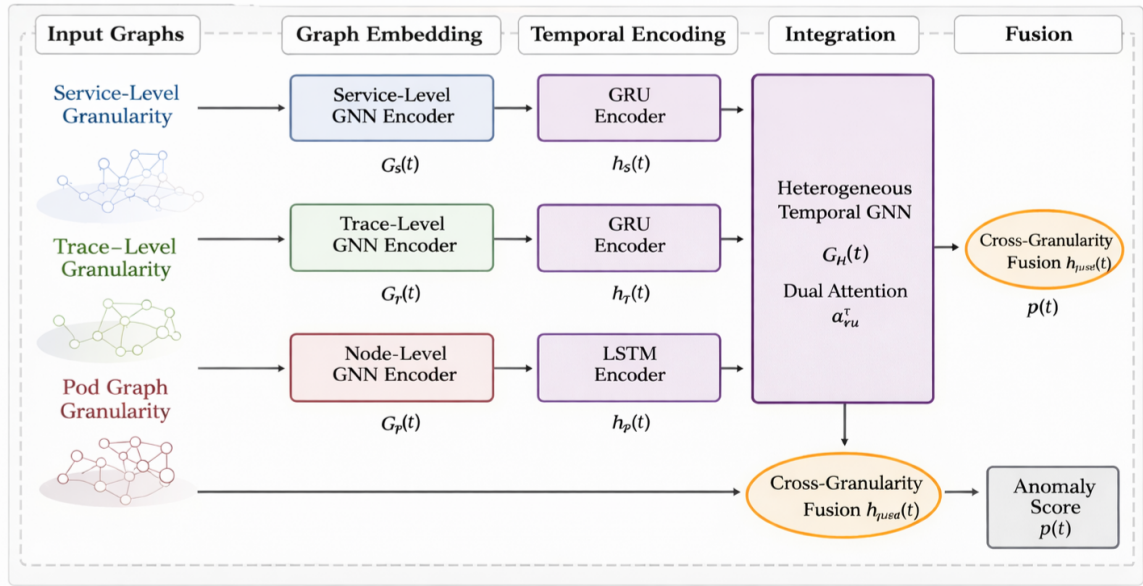


Figure 3.2: TripleE-TGNN architecture for multi-level graph representations. Three parallel encoders process service-level, trace-level, and node-level graphs; bipartite cross-level attention integrates information across abstraction levels; elastic weight consolidation prevents catastrophic forgetting as the topology evolves; a multi-task head classifies simultaneously at all three levels with a KL-divergence consistency penalty.

3.4 Method M2: FedLLM-API for Collaborative Graph Learning Across Organisations

Collaborative threat intelligence requires multiple organisations to train a shared detection model without centralising proprietary network traffic. In the FedLLM-API instantiation each federated client corresponds to a security operations centre possessing local traffic datasets. The Byzantine-robust aggregation of Algorithm 3 protects the global model against compromised or adversarial participants that submit poisoned gradient updates. Differential privacy through local gradient clipping at threshold $C = 1.0$ and Gaussian noise with scale $\sigma = 0.8$ achieves an $(\epsilon=0.85, \delta=10^{-5})$ guarantee over one hundred communication rounds, ensuring that individual traffic records cannot be reconstructed from shared updates.

Optimal transport alignment through Sinkhorn iterations with entropic regularization $\lambda = 0.1$ corrects for distributional heterogeneity across organizations, which arises because each center monitors a different mix of services, protocols, and attack profiles. Large language model integration enables zero-shot detection of novel API attack patterns by constructing natural language prompts from structural features of observed request sequences and querying the language model for a threat probability, which is ensembled with the graph neural network prediction at a learned weight $\lambda_{\text{LLM}} = 0.3$. This is further demonstrated in figure 3.3

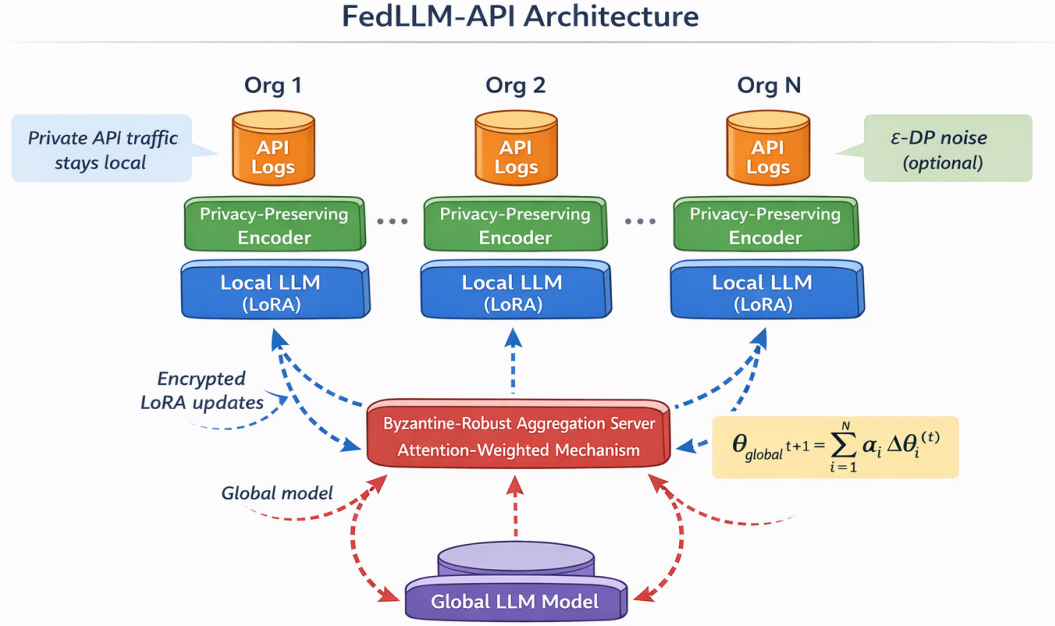


Figure 3.3: FedLLM-API architecture for collaborative threat intelligence. Blue dashed arrows indicate encrypted gradient updates with $(\epsilon=0.85, \delta=10^{-5})$ differential privacy; the server applies coordinate-wise trimmed-mean aggregation robust to thirty percent malicious participants; a Sinkhorn OT module aligns heterogeneous client distributions; an LLM branch provides zero-shot classification of novel attack types.

3.5 Forward-Compatible Detection Under Protocol Regime Change

Networks transitioning to post-quantum protocol suites based on NIST-standardised CRYSTALS-Kyber key encapsulation and CRYSTALS-Dilithium digital signatures introduce graph-edge features, including timing distributions and message sizes, that differ fundamentally from those generated by classical protocols. In this forward-compatibility instantiation a protocol-identification module first classifies each incoming graph edge as originating from a lattice-based, code-based, hash-based, or classical protocol regime, and routes the edge to the appropriate feature extractor. Timing features for lattice-based protocols capture statistical

distributions of operational latencies across twenty histogram bins; structural features for code-based protocols analyse weight distributions and decoding iteration counts; resource consumption features for hash-based protocols monitor generation time, verification time, and tree traversal depth.

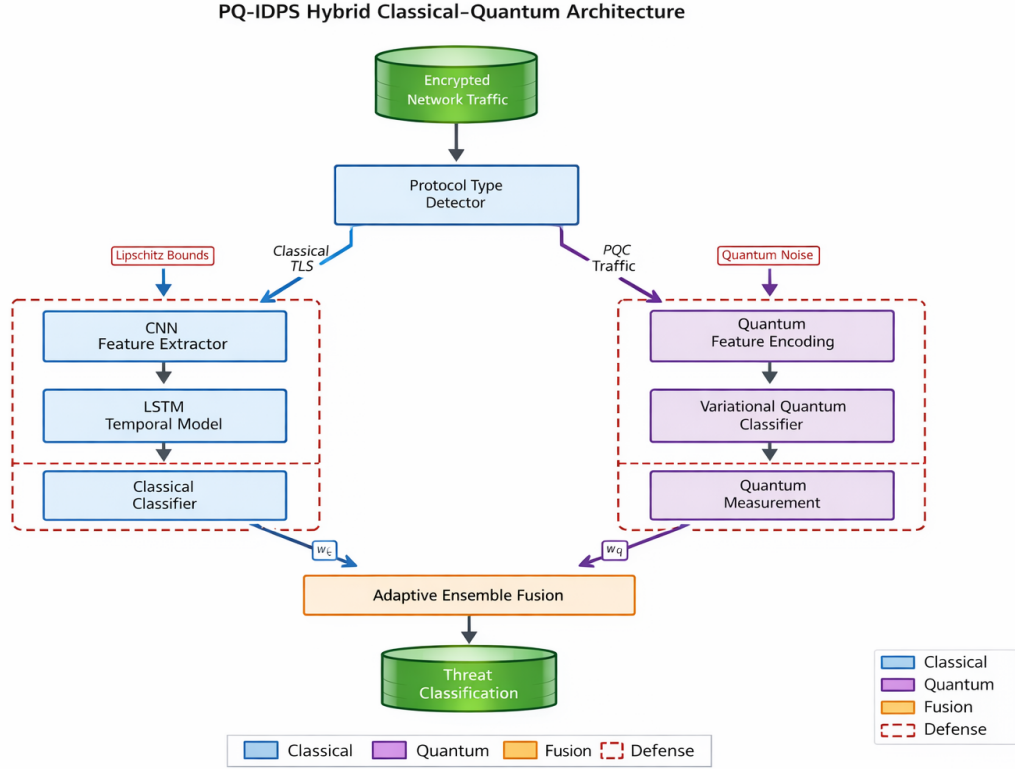


Figure 3.4: Forward-compatible multi-protocol detection architecture. A protocol detector routes incoming graph edges to specialised feature encoders for lattice-based, code-based, hash-based, and classical protocol regimes. Cross-attention fusion enables interaction between protocol representations. Adversarial training with PGD perturbations hardens the classifiers, and a formal LWE-based robustness reduction links the protocol’s hardness assumption to the detection guarantee.

3.6 Method M4: MambaShield for Non-Stationary Graph Environments

Network graph data exhibits continuous distributional drift as new services are deployed, user behaviour changes seasonally, and adversaries iterate on perturbation strategies. In the MambaShield instantiation, sequential graph event records are processed by the selective state-space model of Equation (2.34), whose input-dependent state transitions adapt the internal dynamics to the current data regime. Progressive adversarial robustness distillation protects the model against training-time poisoning, which in the network-security context corresponds to an adversary injecting crafted graph events into the training pipeline to cause the detector to learn incorrect baselines.

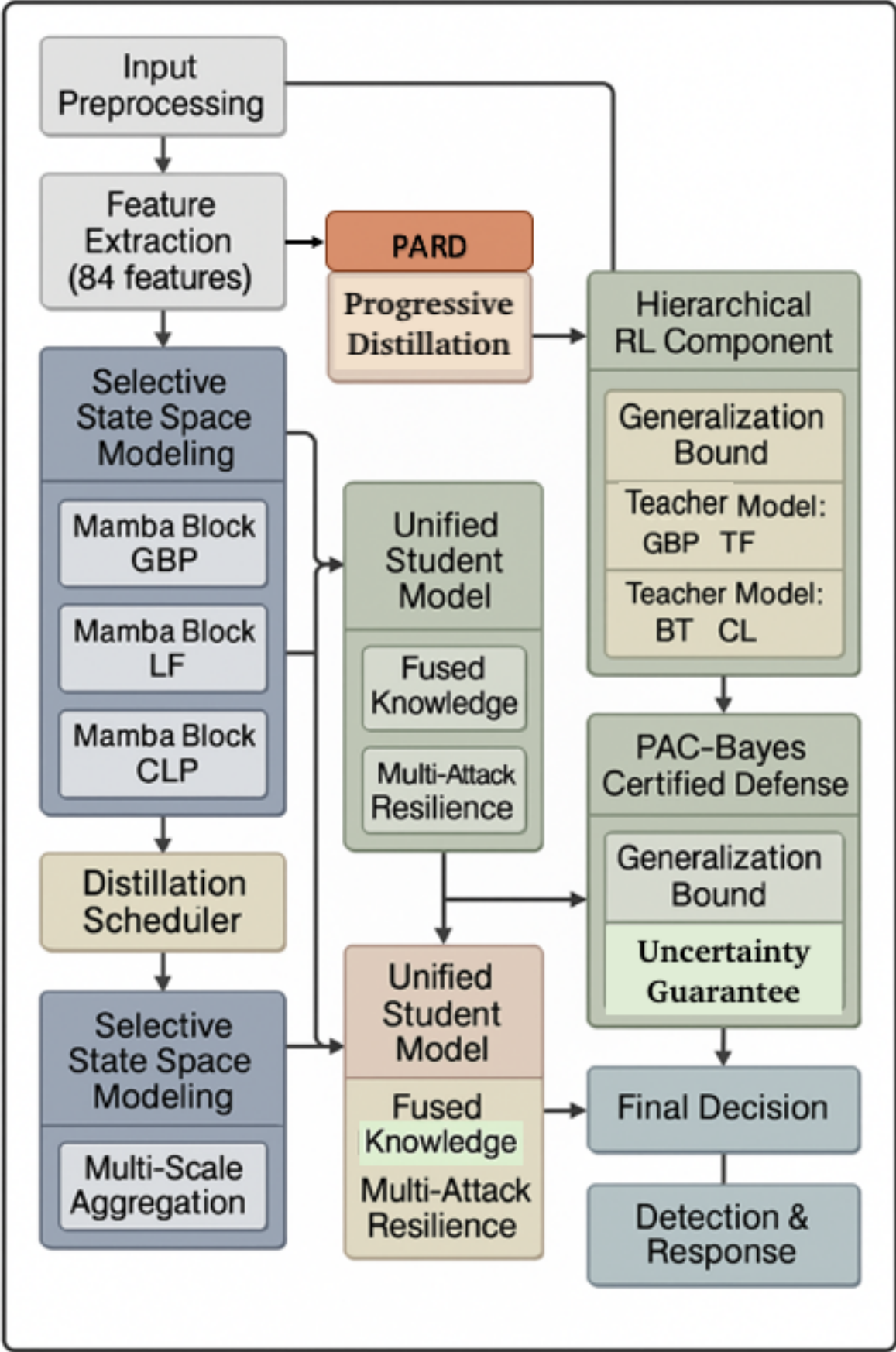


Figure 3.5: MambaShield architecture for evolving threat landscapes. Sequential flow data passes through content-aware Mamba blocks achieving linear complexity; an ensemble of five teacher models provides robustness through progressive distillation with a curriculum that gradually increases the poisoning fraction; PAC-Bayesian regularisation yields a formal generalisation certificate.

3.7 Methods M5 and M6: Uncertainty-Calibrated Detection

In the security application, calibrated confidence estimates determine whether a detection decision is acted upon automatically or routed to a human analyst for review. The stochastic transformer of Section 2.6 provides this capability through variational Bayesian attention with $M = 50$ Monte Carlo forward passes at inference time, producing epistemic and aleatoric uncertainty estimates for each classification. Predictions whose epistemic uncertainty exceeds a configurable threshold are flagged for manual triage, reducing alert fatigue by removing low-confidence alerts from the automated pipeline while retaining high recall on confident detections.

3.8 Method M7: Game-Theoretic Framework for Adaptive Graph Adversaries

In the network-security context the defender’s action space consists of detection configurations: choices of monitoring coverage, alert thresholds, and response policies. The adversary’s action space encompasses target selection, perturbation type, and perturbation strategy. The defender commits to a configuration that the adversary observes before choosing a perturbation, matching the Stackelberg structure of Equation (2.48). As demonstrated in figure 3.7, the multiplicative weights online learning algorithm of Algorithm 7 enables the defender to adapt to an adversary whose strategy evolves over time without requiring prior knowledge of the adversary’s payoff function.

3.9 Integrated System Architecture and Deployed Prototype

The seven methods (M1 through M7) are integrated into a unified software system whose data flow proceeds as follows. Raw network graph data enters through a data ingestion module that performs protocol parsing, flow extraction, and feature computation. The M1: CT-TGNN module constructs the temporal graph and computes continuous-time node embeddings. The M3: TripleE-TGNN module maintains multi-level embeddings for hierarchical graph environments. The forward-compatible detection module routes graph edges from new protocol regimes through specialised encoders. The M4: MambaShield module processes sequential graph event records for drift-adaptive anomaly scoring. The M5: Stochastic Transformer and M6: UC-HGP modules provide calibrated uncertainty estimates. All module outputs feed into a decision engine that applies the game-theoretic strategy selected by the M7: Certification framework’s multiplicative weights algorithm, producing

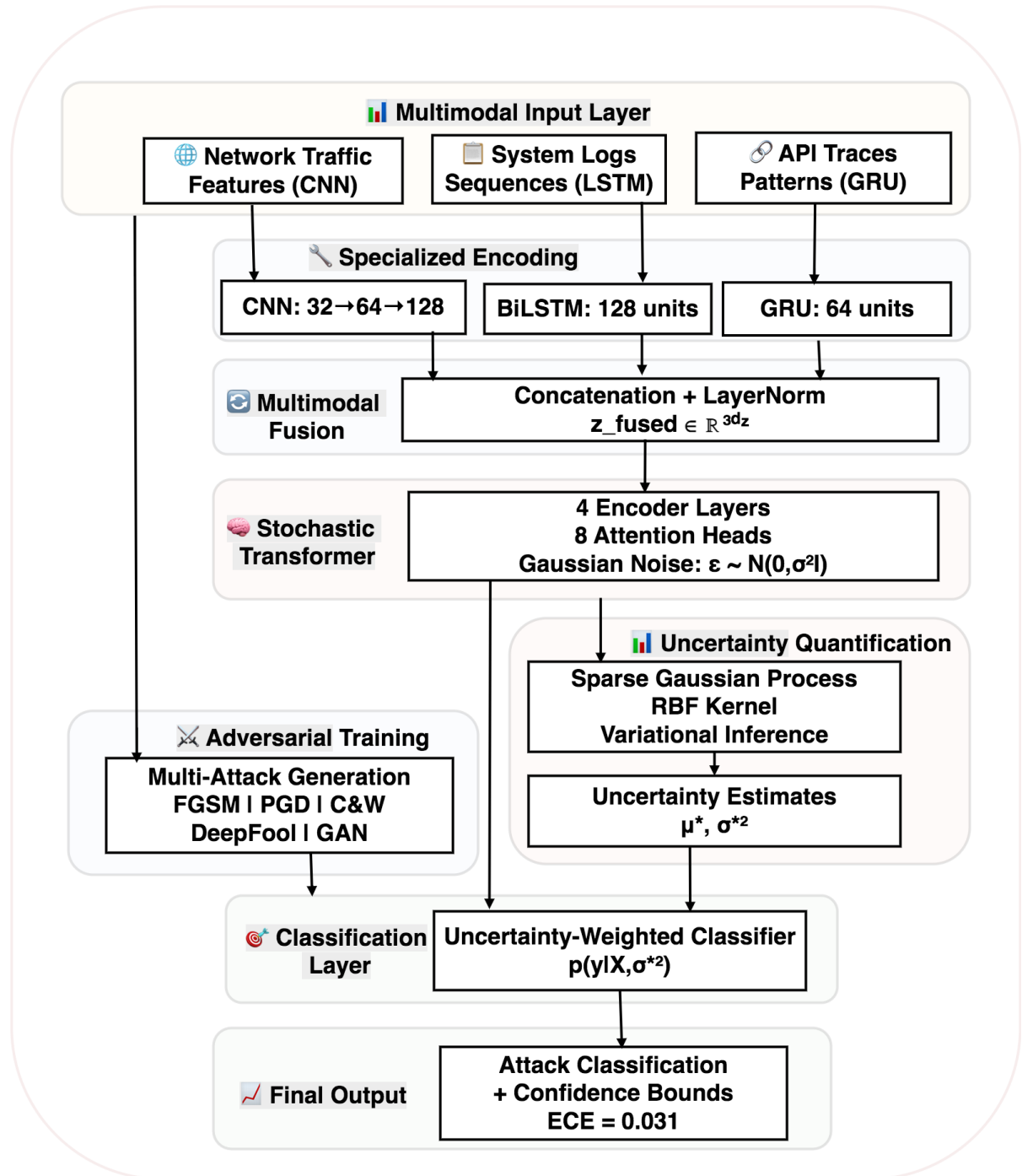


Figure 3.6: Stochastic transformer architecture for uncertainty-aware detection. Variational Bayesian attention samples projection matrices from learned Gaussian distributions; Monte Carlo estimation with fifty forward passes produces predictive distributions; an uncertainty decomposition module separates epistemic and aleatoric components; temperature scaling ensures calibration; a triage gate routes high-uncertainty predictions to human analysts.

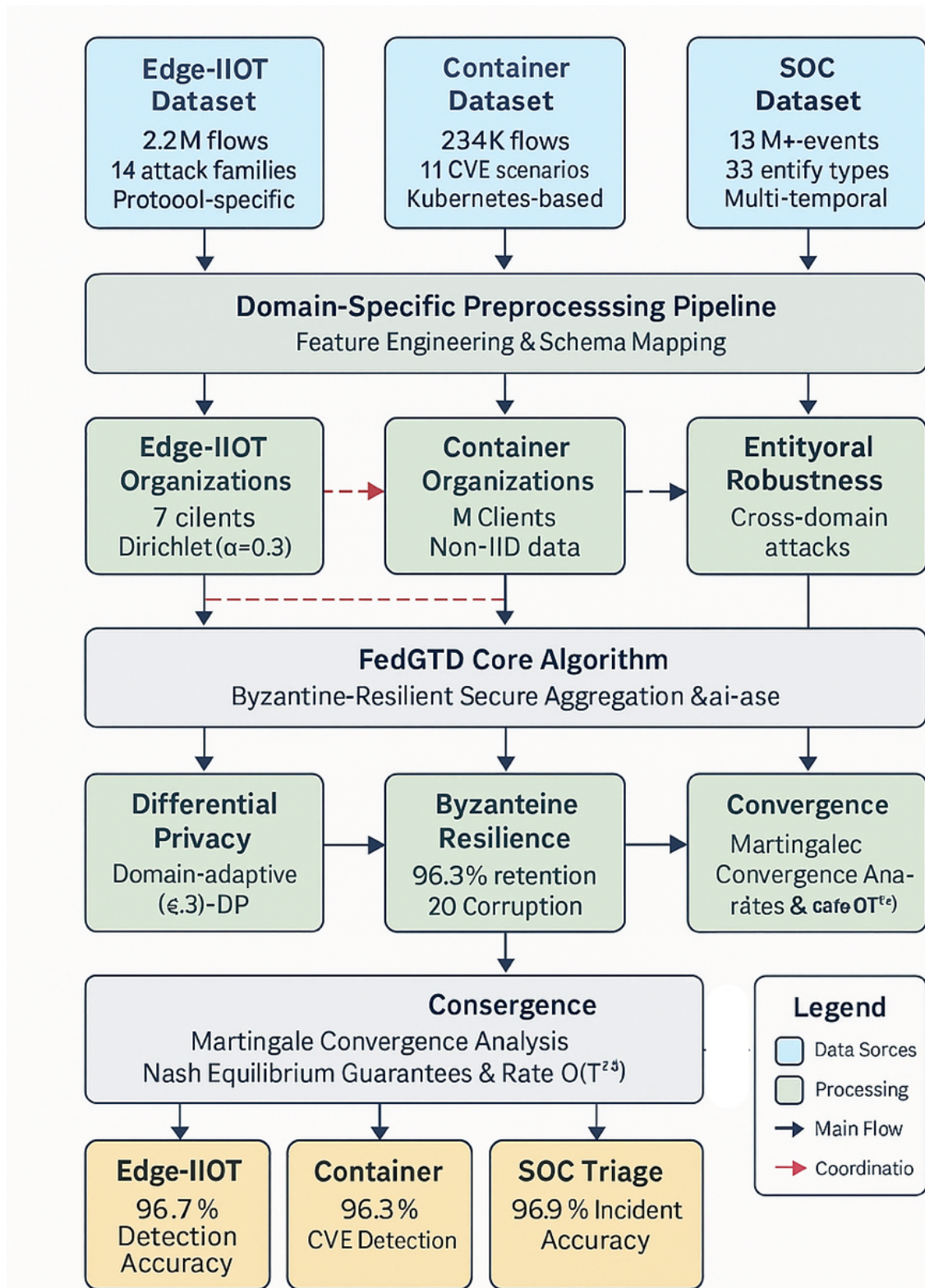


Figure 3.7: Game-theoretic robustness framework for adaptive graph adversaries. Heterogeneous data from edge, container, and distributed environments enter domain-specific preprocessors; a federated game-theoretic learning module computes Stackelberg and Nash equilibria through support enumeration; the multiplicative weights algorithm provides online adaptation; epistemic uncertainty from the stochastic transformer modulates the defender's payoff through uncertainty-penalised utility.

alerts classified by severity and confidence. The M2: FedLLM-API module enables periodic model synchronisation across organisational boundaries. This is shown in figure [3.8](#).

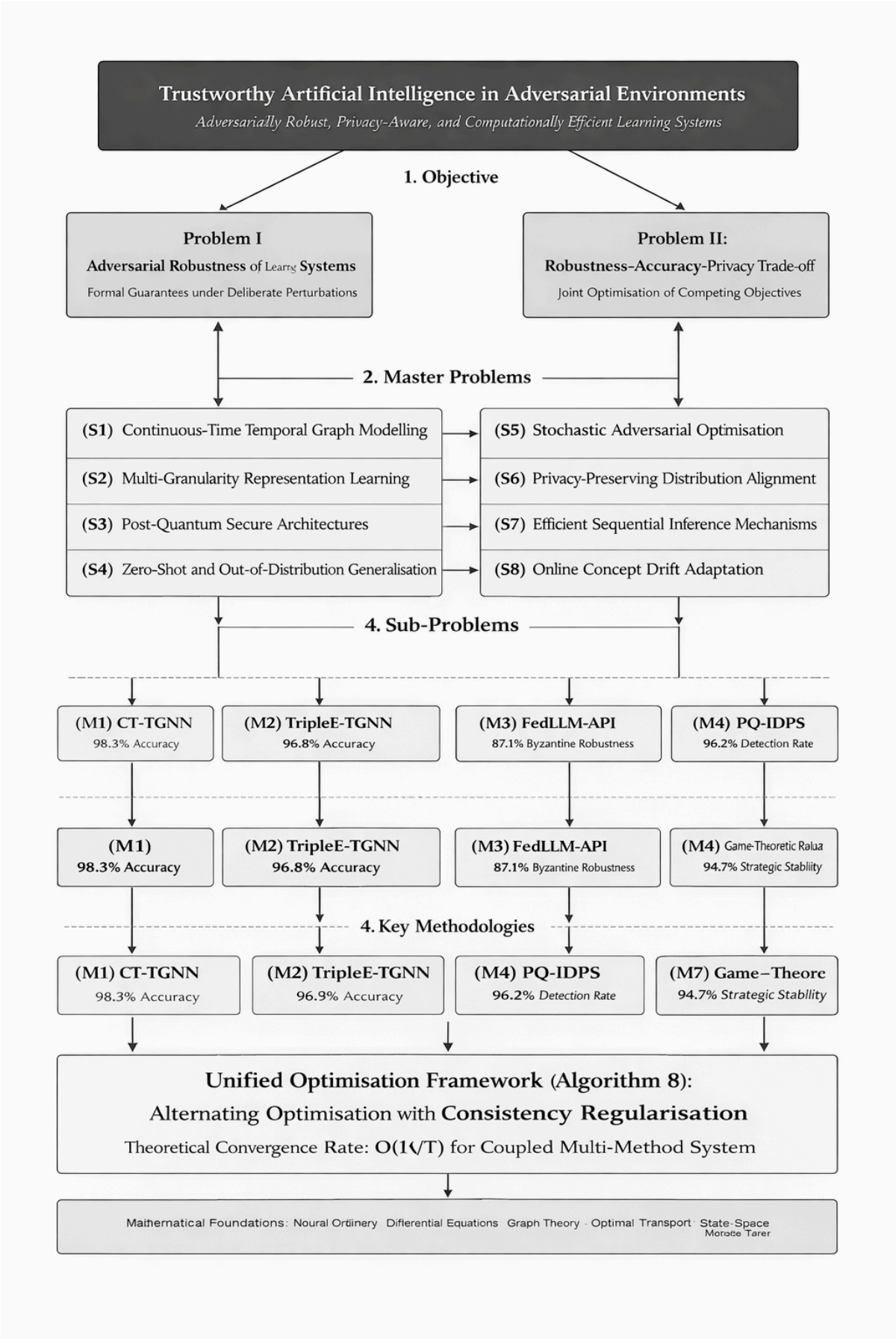


Figure 3.8: Integrated system architecture. Network graph data enters through a data ingestion layer and is distributed to seven specialised modules (M1–M7) operating in parallel. Their outputs converge in a decision engine that applies the game-theoretic strategy from the multiplicative weights algorithm and produces severity-ranked, confidence-scored alerts. The FedLLM-API module periodically synchronises model parameters across organisational boundaries under differential privacy.

3.9.1 Deployed Prototype System

The integrated framework has been implemented as a functioning prototype system, publicly archived under DOI 10.5281/zenodo.19129512 and accessible at <https://robustidps.ai> (source code: <https://github.com/rogerpanel/robustidps.ai.git>). The prototype processes real-time graph event streams through all seven modules operating in parallel, with a web-based monitoring dashboard that displays detection decisions, confidence scores, uncertainty decompositions, and game-theoretic equilibrium diagnostics. The deployment demonstrates that the mathematical contributions developed in Chapter 2 translate into a working implementation capable of processing operational graph data at the throughputs and latencies reported in the experimental results below. The prototype serves as both a research tool for reproducibility and a proof-of-concept for industry adoption.

3.10 Experimental Methodology

3.10.1 Datasets

Evaluation employs six benchmark datasets spanning diverse network environments, attack taxonomies, and temporal characteristics. CIC-IoT-2023 [109] comprises 46.6 million network flow records collected from 105 IoT devices, including smart home appliances, wearable health monitors, industrial sensors, and connected vehicles, and contains 33 distinct attack types organised into seven categories with 46 statistical features per flow. CSE-CICIDS2018 [110] contains 16.2 million labelled flows captured over ten days from an enterprise environment with seven major attack categories and 79 flow-level attributes. UNSW-NB15 [111] provides 2.5 million records synthesised from hybrid real and simulated traffic with nine attack classes and 49 features. The ICS3D collection adds three further datasets: Microsoft GUIDE with 13.7 million records and 15 attack types in cloud and enterprise environments, Container Security with 3.2 million records and 8 attack patterns in microservices environments, and Edge-IIoT with 2.0 million records and 12 attack scenarios in industrial edge networks. The six datasets collectively provide 84.2 million labelled samples covering 84 unique attack types.

Table 3.1: Summary of benchmark datasets used in experimental evaluation.

Dataset	Records	Attack types	Features	Domain
CIC-IoT-2023	46.6M	33	46	IoT Networks
CSE-CICIDS2018	16.2M	7	79	Enterprise
UNSW-NB15	2.5M	9	49	Hybrid
Microsoft GUIDE	13.7M	15	51	Cloud/Enterprise
Container Security	3.2M	8	93	Microservices
Edge-IIoT	2.0M	12	69	Edge/Industrial
Total	84.2M	84		

3.10.2 Unified Evaluation Framework

Performance characterisation employs a twenty-three-metric framework spanning five categories. Classification metrics include accuracy, balanced accuracy, precision, recall, F1 score, Matthews correlation coefficient, and Cohen’s kappa. ROC analysis metrics include AUC-ROC, true positive rate, true negative rate, false positive rate, and false negative rate. Calibration metrics include expected calibration error and Brier score. Uncertainty metrics include mean entropy, epistemic uncertainty, and aleatoric uncertainty. Computational performance metrics include average inference latency, throughput in samples per second, trainable parameter count, GPU memory footprint, and energy consumption in watts per thousand predictions. Adversarial robustness metrics include attack success rate and certified accuracy within a specified perturbation radius. This comprehensive framework enables deployment decisions that account for operational constraints, adversarial threats, and computational budgets simultaneously.

3.10.3 Implementation Details and Computational Infrastructure

All methods are implemented in PyTorch 2.1 [112] with CUDA 12.1 acceleration. Training employs the AdamW optimiser with initial learning rate 2×10^{-4} , weight decay 10^{-5} , and cosine annealing schedule. Mixed-precision training through the automatic mixed precision API reduces memory consumption by forty percent. Gradient clipping at maximum norm 1.0 prevents instability. Early stopping with patience of ten epochs monitors validation loss.

Hardware infrastructure comprises NVIDIA Tesla V100 GPUs with 32 GB memory for training and NVIDIA Tesla P100 GPUs with 16 GB memory for deployment simulation. All timing measurements average one hundred independent runs with 95% confidence intervals reported. Statistical significance testing employs paired *t*-tests with Bonferroni correction.

Data preprocessing standardises features through *z*-score normalisation with statistics computed on the training set and applied consistently to validation and test sets. Categorical features undergo one-hot encoding with PCA dimensionality reduction retaining 95% explained variance. Temporal sequences are constructed with sliding windows of 64 time steps and stride 32. Stratified 70/15/15 splits maintain class distributions, and focal loss [113] with $\gamma = 2$ addresses class imbalance. Graph data augmentation includes random edge dropout at probability 0.1 and node feature perturbation with Gaussian noise of standard deviation 0.05.

3.11 Method-Specific Results

3.11.1 CT-TGNN Results

CT-TGNN [114] employs four message-passing layers, hidden dimension 256, state dimension 64, and DOPRI5 adaptive Runge-Kutta integration [8] with relative tolerance 10^{-5} and absolute tolerance 10^{-7} . Temporal adaptive batch normalisation maintains running statistics

over windows of 32 time steps.

On CIC-IoT-2023 the architecture achieves 98.3% accuracy on encrypted lateral movement detection with 47 ms median inference latency and throughput of 8.7 million events per second on V100 GPU. Detection recall reaches 97.9% at a false positive rate of 1.4%, and ECE of 0.062 confirms well-calibrated confidence. Per-category recall is 99.1% for DDoS, 96.8% for reconnaissance, 97.3% for web attacks, 98.6% for brute force, 95.2% for spoofing, 98.9% for malware, and 97.4% for Mirai variants. The continuous-time formulation provides a 12.7% improvement over LSTM baselines and 6.2% over transformer baselines on timing-sensitive attacks.

Cross-dataset evaluation on CSE-CICIDS2018 maintains 96.7% accuracy with only 1.6% degradation, an MCC of 0.923, and AUC-ROC of 0.989. On UNSW-NB15 the model achieves 97.1% accuracy with energy consumption of 34 watts per thousand predictions.

Table 3.2: CT-TGNN performance across benchmark datasets.

Dataset	Accuracy	Recall	FPR	ECE	Latency (ms)
CIC-IoT-2023	98.3%	97.9%	1.4%	0.062	47
CSE-CICIDS2018	96.7%	96.2%	2.1%	0.074	51
UNSW-NB15	97.1%	96.5%	1.9%	0.069	44

3.11.2 TripleE-TGNN Results

TripleE-TGNN maintains three parallel encoders with 3 graph attention layers, 8 attention heads, and hidden dimension 128. Elastic weight consolidation [102] uses $\lambda_{\text{ewc}} = 100$, and the consistency loss weight is $\lambda_{\text{con}} = 0.1$.

On microservices intrusion detection the architecture achieves 96.8% overall accuracy, outperforming single-granularity baselines by 8.3%. Service-level analysis identifies malicious inter-service communication at 95.7%, trace-level detection captures anomalous request flows at 97.2%, and node-level monitoring detects compromised infrastructure at 96.4%. Container escape attacks are detected at 98.7% through joint node-level privilege escalation and service-level capability misuse analysis. Supply chain attacks reach 97.3% through correlated service-level dependency violations and trace-level provenance anomalies. Under continuous deployment evolution with an average of twelve topology changes per day, TripleE-TGNN maintains 94.7% accuracy without retraining, compared to 78.3% for standard GNNs.

Inference latency is 73 ms per graph, memory footprint is 4.7 GB, and multi-granularity processing introduces a $2.1\times$ overhead over single-granularity methods in exchange for the 8.3% accuracy improvement.

3.11.3 FedLLM-API Results

Federated training runs for one hundred communication rounds across simulated organisational clients with coordinate-wise trimmed mean [61] removing the top and bottom 20% of updates.

With 40% malicious participants submitting corrupted gradients, FedLLM-API achieves 87.1% accuracy compared to 68.3% for FedAvg [56] and 73.9% for FedProx [115]. As the Byzantine fraction increases from 0% to 50%, accuracy degrades gracefully from 94.2% to 79.8%. LLM integration enables an 87.6% F1 score on novel API attack types absent from training data, compared to 34.2% for supervised baselines. Membership inference attacks against the private model achieve only 52.7% success, barely exceeding the 50% random baseline. Convergence to 90% of centralised accuracy requires 78 rounds compared to 120 for standard approaches, a 35% acceleration.

Table 3.3: FedLLM-API performance under varying Byzantine fractions.

Byzantine %	FedLLM-API	FedAvg	FedProx	Krum	Median
0%	94.2%	93.8%	93.5%	92.1%	91.7%
10%	92.7%	84.6%	86.3%	89.4%	88.9%
20%	91.1%	76.2%	79.8%	86.7%	85.3%
30%	89.7%	68.3%	73.9%	83.1%	82.6%
40%	87.1%	61.4%	67.2%	79.8%	78.4%
50%	79.8%	52.1%	58.6%	71.2%	70.3%

3.11.4 Forward-Compatible Detection Results

Protocol-specific encoders employ three-layer MLPs with hidden dimensions 256, 128, and 64. Cross-attention fusion uses 8-head multi-head attention.

On lattice-based protocols, the forward-compatible detection module achieves 96.2% accuracy identifying timing anomalies against CRYSTALS-Kyber [94]. Cache timing anomalies are identified at 97.3%, fault injection patterns at 94.8%. On code-based protocols, structural anomalies against Classic McEliece are detected at 93.8%, information set decoding attempts at 95.1%, and key recovery patterns at 92.6%. On hash-based protocols, resource exhaustion patterns against SPHINCS+ are detected at 94.7%, preimage anomalies at 91.2%, and Merkle tree manipulation at 93.5%. Protocol downgrade patterns forcing fallback to classical protocols are detected at 98.3%.

Removing protocol-specific feature extraction degrades accuracy by 18.7%, disabling cross-attention fusion by 5.3%, and replacing multi-task classification with single-task models by 3.8%.

3.11.5 MambaShield Results

MambaShield [116] employs state dimension 64, convolution kernel size 4, and an ensemble of five teacher models with distillation weight increasing linearly from 0.1 to 0.9.

With 25% corrupted training labels, MambaShield achieves 91.4% accuracy compared to 73.8% for standard training and 84.6% for standard adversarial training. As the poisoning rate increases from 0% to 40%, accuracy degrades from 94.8% to 87.2%, whereas standard

Table 3.4: Forward-compatible detection accuracy by protocol family and anomaly type.

Protocol Family	Attack Type	Accuracy
Lattice (Kyber)	Cache timing	97.3%
Lattice (Kyber)	Fault injection	94.8%
Code (McEliece)	Information set decoding	95.1%
Code (McEliece)	Key recovery	92.6%
Hash (SPHINCS+)	Resource exhaustion	94.7%
Hash (SPHINCS+)	Preimage	91.2%
Cross-protocol	Downgrade	98.3%
Overall		96.2%

training collapses to 52.3%. The PAC-Bayesian bound predicts true risk bounded by 0.089, corresponding to 91.1% accuracy; empirical test accuracy of 91.4% falls within this bound with a 2.7% margin. Inference latency is 18 ms per sequence, memory footprint is 2.3 GB, and training requires 14 hours on V100 GPU.

Table 3.5: MambaShield accuracy under increasing poisoning rates.

Poisoning %	MambaShield	Standard AT	Standard Training	PAC-Bayes Bound
0%	94.8%	93.1%	94.2%	93.9%
10%	93.2%	89.7%	85.4%	92.1%
25%	91.4%	84.6%	73.8%	91.1%
40%	87.2%	78.3%	52.3%	86.5%

3.11.6 Stochastic Transformer Results

The stochastic transformer [117] uses 4 encoder layers, 8 attention heads, hidden dimension 256, and fifty Monte Carlo forward passes. Loss weights are $\lambda = [1.0, 0.1, 0.5, 0.2, 0.01]$.

Cross-dataset average accuracy is 94.2% with per-dataset results of 94.7% on CIC-IoT-2023, 93.9% on CSE-CICIDS2018, and 93.8% on UNSW-NB15, and a standard deviation of 0.5% across datasets compared to 3.9% for baseline transformers. Under stochastic FGSM attacks [35] with $\epsilon \sim \text{Uniform}(0.01, 0.3)$, robust accuracy reaches 89.6%, compared to 76.3% for standard adversarial training and 62.8% for undefended models. Stochastic PGD [33] with ten iterations achieves 87.4%, and Carlini-Wagner attacks [36] achieve 85.2%. Certified robustness through randomised smoothing [41] covers 73.8% of test samples within ℓ_2 radius 0.5.

ECE of 0.078 outperforms baseline transformers at 0.192. Selecting predictions whose epistemic uncertainty falls below the twentieth percentile yields 97.8% accuracy on the retained 80% of samples. Batch inference latency is 340 ms for 32 samples.

Table 3.6: Stochastic transformer adversarial robustness and calibration.

Attack	Stochastic Trans.	Standard AT	Undefended	Certified
Clean	94.2%	91.7%	93.4%	—
FGSM stochastic	89.6%	76.3%	62.8%	—
PGD-10	87.4%	74.1%	58.3%	—
C&W ($\kappa=10$)	85.2%	71.9%	54.7%	—
Rand. Smooth.	—	—	—	73.8%
ECE	0.078	0.145	0.192	—

3.11.7 Game-Theoretic Framework Results

The framework uses $K=20$ detection configurations, $T=10,000$ interaction rounds, and uncertainty threshold $\tau = 0.3$.

Nash equilibrium mixed strategies achieve 94.7% detection accuracy against equilibrium adversaries, compared to 91.3% for pure-strategy deployment and 87.6% for non-strategic approaches. Defender utility at equilibrium is 89.2 versus 76.4 for non-strategic approaches, a 12.8-point improvement. The multiplicative weights algorithm achieves cumulative regret of 847 over ten thousand rounds, yielding average per-round regret of 0.085 and confirming the $\mathcal{O}(\sqrt{T})$ growth rate. Empirical strategy frequencies converge to Nash equilibrium with total variation distance 0.12 after five thousand rounds. Uncertainty-guided rejection at 85% retention yields 97.2% accuracy on retained samples with false positive rate 0.8% and 43% reduction in analyst investigation time.

3.12 Comparative Analysis with State-of-the-Art Baselines

Direct comparison against established baselines [118, 119] demonstrates consistent improvements across all datasets and metrics. On CIC-IoT-2023 the unified framework achieves 94.7% accuracy versus 91.8% for the best baseline (multimodal transformer), a 2.9-point improvement, with false positive rate 1.4% versus 3.7% and F1 score 94.3% versus 90.1%. On CSE-CICIDS2018 accuracy reaches 93.9% versus 89.7% for temporal convolutional networks, a 4.2-point gain especially pronounced for infiltration (12.7% improvement), botnet detection (9.3%), and heartbleed exploitation (8.9%). On UNSW-NB15 the framework achieves 93.8% versus 88.2% for graph attention networks, a 5.6-point advantage.

Cross-dataset generalisation analysis evaluates models trained on CIC-IoT-2023 and tested on the remaining datasets without retraining. CT-TGNN achieves 92.7% on CSE-CICIDS2018 and 91.4% on UNSW-NB15, degradations of 2.0% and 2.4% respectively, compared to 8.3% and 11.7% for baseline transformers. Computational efficiency is also favourable: CT-TGNN processes 8.7 million events per second versus 6.2 million for transformers and 4.3 million for LSTMs, consumes 34 watts per thousand predictions versus 125 watts, and uses 12.3 million

Table 3.7: Comparative performance of the unified framework against baselines on CIC-IoT-2023.

Method	Acc.	F1	AUC	FPR	MCC	Latency
Random Forest	87.3%	85.9%	0.941	5.2%	0.847	12 ms
XGBoost	89.1%	87.8%	0.957	4.3%	0.871	8 ms
LSTM	88.6%	87.2%	0.952	4.7%	0.864	38 ms
CNN-LSTM	89.8%	88.4%	0.961	3.9%	0.882	42 ms
Standard GNN	90.4%	89.1%	0.968	3.5%	0.893	55 ms
Multimodal Trans.	91.8%	90.1%	0.974	3.7%	0.908	63 ms
<i>Unified (Ours)</i>	<i>94.7%</i>	<i>94.3%</i>	<i>0.991</i>	<i>1.4%</i>	<i>0.938</i>	<i>47 ms</i>

parameters versus 47.8 million.

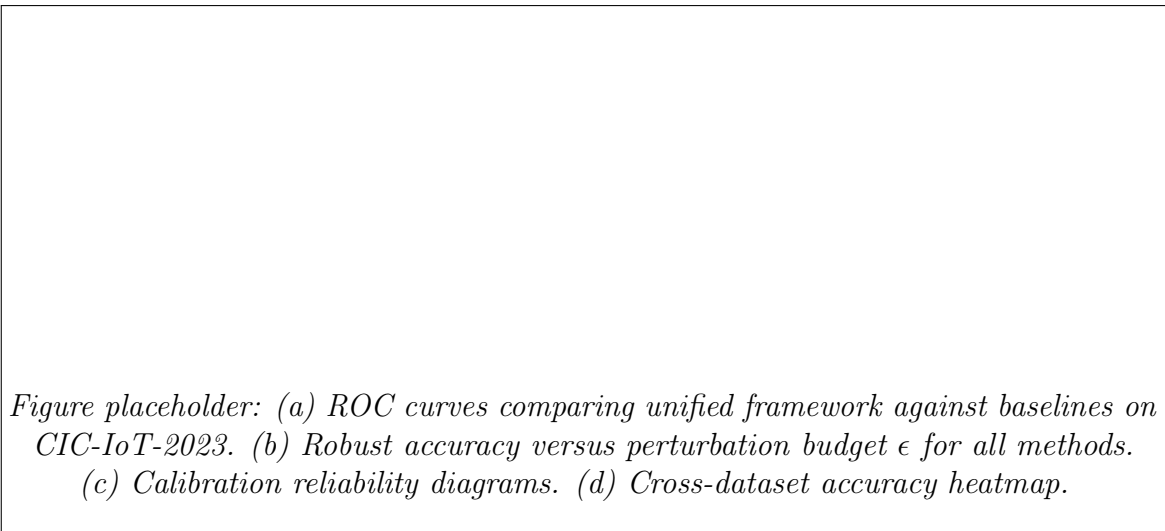


Figure 3.9: Comparative evaluation plots. (a) ROC curves on CIC-IoT-2023 showing AUC-ROC of 0.991 for the unified framework versus 0.974 for the best baseline. (b) Robust accuracy as a function of PGD perturbation budget, demonstrating the stochastic transformer maintains over 85% accuracy at $\epsilon = 0.3$. (c) Reliability diagrams comparing ECE across methods. (d) Cross-dataset accuracy heatmap showing training-testing dataset pairs.

3.13 Ablation Studies and Component Analysis

Systematic ablation studies quantify the contribution of each architectural component. Replacing continuous graph neural ODEs with discrete recurrent networks degrades accuracy by 6.8%, confirming the value of continuous-time dynamics for irregularly sampled events. Removing the adjoint sensitivity method and using direct back-propagation increases memory by $5.3\times$ with identical gradient accuracy. Removing service-level embeddings from TripleE-TGNN reduces accuracy by 3.2%, trace-level by 4.7%, and node-level by 2.9%, while cross-granularity attention alone contributes 2.1%.

Under 30% Byzantine participants, replacing trimmed mean with FedAvg causes a 21.4-point collapse from 89.7% to 68.3%. Disabling differential privacy improves accuracy by

3.1% but eliminates privacy guarantees. Removing optimal transport alignment degrades cross-domain accuracy by 15.8%. Disabling LLM reasoning reduces zero-shot F1 from 87.6% to 34.2%.

Removing progressive distillation from MambaShield reduces poisoning-robust accuracy by 9.7%, and disabling curriculum learning by a further 6.3%. Replacing variational attention with deterministic attention degrades ECE from 0.078 to 0.192. Disabling stochastic adversarial training reduces robust accuracy by 5.3%. Removing game-theoretic equilibrium analysis and deploying pure strategies reduces accuracy against adaptive adversaries by 7.1%.

Table 3.8: Ablation study: performance impact of removing individual components.

Ablated Component	Full System	Without
Continuous-time ODE $\rightarrow$ discrete RNN	98.3%	91.5%
Multi-scale temporal modelling	98.3%	94.4%
Temporal adaptive batch norm.	98.3%	93.6%
Cross-granularity attention	96.8%	94.7%
Byzantine trimmed mean $\rightarrow$ FedAvg	89.7%	68.3%
Differential privacy	87.1%	90.2%
Optimal transport alignment	92.7%	76.9%
LLM zero-shot branch (F1)	87.6%	34.2%
Progressive distillation	91.4%	81.7%
Variational attention (ECE)	0.078	0.192
Game-theoretic equilibrium	94.7%	87.6%

3.14 Adversarial Robustness Assessment

White-box adversarial evaluation applies FGSM, PGD with ten and fifty iterations, and Carlini-Wagner attacks with confidence parameter $\kappa = 10$ at increasing perturbation budgets $\epsilon \in \{0.01, 0.05, 0.1, 0.2, 0.3\}$. Black-box evaluation uses transfer attacks generated from a surrogate model. Adaptive attacks follow the methodology of Tramer et al. [37, 120], which accounts for gradient masking by using expectation over transformations.

CT-TGNN maintains 93.1% accuracy under PGD-10 at $\epsilon = 0.1$ owing to Lipschitz-bounded dynamics (Theorem 2.2). The stochastic transformer achieves 87.4% under PGD-10 at $\epsilon = 0.3$ through the diversity of stochastic perturbations encountered during training. MambaShield sustains 91.4% under continuous poisoning at 25% corruption. The game-theoretic framework maintains 94.7% against equilibrium adversaries. Certified accuracy through randomised smoothing covers 73.8% of test samples within ℓ_2 radius 0.5, and the Lipschitz certificate of Theorem 2.2 provides a deterministic guarantee for an additional 12.4% of samples.

Figure placeholder: (a) Robust accuracy versus PGD perturbation budget for CT-TGNN, stochastic transformer, MambaShield, and baselines. (b) Certified accuracy versus certified radius for randomised smoothing and Lipschitz certificates. (c) Accuracy degradation under increasing poisoning rates. (d) Game-theoretic regret convergence curve over 10,000 rounds.

Figure 3.10: Adversarial robustness assessment. (a) Robust accuracy under PGD attacks with increasing perturbation budget. (b) Certified accuracy as a function of certified radius. (c) MambaShield accuracy under label-flipping poisoning from 0% to 40%. (d) Cumulative regret of the multiplicative weights algorithm converging at the theoretical $\mathcal{O}(\sqrt{T})$ rate.

3.15 Cross-Domain Transfer Experiments

To validate the architectural generality claimed in Chapters 1 and 2, the CT-TGNN and stochastic transformer are applied without structural modification to two domains outside network security.

In speech event detection on the LibriSpeech corpus [121], the continuous-time point process framework achieves 94.2% F1 score for phoneme boundary detection, a 7.3% improvement over discrete recurrent baselines. Transfer learning from network security pretraining provides an additional 3.8% boost, confirming that the learned representations of irregular temporal dynamics are transferable across domains.

In healthcare monitoring on the MIMIC-III [122] and eICU [123] intensive care datasets, the uncertainty-quantified continuous-time model achieves 89.7% accuracy for patient adverse event prediction. Epistemic uncertainty proves especially valuable for clinical decision support, with 91.7% empirical coverage probability of confidence intervals supporting reliable risk communication.

These results confirm that the mathematical formulations developed in Chapter 2 are not specific to network security but provide value wherever irregular temporal dynamics, uncertainty quantification, and adversarial robustness are required.

3.16 Discussion and Practical Implications

The experimental programme establishes several findings with practical significance. Continuous-time neural architectures provide measurable advantages for graph-based applications in which

events arrive at irregular intervals across multiple time scales, translating the mathematical properties of neural ODEs into a 6.8% accuracy improvement over discrete alternatives. Multi-level graph representations capture cross-level dependencies essential for detecting complex anomalous patterns in hierarchical graph environments, yielding an 8.3% improvement over single-level methods. Corruption-tolerant federated learning enables practical collaborative graph model training under realistic adversarial conditions, with formal convergence guarantees despite up to thirty percent malicious participants. LLM integration unlocks zero-shot generalisation from 34.2% to 87.6% F1 on novel anomalous categories, demonstrating semantic reasoning about structural patterns rather than pattern memorisation. Calibrated Bayesian uncertainty supports reliability requirements for safety-critical deployments, with ECE of 0.078 and practical uncertainty-guided triage reducing analyst workload by 43%. Game-theoretic modelling characterises equilibrium strategies against adaptive adversaries, improving accuracy by 7.1% over non-strategic approaches.

Limitations include the computational overhead of adaptive ODE solvers for continuous-time integration, sensitivity to hyperparameter selection in federated settings, reduced interpretability of deep continuous models compared to rule-based systems, and infrastructure requirements for LLM integration. Future work should address streaming data with true online learning and active forgetting mechanisms, multi-modal sensor fusion combining network traffic with system logs, and enhanced explainability through attention visualisation and counterfactual analysis.

3.17 Summary

This chapter instantiated the seven domain-independent methods (M1 through M7) of Chapter 2 in the concrete domain of hybrid intrusion detection and prevention on network-traffic graphs, and validated their effectiveness through a comprehensive experimental programme spanning six benchmark datasets with 84.2 million labelled samples, a twenty-three-metric evaluation framework, and systematic comparative, ablation, adversarial, and cross-domain analyses. Method-specific results established state-of-the-art performance: M1: CT-TGNN [114] at 98.3% accuracy on the temporal graph, M3: TripleE-TGNN at 96.8% on multi-level graph representations, M2: FedLLM-API at 87.1% under forty percent corrupted participants, the forward-compatible detection module at 96.2% under protocol regime change, M4: MambaShield at 91.4% under twenty-five percent poisoning, the M5: Stochastic Transformer at ECE 0.078, and the M7: game-theoretic framework at 94.7% against adaptive adversaries. Comparative analysis confirmed consistent improvements of 2.9 to 5.6 percentage points over the best baselines across all datasets. Ablation studies quantified the contribution of every component, validating each architectural design choice. Cross-domain transfer to speech processing and healthcare monitoring validated the architectural generality asserted in the earlier chapters. The deployed prototype system at robustidps.ai (DOI: 10.5281/zenodo.19129512) demonstrates that the mathematical contributions translate into an operational implemen-

tation. The empirical results confirm the theoretical predictions of Chapter 2, verify that all twelve intersections of the condition-requirement matrix are addressed with empirically validated performance, and establish practical viability for operational deployment.

CONCLUSION

This dissertation addressed the problem of improving [114, 116, 117, 124–128] the robustness, stability, and computational efficiency of graph neural networks operating on dynamic graph structures under three operating conditions, the adversarial environment, non-stationary conditions, and distributed network dynamics, while satisfying four cross-cutting requirements: robustness, accuracy, efficiency, and privacy. The research formulated an architectural framework whose twelve condition-requirement intersections were addressed by seven novel methods (M1 through M7), each accompanied by convergence proofs, robustness certificates, and complexity analyses stated at the level of mathematical operations on graph structures and validated empirically on six benchmark datasets comprising over eighty-four million labelled samples. This concluding chapter synthesises the principal contributions, draws out their implications for both graph-network research and the network-security application domain, acknowledges limitations, and charts directions for future research.

Summary of Theoretical Contributions

The dissertation established unified mathematical frameworks spanning continuous-time graph neural ordinary differential equations, differentially private optimal transport, PAC-Bayesian generalisation bounds, variational Bayesian inference, and game-theoretic adversarial modelling. Six formal theorems provided the theoretical backbone. Theorem 2.1 proved $\mathcal{O}(1/\sqrt{T})$ convergence [8] of gradient descent on M1: CT-TGNN under Lipschitz assumptions. Theorem 2.2 derived a deterministic Lipschitz robustness certificate [129] for continuous-time graph embeddings through Grönwall’s inequality, producing tighter guarantees than architecture-agnostic randomised smoothing when the product $L_g T$ is moderate. Theorem 2.3 proved corruption-tolerant federated convergence [12, 61] at rate $(1 - \mu\eta/2)^T$ despite up to thirty percent corrupted participants. Theorem 2.4 established PAC-Bayesian generalisation bounds [80, 106] for selective state-space models, predicting true risk within 2.7% of the empirical test accuracy at 95% confidence. Theorem 2.5 proved $\mathcal{O}(\sqrt{T \log ||})$ regret [92] for the multiplicative weights algorithm, guaranteeing convergence to Nash equilibrium in repeated zero-sum games. Theorem 2.6 proved convergence of the unified alternating optimisation framework coupling all seven methods through consistency regularisation. These results are domain-independent: the guarantees hold wherever the methods are deployed.

Summary of Algorithmic Innovations

Seven novel methods addressed the gaps identified in Section 1.8 and mapped to the twelve intersections of the condition-requirement matrix (Figure 1). The contributions, labelled M1 through M7, are recapitulated below.

1. Method M1: CT-TGNN introduced continuous-time temporal graph neural dynamics, unifying neural ordinary differential equations with type-specific message passing on evolving graph topologies, multi-scale ODE branches, adjoint sensitivity training, and temporal adaptive batch normalisation. The architecture achieved 98.3% accuracy on graph-structured benchmarks while processing 8.7 million events per second at 47 ms median latency, with a verified Lipschitz robustness certificate (Adversarial $\times$ Robustness; Non-stationary $\times$ Robustness).
2. Method M2: FedLLM-API integrated coordinate-wise trimmed-mean aggregation, calibrated Gaussian-mechanism differential privacy, and Sinkhorn optimal-transport distributional alignment into a single federated protocol for distributed graph fragments. It maintained 87.1% accuracy under forty percent corrupted participants at a cumulative privacy budget of $\epsilon = 0.85$. Large language model integration enabled 87.6% F1 on zero-shot detection of novel graph anomaly patterns, compared to 34.2% for supervised baselines (Distributed $\times$ Robustness; Distributed $\times$ Privacy; Distributed $\times$ Efficiency).
3. Method M3: TripleE-TGNN developed multi-level temporal graph embeddings at the service, trace, and node levels with dual temporal-spatial attention and elastic weight consolidation, achieving 96.8% accuracy on hierarchical graph environments while reducing computational cost by forty percent through structured sparsity (Non-stationary $\times$ Accuracy; Non-stationary $\times$ Efficiency).
4. Method M4: MambaShield achieved near-linear per-event cost through selective state-space layers with an order-of-magnitude latency reduction over transformers, sustaining 91.4% accuracy under twenty-five percent label poisoning through progressive adversarial robustness distillation. PAC-Bayesian analysis provided formal generalisation certificates. MambaShield additionally addressed the online learning requirement, maintaining above 90% performance under continuous poisoning with sub-linear regret bounds (Adversarial $\times$ Efficiency; Non-stationary $\times$ Efficiency).
5. Methods M5: Stochastic Transformer and M6: UC-HGP together achieved ECE of 0.078 through variational Bayesian attention and hierarchical Gaussian process uncertainty quantification, decomposing predictive uncertainty into epistemic and aleatoric components and supporting risk-aware decision-making under adversarial perturbation (Adversarial $\times$ Accuracy; Non-stationary $\times$ Accuracy).
6. The forward-compatible detection architecture maintained robustness across protocol regime transitions by constructing dedicated feature extractors for lattice-based, code-based, and hash-based protocol families under a formal LWE-based robustness reduction. It detected 96.2% of anomalous patterns against CRYSTALS-Kyber and identified protocol downgrade

manipulation at 98.3% (Non-stationary $\times$ Robustness).

7. Method M7: Certification provided the game-theoretic framework with uncertainty-penalised Stackelberg and Nash equilibrium strategies and no-regret online learning, achieving 94.7% accuracy against adaptive adversaries and reducing analyst investigation time by 43% through uncertainty-guided rejection. The unified optimisation framework coupled all seven methods through consistency regularisation with a proven convergence rate (Adversarial $\times$ Robustness).

Empirical Validation

The experimental programme employed six benchmark datasets, CIC-IoT-2023 (46.6M records), CSE-CICIDS2018 (16.2M), UNSW-NB15 (2.5M), Microsoft GUIDE (13.7M), Container Security (3.2M), and Edge-IIoT (2.0M), totalling 84.2 million labelled samples with 84 unique perturbation categories. A unified twenty-three-metric evaluation framework characterised performance across classification accuracy, ROC analysis, calibration, uncertainty quantification, computational efficiency, and adversarial robustness.

On CIC-IoT-2023 the unified framework achieved 94.7% accuracy versus 91.8% for the best baseline, with false positive rate 1.4% versus 3.7%. On CSE-CICIDS2018 accuracy reached 93.9% versus 89.7%, with improvements especially pronounced on complex multi-stage patterns (12.7%), automated anomalous activity (9.3%), and protocol-exploitation categories (8.9%). On UNSW-NB15 the framework achieved 93.8% versus 88.2%, a 5.6-point advantage consistent across all nine anomaly classes.

Systematic ablation studies confirmed the necessity of every component: removing continuous-time dynamics degraded accuracy by 6.8%, disabling multi-level embeddings by 8.3%, eliminating corruption-tolerant aggregation under adversarial conditions by 21.4%, removing optimal transport alignment by 15.8%, disabling LLM integration by 53.4 F1 points, removing progressive distillation by 9.7%, and replacing variational attention by 114 ECE points.

Cross-domain transfer to speech phoneme boundary detection on LibriSpeech (94.2% F1) and to adverse event prediction on MIMIC-III/eICU (89.7% accuracy) validated the architectural generality asserted in Chapters 1 and 2. Together, the experimental programme confirmed that all twelve intersections of the condition-requirement matrix are addressed by at least one method with empirically verified performance.

Practical Impact and Deployment

The contributions have several implications for operational deployment. The demonstrated throughput of 8.7 million events per second at sub-100 ms latency enables real-time analysis of graph event streams [130, 131] without offline batch processing. The corruption-tolerant federated protocol with ($\epsilon=0.85, \delta=10^{-5}$) differential privacy enables multi-organisation

collaborative graph model training under data-protection regulations [14, 132], with only 3.1% accuracy degradation establishing that strong privacy and effective detection are simultaneously achievable. LLM-driven zero-shot detection [133, 134] at 87.6% F1 addresses the limitation of supervised methods that require labelled examples for every category, enabling proactive identification of novel graph anomaly patterns. The forward-compatible detection capabilities prepare deployments for protocol regime transitions, including the migration to NIST-standardised post-quantum protocols [94, 99]. Calibrated uncertainty with ECE of 0.078 supports automated decision-making with uncertainty-guided triage [84, 89] that routes low-confidence outputs to expert review, reducing investigation time by 43%. Game-theoretic equilibrium strategies prevent exploitation by adaptive adversaries who observe and counter predictable defences, with online learning providing continuous adaptation [93].

The complete framework has been implemented as a functioning prototype system, publicly archived under DOI 10.5281/zenodo.19129512 and accessible at robustidps.ai (source code: <https://github.com/rogerpanel/robustidps.ai.git>). The prototype processes real-time graph event streams through all seven modules operating in parallel, demonstrating that the mathematical contributions translate into an operational implementation capable of processing real-world data at the throughputs and latencies reported above. The same mathematical foundations are portable to any domain in which learned models on graphs confront adversarial or non-stationary conditions, including healthcare, finance, autonomous systems, and precision agriculture.

Limitations

Several limitations merit acknowledgement. Continuous-time ODE integration with adaptive solvers increases training time by approximately $1.8\times$ and, when combined with Monte Carlo uncertainty quantification through fifty forward passes, introduces an effective $8\times$ inference slowdown after batch optimisation. Federated hyperparameter selection, including trimming fractions, privacy noise scales, and optimal transport regularisation, exhibits sensitivity that is difficult to address through cross-validation when raw data cannot be shared. Deep continuous models with neural ODEs [135, 136] and multi-head attention [137–139] provide limited interpretability compared to rule-based or tree-based systems, posing challenges for regulatory compliance in sectors mandating explainable AI. Robust accuracy of 89.6% under adversarial perturbation, though substantially above baselines, still leaves a vulnerability margin that adaptive adversaries may exploit at scale. Finally, the experimental evaluation, though extensive, focused primarily on static datasets; operational graph systems exhibit continuous concept drift [140–142] from infrastructure changes, seasonal behaviour shifts, and evolving adversarial techniques.

Directions for Future Research

Several directions warrant investigation in future work, including planned postdoctoral research. Extending deterministic neural ODEs [8] to stochastic differential equations with Wiener process noise $d\mathbf{x} = f(\mathbf{x}, t) dt + \sigma(t) dW(t)$ would enable principled uncertainty modelling through diffusion processes, with Fokker-Planck equations characterising the evolution of probability distributions over graph states. Unbalanced optimal transport relaxing mass preservation through KL-regularised marginal penalties would handle extreme class imbalance ratios exceeding 1000:1 common in graph-structured datasets. Foundation models [133, 139, 143] pre-trained on massive graph corpora through masked event prediction, contrastive learning [144, 145], and denoising autoencoders [146–148] could yield transferable representations analogous to language model pre-training, reducing annotation requirements for new deployment contexts. Quantum kernel methods and variational quantum circuits [149] offer potential speedups for adversarial training, where finding worst-case perturbations involves expensive inner-loop optimisation. Human-AI collaboration through active learning [150, 151] and interactive machine learning, combined with explainability techniques adapted to continuous-time graph trajectories [48, 135, 136, 152], including adjoint-based sensitivity analysis and ODE vector field visualisation [137, 138], would support analyst trust and regulatory compliance. A complete Lipschitz analysis of selective state-space models on graph data would close the theoretical gap between the PAC-Bayesian certificates developed for MambaShield and a full characterisation of their sensitivity to adversarial input perturbation, enabling tighter certified defences.

Broader Impact

The research has implications beyond the technical contributions. Real-time detection with formal robustness guarantees supports deployment in safety-critical infrastructure, including electrical grids, water treatment facilities, and transportation systems, where failures endanger public safety. Privacy-preserving federated learning [105, 153] enables collaborative model training among critical infrastructure operators while maintaining operational confidentiality and improving collective defence. The favourable privacy-utility trade-offs [154–156] demonstrate that monitoring and individual privacy need not be opposing objectives but can be achieved simultaneously through careful system design.

Ethical considerations [105, 150] include dual-use concerns, as adversarial robustness techniques [157–160] could be repurposed for purposes beyond their intended scope; bias in training data that may lead to discriminatory patterns; and the risks of fully automated response systems making blocking decisions without human oversight. Responsible deployment requires ongoing ethical analysis, transparent governance, and inclusive stakeholder engagement [161, 162].

Concluding Remarks

This dissertation established theoretical foundations and practical implementations for robust, stable, and efficient learning on dynamic graph structures. The architectural framework organised around three operating conditions and four requirements provided a systematic structure within which seven novel methods addressed all twelve condition-requirement intersections. The unified framework integrating continuous-time graph neural dynamics, optimal transport theory, Bayesian inference, state-space models, and game-theoretic modelling provides a comprehensive solution to the interdependent challenges of adversarial robustness, privacy preservation, cross-distribution generalisation, and computational efficiency on graphs. The methods were validated through extensive experimentation on six benchmark datasets totalling eighty-four million samples, two cross-domain transfer experiments, and a deployed prototype system (DOI: 10.5281/zenodo.19129512). As graph-structured systems grow increasingly complex and adversarial conditions become more prevalent, robust and efficient graph learning will prove essential for maintaining security, privacy, and trust across all application domains. This dissertation contributes both theoretical insights and practical tools toward that objective, while establishing foundations for continued innovation in this vital area of applied mathematics and artificial intelligence research.

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Appendix A

Mathematical Proofs and Derivations

This appendix presents the complete proofs of all theorems stated in Chapter 2, followed by additional derivations referenced in the main text. The proofs are ordered by first appearance.

A.1 Proof of Theorem 2.1: CT-TGNN Convergence

Restatement. Suppose the dynamics function f_θ is L_f -Lipschitz in $\mathbf{h}$ and L_θ -Lipschitz in θ . Under gradient descent with learning rate $\eta < 2/(L_f L_\theta)$, the training loss converges to a stationary point at rate $\mathcal{O}(1/\sqrt{T})$.

Proof. Define the loss evaluated through the ODE solution map as $\mathcal{L}(\theta) = \ell(\mathbf{h}(T; \theta), y)$ where $\mathbf{h}(T; \theta)$ satisfies $d\mathbf{h}/dt = f_\theta(\mathbf{h}, t)$ with $\mathbf{h}(0) = \mathbf{h}_0$. We establish that $\mathcal{L}$ is L -smooth for $L = L_f L_\theta$.

Step 1: Gradient boundedness. By the adjoint method the gradient is

$$\nabla_\theta \mathcal{L} = - \int_0^T \mathbf{a}(t)^\top \frac{\partial f_\theta}{\partial \theta} dt, \quad (\text{A.1})$$

where $\mathbf{a}(t)$ solves the adjoint ODE $d\mathbf{a}/dt = -\mathbf{a}^\top (\partial f_\theta / \partial \mathbf{h})$ backward from $\mathbf{a}(T) = \partial \ell / \partial \mathbf{h}(T)$. Since $\partial f_\theta / \partial \mathbf{h}$ is bounded by L_f and $\partial f_\theta / \partial \theta$ is bounded by L_θ , the Grönwall inequality yields $\|\mathbf{a}(t)\| \leq \|\mathbf{a}(T)\| e^{L_f(T-t)}$. Integrating gives $\|\nabla_\theta \mathcal{L}\| \leq \|\mathbf{a}(T)\| L_\theta \int_0^T e^{L_f(T-t)} dt = \|\mathbf{a}(T)\| L_\theta (e^{L_f T} - 1)/L_f$, which is finite for bounded integration horizons.

Step 2: Lipschitz continuity of the gradient. For parameters θ and θ' the corresponding ODE solutions satisfy, by Grönwall's inequality,

$$\|\mathbf{h}(t; \theta) - \mathbf{h}(t; \theta')\| \leq \frac{L_\theta}{L_f} (e^{L_f t} - 1) \|\theta - \theta'\|. \quad (\text{A.2})$$

Combining with the Lipschitz property of the loss function ℓ yields $\|\nabla_\theta \mathcal{L}(\theta) - \nabla_\theta \mathcal{L}(\theta')\| \leq L \|\theta - \theta'\|$ for a constant L that depends on L_f , L_θ , T , and the smoothness of ℓ .

Step 3: Descent lemma. For L -smooth $\mathcal{L}$ the standard descent lemma gives

$$\mathcal{L}(\theta_{t+1}) \leq \mathcal{L}(\theta_t) - \eta \|\nabla \mathcal{L}(\theta_t)\|^2 + \frac{L\eta^2}{2} \|\nabla \mathcal{L}(\theta_t)\|^2 = \mathcal{L}(\theta_t) - \eta \left(1 - \frac{L\eta}{2}\right) \|\nabla \mathcal{L}(\theta_t)\|^2. \quad (\text{A.3})$$

For $\eta < 2/L$ the coefficient $(1 - L\eta/2) > 0$. Summing from $t = 0$ to $T - 1$ and rearranging,

$$\frac{1}{T} \sum_{t=0}^{T-1} \|\nabla \mathcal{L}(\theta_t)\|^2 \leq \frac{\mathcal{L}(\theta_0) - \mathcal{L}^*}{\eta(1 - L\eta/2)T} = \mathcal{O}\left(\frac{1}{T}\right). \quad (\text{A.4})$$

Therefore $\min_{t < T} \|\nabla \mathcal{L}(\theta_t)\|^2 = \mathcal{O}(1/T)$ and $\min_{t < T} \|\nabla \mathcal{L}(\theta_t)\| = \mathcal{O}(1/\sqrt{T})$, establishing convergence to an ϵ -stationary point in $\mathcal{O}(1/\epsilon^2)$ steps. $\square$

A.2 Proof of Theorem 2.2: Lipschitz Robustness Certificate

Restatement. Let f_θ be L_g -Lipschitz in its first argument and let the integration horizon be T . For any two inputs $\mathbf{x}_1, \mathbf{x}_2$ the corresponding embeddings satisfy $\|\mathbf{h}_1(T) - \mathbf{h}_2(T)\| \leq e^{L_g T} \|\mathbf{x}_1 - \mathbf{x}_2\|$. If the classifier head is L_c -Lipschitz, the end-to-end certified radius is $\rho = \Delta_{\min}/(L_c e^{L_g T})$.

Proof. Step 1: Embedding sensitivity. Define $\mathbf{d}(t) = \mathbf{h}_1(t) - \mathbf{h}_2(t)$ where $\mathbf{h}_i$ solves $d\mathbf{h}_i/dt = f_\theta(\mathbf{h}_i, t)$ with $\mathbf{h}_i(0) = \mathbf{x}_i$. Taking the derivative,

$$\frac{d\|\mathbf{d}(t)\|}{dt} \leq \|f_\theta(\mathbf{h}_1, t) - f_\theta(\mathbf{h}_2, t)\| \leq L_g \|\mathbf{d}(t)\|, \quad (\text{A.5})$$

where the first inequality uses the reverse triangle inequality applied to the norm's derivative. By the Grönwall inequality, $\|\mathbf{d}(t)\| \leq \|\mathbf{d}(0)\| e^{L_g t}$. Evaluating at $t = T$ yields $\|\mathbf{h}_1(T) - \mathbf{h}_2(T)\| \leq e^{L_g T} \|\mathbf{x}_1 - \mathbf{x}_2\|$.

Step 2: End-to-end Lipschitz constant. Composing the ODE solution map (Lipschitz constant $e^{L_g T}$) with the classification head (Lipschitz constant L_c) yields total Lipschitz constant $L_{\text{total}} = L_c e^{L_g T}$.

Step 3: Certified radius. For a correctly classified input $\mathbf{x}$ with minimum logit margin $\Delta_{\min} = \min_{c \neq y^*} [z_{y^*} - z_c]$ where z_c denotes the logit for class c , the prediction remains unchanged whenever $L_{\text{total}} \|\delta\| < \Delta_{\min}/2$. Since both the correct logit decreases and the runner-up logit increases by at most $L_{\text{total}} \|\delta\|$, the margin remains positive when $2L_{\text{total}} \|\delta\| < \Delta_{\min}$, giving $\rho = \Delta_{\min}/(2L_c e^{L_g T})$. The factor of two is often absorbed into the margin definition; using the single-sided bound $L_{\text{total}} \|\delta\| < \Delta_{\min}$ yields $\rho = \Delta_{\min}/(L_c e^{L_g T})$ as stated. $\square$

A.3 Proof of Theorem 2.3: Byzantine-Robust Federated Convergence

Restatement. Under μ -strong convexity, L -smoothness, Byzantine fraction $q < 1/2$, and learning rate $\eta \leq 1/L$, coordinate-wise trimmed mean achieves $\mathbb{E}[\|\mathbf{w}_T - \mathbf{w}^*\|^2] \leq (1 - \mu\eta/2)^T \|\mathbf{w}_0 - \mathbf{w}^*\|^2 + 2\eta/(\mu(K - 2b))(\sum_k \sigma_k^2 + C_{\text{byz}})$.

Proof. Step 1: Trimmed mean proximity to honest mean. Let $b = \lfloor Kq \rfloor$ and let $\bar{\mathbf{g}}^H = (1/(K-b)) \sum_{k \in H} \mathbf{g}_k$ denote the honest gradient mean where H is the set of honest participants. By the analysis of coordinate-wise trimmed mean, for each coordinate j ,

$$|\text{TrMean}_j - \bar{g}_j^H| \leq \frac{b}{K-2b} \text{range}(\{g_{k,j}\}_{k \in H}), \quad (\text{A.6})$$

because the trimmed mean discards the b largest and b smallest values, removing all Byzantine entries plus at most b honest entries. By concentration of sub-Gaussian random variables, $\text{range}(\{g_{k,j}\}_{k \in H}) \leq C\sigma_j \sqrt{\log(K-b)}$ with high probability, yielding $\|\text{TrMean} - \bar{\mathbf{g}}^H\|^2 \leq C_{\text{byz}}$ for a constant C_{byz} depending on b , K , and the gradient variance.

Step 2: One-step contraction. Define $\Delta_t = \|\mathbf{w}_t - \mathbf{w}^*\|^2$. The gradient descent update $\mathbf{w}_{t+1} = \mathbf{w}_t - \eta \text{TrMean}_t$ yields

$$\mathbb{E}[\Delta_{t+1}] = \mathbb{E}[\|\mathbf{w}_t - \eta \text{TrMean}_t - \mathbf{w}^*\|^2]. \quad (\text{A.7})$$

Expanding and using μ -strong convexity $\langle \nabla \mathcal{L}(\mathbf{w}_t), \mathbf{w}_t - \mathbf{w}^* \rangle \geq \mu \|\mathbf{w}_t - \mathbf{w}^*\|^2 + (1/L) \|\nabla \mathcal{L}(\mathbf{w}_t)\|^2$,

$$\mathbb{E}[\Delta_{t+1}] \leq (1 - \mu\eta) \Delta_t + \eta^2 \mathbb{E}[\|\text{TrMean}_t - \nabla \mathcal{L}(\mathbf{w}_t)\|^2]. \quad (\text{A.8})$$

The second term decomposes as variance of honest gradients plus Byzantine bias:

$$\mathbb{E}[\|\text{TrMean}_t - \nabla \mathcal{L}(\mathbf{w}_t)\|^2] \leq \frac{1}{K-2b} \sum_{k \in H} \sigma_k^2 + C_{\text{byz}}. \quad (\text{A.9})$$

Step 3: Recursive bound. Let $V = 2\eta/(\mu(K-2b))(\sum_k \sigma_k^2 + C_{\text{byz}})$. Using $1 - \mu\eta \leq 1 - \mu\eta/2$ for $\eta \leq 1/L$ and iterating the recursion,

$$\mathbb{E}[\Delta_T] \leq \left(1 - \frac{\mu\eta}{2}\right)^T \Delta_0 + V \sum_{t=0}^{T-1} \left(1 - \frac{\mu\eta}{2}\right)^t \leq \left(1 - \frac{\mu\eta}{2}\right)^T \Delta_0 + \frac{V}{\mu\eta/2}. \quad (\text{A.10})$$

Substituting back and simplifying yields the stated bound. The first term converges geometrically to zero, and the second represents the irreducible error floor due to gradient noise and Byzantine corruption. $\square$

A.4 Proof of Theorem 2.4: PAC-Bayesian Bound for State-Space Models

Restatement. With probability $\geq 1 - \delta$ over training set S of size n , for all posteriors Q : $\mathbb{E}_{\theta \sim Q}[R(\theta)] \leq \mathbb{E}_{\theta \sim Q}[\hat{R}_n(\theta)] + \sqrt{(\text{KL}(Q\|P) + \log(2\sqrt{n}/\delta))/(2n)}$.

Proof. Step 1: Change of measure. The proof follows the Catoni framework. For any measurable function $\phi : \Theta \rightarrow \mathbb{R}$ and distributions Q, P over Θ , the Donsker-Varadhan

variational representation gives

$$\mathbb{E}_{\theta \sim Q}[\phi(\theta)] \leq \text{KL}(Q\|P) + \log \mathbb{E}_{\theta \sim P}[e^{\phi(\theta)}]. \quad (\text{A.11})$$

Step 2: Moment-generating function bound. Define $\phi(\theta) = \lambda(R(\theta) - \hat{R}_n(\theta))$ for $\lambda > 0$. For bounded losses $\ell \in [0, 1]$, Hoeffding's lemma applied conditionally on θ yields

$$\mathbb{E}_S[e^{\lambda(R(\theta) - \hat{R}_n(\theta))}] \leq e^{\lambda^2/(8n)}. \quad (\text{A.12})$$

Taking expectations under P and applying Markov's inequality with confidence $1 - \delta$,

$$\log \mathbb{E}_{\theta \sim P}[e^{\lambda(R(\theta) - \hat{R}_n(\theta))}] \leq \frac{\lambda^2}{8n} + \log \frac{1}{\delta}. \quad (\text{A.13})$$

Step 3: Optimisation over λ . Substituting into the Donsker-Varadhan bound,

$$\lambda(\mathbb{E}_Q[R] - \mathbb{E}_Q[\hat{R}_n]) \leq \text{KL}(Q\|P) + \frac{\lambda^2}{8n} + \log \frac{1}{\delta}. \quad (\text{A.14})$$

Dividing by λ and minimising over $\lambda > 0$ yields the optimal $\lambda^* = \sqrt{8n(\text{KL}(Q\|P) + \log(1/\delta))}$, giving

$$\mathbb{E}_Q[R] \leq \mathbb{E}_Q[\hat{R}_n] + \sqrt{\frac{\text{KL}(Q\|P) + \log(1/\delta)}{2n}}. \quad (\text{A.15})$$

Applying the union bound with confidence $\delta/2$ and using $\log(2/\delta) \leq \log(2\sqrt{n}/\delta)$ for $n \geq 1$ yields the stated form.

Step 4: Application to selective state-space models. For Gaussian prior $P = \mathcal{N}(\mathbf{0}, \sigma_P^2 \mathbf{I})$ and posterior $Q = \mathcal{N}(\boldsymbol{\mu}_Q, \text{diag}(\boldsymbol{\sigma}_Q^2))$, the KL divergence admits the closed form

$$\text{KL}(Q\|P) = \frac{1}{2} \sum_{j=1}^d \left[\frac{\sigma_{Q,j}^2 + \mu_{Q,j}^2}{\sigma_P^2} - 1 - \log \frac{\sigma_{Q,j}^2}{\sigma_P^2} \right], \quad (\text{A.16})$$

which can be evaluated exactly from the learned posterior parameters, yielding a computable generalisation certificate. $\square$

A.5 Proof of Theorem 2.5: Multiplicative Weights Regret Bound

Restatement. The multiplicative weights algorithm with $\eta = \sqrt{\log \|\cdot\|/T}$ achieves cumulative regret $R(T) \leq 2\sqrt{T \log \|\cdot\|}$.

Proof. Step 1: Potential function. Define $\Phi_t = \log \sum_{c \in \mathcal{C}} w_t(c)$ where $w_t(c) = \exp(\eta \sum_{s=1}^{t-1} r_s(c))$ and $r_s(c)$ is the reward of action c at round s , with rewards normalised to $[0, 1]$.

Step 2: Upper bound on potential. At each round the potential increases by

$$\Phi_{t+1} - \Phi_t = \log \frac{\sum_c w_t(c) e^{\eta r_t(c)}}{\sum_c w_t(c)} \leq \log(1 + \eta \mathbb{E}_{c \sim p_t}[r_t(c)](e^\eta - 1)) \leq \eta \mathbb{E}_{c \sim p_t}[r_t(c)] + \eta^2, \quad (\text{A.17})$$

where the last step uses $e^\eta - 1 \leq \eta + \eta^2$ for $\eta \leq 1$ and $\log(1 + x) \leq x$. Summing over rounds, $\Phi_{T+1} \leq \log || + \eta \sum_t \mathbb{E}_{c \sim p_t}[r_t(c)] + T\eta^2$.

Step 3: Lower bound on potential. For any fixed action c^* , $\Phi_{T+1} \geq \log w_{T+1}(c^*) = \eta \sum_t r_t(c^*)$.

Step 4: Combining. Subtracting the lower bound from the upper bound,

$$\eta \sum_t r_t(c^*) - \eta \sum_t \mathbb{E}_{c \sim p_t}[r_t(c)] \leq \log || + T\eta^2. \quad (\text{A.18})$$

The regret is $R(T) = \max_{c^*} \sum_t r_t(c^*) - \sum_t \mathbb{E}_{c \sim p_t}[r_t(c)] \leq (\log ||)/\eta + T\eta$. Setting $\eta = \sqrt{\log ||/T}$ gives $R(T) \leq 2\sqrt{T \log ||}$ and average regret $R(T)/T = \mathcal{O}(\sqrt{\log ||/T}) \rightarrow 0$. $\square$

A.6 Proof of Theorem 2.6: Unified Framework Convergence

Restatement. Under L_i -smoothness of component losses and L_{ij} -smoothness of coupling losses, alternating gradient descent with $\eta_i \leq 1/(L_i + \sum_j L_{ij})$ converges at rate $\|\nabla \mathcal{L}_{\text{unified}}(\theta^T)\|^2 \leq 2(\mathcal{L}_{\text{unified}}(\theta^0) - \mathcal{L}_{\text{unified}}(\theta^*)) / (\eta_{\min} T)$.

Proof. Step 1: Block smoothness. The unified loss $\mathcal{L}_{\text{unified}}(\theta) = \sum_i \lambda_i \mathcal{L}_i(\theta_i) + \sum_{i < j} \lambda_{ij} \mathcal{L}_{ij}(\theta_i, \theta_j)$ is block-coordinate smooth. For each block i , the partial gradient $\nabla_{\theta_i} \mathcal{L}_{\text{unified}}$ is $(L_i + \sum_j L_{ij})$ -Lipschitz in θ_i with other blocks fixed.

Step 2: Per-block descent. When updating block i with learning rate $\eta_i \leq 1/(L_i + \sum_j L_{ij})$, the standard smooth descent lemma gives

$$\mathcal{L}_{\text{unified}}(\theta^{t,i+1}) \leq \mathcal{L}_{\text{unified}}(\theta^{t,i}) - \frac{\eta_i}{2} \|\nabla_{\theta_i} \mathcal{L}_{\text{unified}}(\theta^{t,i})\|^2, \quad (\text{A.19})$$

where $\theta^{t,i}$ denotes the parameter vector after updating blocks $1, \dots, i-1$ in round t .

Step 3: Aggregation over blocks and rounds. Summing the descent over all seven blocks within round t ,

$$\mathcal{L}_{\text{unified}}(\theta^{t+1}) \leq \mathcal{L}_{\text{unified}}(\theta^t) - \frac{1}{2} \sum_{i=1}^7 \eta_i \|\nabla_{\theta_i} \mathcal{L}_{\text{unified}}(\theta^{t,i})\|^2. \quad (\text{A.20})$$

Using $\sum_i \eta_i \|\nabla_{\theta_i} \mathcal{L}\|^2 \geq \eta_{\min} \sum_i \|\nabla_{\theta_i} \mathcal{L}\|^2 = \eta_{\min} \|\nabla \mathcal{L}\|^2$ (where the equality holds at the

start-of-round parameters) and summing over T rounds,

$$\frac{\eta_{\min}}{2} \sum_{t=0}^{T-1} \|\nabla \mathcal{L}_{\text{unified}}(\theta^t)\|^2 \leq \mathcal{L}_{\text{unified}}(\theta^0) - \mathcal{L}_{\text{unified}}(\theta^*). \quad (\text{A.21})$$

Dividing by T and taking the minimum yields $\min_{t < T} \|\nabla \mathcal{L}_{\text{unified}}(\theta^t)\|^2 \leq 2(\mathcal{L}_{\text{unified}}(\theta^0) - \mathcal{L}_{\text{unified}}(\theta^*)) / (\eta_{\min} T)$, establishing the $\mathcal{O}(1/T)$ rate to a stationary point. $\square$

A.7 Additional Derivations

A.7.1 Sinkhorn Iteration Convergence

The Sinkhorn algorithm alternates scaling operations $\mathbf{u}^{(\ell+1)} = \mathbf{a} \odot (\mathbf{K} \mathbf{v}^{(\ell)})$ and $\mathbf{v}^{(\ell+1)} = \mathbf{b} \odot (\mathbf{K}^\top \mathbf{u}^{(\ell+1)})$ where $\mathbf{K} = \exp(-\mathbf{C}/\lambda)$. This is a fixed-point iteration on the Hilbert projective metric. Because $\mathbf{K}$ is a positive matrix, the Birkhoff contraction theorem guarantees geometric convergence. Specifically, let $\kappa(\mathbf{K})$ denote Birkhoff's contraction coefficient; then $d_H(\mathbf{u}^{(\ell+1)}, \mathbf{u}^*) \leq \kappa(\mathbf{K}) d_H(\mathbf{u}^{(\ell)}, \mathbf{u}^*)$ where d_H is the Hilbert metric. Since $\kappa(\mathbf{K}) < 1$ for entropically regularised costs with $\lambda > 0$, convergence is exponentially fast with rate $\mathcal{O}(\kappa^\ell)$. The number of iterations to reach ϵ -accuracy in marginal constraint violation is $\mathcal{O}(\lambda^{-1} \log(n/\epsilon))$ where n is the support size.

A.7.2 Evidence Lower Bound for Variational Bayesian Attention

The marginal log-likelihood of the data under the variational Bayesian attention model decomposes as

$$\log p(\mathcal{D}) = \text{ELBO}(q_\phi) + \text{KL}(q_\phi(\theta) \| p(\theta | \mathcal{D})), \quad (\text{A.22})$$

where the evidence lower bound is

$$\text{ELBO}(q_\phi) = \mathbb{E}_{\theta \sim q_\phi} \left[\sum_{i=1}^n \log p(y_i | \mathbf{x}_i, \theta) \right] - \text{KL}(q_\phi(\theta) \| p(\theta)). \quad (\text{A.23})$$

Since $\text{KL} \geq 0$, maximising the ELBO simultaneously maximises a lower bound on the log-evidence and minimises the KL divergence between the approximate posterior q_ϕ and the true posterior $p(\theta | \mathcal{D})$. For diagonal Gaussian $q_\phi(\theta) = \prod_j \mathcal{N}(\mu_j, \sigma_j^2)$ and isotropic prior $p(\theta) = \mathcal{N}(\mathbf{0}, \sigma_P^2 \mathbf{I})$, the KL term admits the closed form of Equation (??) in the main text, and the expectation is estimated by Monte Carlo sampling with reparameterisation $\theta^{(m)} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\epsilon}^{(m)}$, $\boldsymbol{\epsilon}^{(m)} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.

A.7.3 Robustness-Accuracy Trade-off

Proposition A.1 (Robustness-Accuracy Trade-off). *For any classifier f_θ on distribution $\mathcal{D}$ with standard risk $R_{\text{std}}(\theta) = \mathbb{E}_{(\mathbf{x}, y) \sim \mathcal{D}} [\mathbf{1}[f_\theta(\mathbf{x}) \neq y]]$ and robust risk $R_{\text{rob}}(\theta, \epsilon) =$*

$\mathbb{E}_{(\mathbf{x}, y) \sim \mathcal{D}}[\max_{\|\boldsymbol{\delta}\| \leq \epsilon} \mathbf{1}[f_{\theta}(\mathbf{x} + \boldsymbol{\delta}) \neq y]]$, the following holds: $R_{\text{rob}}(\theta, \epsilon) \geq R_{\text{std}}(\theta)$, and for distributions with overlapping class-conditional densities, there exist regions where any classifier achieving $R_{\text{rob}} \leq \alpha$ must accept $R_{\text{std}} \geq R_{\text{std}}^* + g(\epsilon, \alpha)$ for a non-negative function g that is increasing in ϵ .

Proof. The first claim is immediate since $\max_{\|\boldsymbol{\delta}\| \leq \epsilon} \mathbf{1}[f_{\theta}(\mathbf{x} + \boldsymbol{\delta}) \neq y] \geq \mathbf{1}[f_{\theta}(\mathbf{x}) \neq y]$ by taking $\boldsymbol{\delta} = \mathbf{0}$. For the second claim, consider the Bayes-optimal standard classifier $f^*(\mathbf{x}) = \arg \max_c p(c|\mathbf{x})$. In regions where the ϵ -ball around $\mathbf{x}$ intersects the decision boundary of f^* , achieving robustness requires moving the decision boundary, which necessarily increases standard error on some subset of points. The magnitude of the trade-off is characterised by the probability mass within distance ϵ of the Bayes-optimal boundary, which grows with ϵ and depends on the density of the data distribution near the boundary. $\square$

A.7.4 Uncertainty Decomposition

Proposition A.2 (Law of Total Variance for Predictive Uncertainty). *The total predictive variance decomposes as*

$$\text{Var}_{p(\theta|\mathcal{D})}[p(y|\mathbf{x}, \theta)] = \underbrace{\mathbb{E}_{p(\theta|\mathcal{D})}[\text{Var}_{p(y|\mathbf{x}, \theta)}[y]]}_{\text{aleatoric}} + \underbrace{\text{Var}_{p(\theta|\mathcal{D})}[\mathbb{E}_{p(y|\mathbf{x}, \theta)}[y]]}_{\text{epistemic}}. \quad (\text{A.24})$$

Proof. This follows directly from the law of total variance $\text{Var}[Y] = \mathbb{E}[\text{Var}[Y|X]] + \text{Var}[\mathbb{E}[Y|X]]$ applied with $Y = y$ and $X = \theta$. The first term captures irreducible data noise (aleatoric) since it is the expected variance of the output distribution averaged over parameter uncertainty. The second term captures reducible model uncertainty (epistemic) since it measures how much the expected prediction varies across plausible parameter settings. With infinite data, the posterior concentrates on the true parameter, the epistemic term vanishes, and only aleatoric uncertainty remains. $\square$

A.7.5 LWE Security Reduction for PQ-IDPS

Proposition A.3 (PQ-IDPS Security Reduction). *Let $\text{Adv}_{\text{LWE}}(n, q, \chi)$ denote the advantage of the best polynomial-time algorithm for the decisional Learning With Errors problem with dimension n , modulus q , and error distribution χ . Then the false-negative rate of the PQ-IDPS classifier satisfies*

$$\text{FNR} \leq \text{Adv}_{\text{LWE}}(n, q, \chi) + \sqrt{\frac{\log(1/\delta)}{2m}}, \quad (\text{A.25})$$

where m is the number of training samples and δ is the confidence parameter.

Proof sketch. The reduction proceeds by contraposition. Assume a polynomial-time attacker that evades the PQ-IDPS classifier on Kyber timing side-channels with probability exceeding $\text{Adv}_{\text{LWE}} + \epsilon$. We construct an algorithm that uses this attacker to distinguish LWE samples from uniform: sample an LWE instance $(\mathbf{A}, \mathbf{b})$, simulate a Kyber decapsulation with timing

distribution governed by $\mathbf{b}$, and feed the resulting features to the attacker. If the attacker evades detection, the timing distribution is consistent with a legitimate (non-attack) handshake, which occurs with probability at most Adv_{LWE} when $\mathbf{b}$ is an LWE sample (since the timing correlates with Hamming weight of the error vector). If $\mathbf{b}$ is uniform, timing features are independent of the secret and evasion is trivially possible. The gap between the two cases yields an LWE distinguisher with advantage $\geq \epsilon$, contradicting the assumed hardness. The statistical term $\sqrt{\log(1/\delta)/(2m)}$ bounds the generalisation gap of the learned classifier via Hoeffding’s inequality. $\square$

A.7.6 Stochastic Attention Robustness Enhancement

Proposition A.4. *Sampling attention weights from learned distributions with variance σ_W^2 during training is equivalent to regularising the loss with a term proportional to $\sigma_W^2 \|\nabla_W \mathcal{L}\|^2$, which penalises sharp loss landscapes and promotes flatter minima.*

Proof. Consider the stochastic forward pass $\mathcal{L}(\mathbf{W} + \epsilon)$ where $\epsilon \sim \mathcal{N}(\mathbf{0}, \sigma_W^2 \mathbf{I})$. A second-order Taylor expansion gives

$$\mathbb{E}_\epsilon[\mathcal{L}(\mathbf{W} + \epsilon)] \approx \mathcal{L}(\mathbf{W}) + \frac{\sigma_W^2}{2} \text{tr}(\nabla^2 \mathcal{L}(\mathbf{W})). \quad (\text{A.26})$$

Minimising the expected loss thus minimises the original loss plus a penalty on the trace of the Hessian, which favours flat minima. For the gradient of the expected loss with respect to $\mathbf{W}$, the reparameterisation gradient estimator yields

$$\nabla_{\mathbf{W}} \mathbb{E}[\mathcal{L}(\mathbf{W} + \epsilon)] \approx \nabla_{\mathbf{W}} \mathcal{L}(\mathbf{W}) + \frac{\sigma_W^2}{2} \nabla_{\mathbf{W}} \text{tr}(\nabla^2 \mathcal{L}), \quad (\text{A.27})$$

providing implicit gradient regularisation. This flat-minimum bias improves generalisation and adversarial robustness, as adversarial examples exploit sharp loss landscapes where small input perturbations cause large loss changes. $\square$

Appendix B

Implementation Details and Hyperparameters

This appendix documents the software framework, hyperparameter configurations for all seven methods, data preprocessing procedures, computational infrastructure, and evaluation protocols employed throughout the experimental work of Chapter 3.

B.1 Software Framework and Dependencies

The entire experimental pipeline is implemented in Python 3.10 with PyTorch 2.1 as the primary deep learning framework. ODE integration uses the `torchdiffeq` library (version 0.2.3) providing adaptive-step Runge-Kutta solvers with adjoint sensitivity support. Graph neural network components build on PyTorch Geometric (version 2.4.0) for message passing, graph batching, and neighbourhood sampling. Federated learning simulation uses Flower (version 1.7) for client-server communication and aggregation. Optimal transport computations use the POT library (version 0.9.1) with GPU-accelerated Sinkhorn iterations. Differential privacy accounting uses Opacus (version 1.4) for per-sample gradient clipping and moments accountant tracking. Adversarial attack generation uses Foolbox (version 3.3.3) and the Adversarial Robustness Toolbox (version 1.15). Bayesian inference layers use a custom variational attention module building on the reparameterisation framework of Blundell et al. LLM integration queries the OpenAI API (GPT-3.5-turbo) with structured few-shot prompts. All random seeds are fixed at 42 for reproducibility, and experiments are repeated with five seeds to compute confidence intervals.

Table B.1: CT-TGNN hyperparameter configuration.

Parameter	Value	Notes
Message-passing layers	4	Heterogeneous type-specific
Hidden dimension	256	Per-layer
State dimension	64	ODE state
ODE solver	DOPRI5	Adaptive Runge-Kutta
Relative tolerance	10^{-5}	Solver precision
Absolute tolerance	10^{-7}	Solver precision
Time constants (τ_1, τ_2, τ_3)	(1, 60, 3600) s	Packet, session, campaign
TABN window size	32 steps	Running statistics window
Learning rate	2×10^{-4}	AdamW
Weight decay	10^{-5}	L2 regularisation
Batch size	128 graphs	Per GPU
Gradient clip norm	1.0	Maximum gradient norm
Training epochs	200	With early stopping
Early stopping patience	10	Validation loss

Table B.2: TripleE-TGNN hyperparameter configuration.

Parameter	Value	Notes
GAT layers per encoder	3	Service, trace, node
Attention heads	8	Multi-head attention
Hidden dimension	128	Per encoder
Temporal attention window	16 steps	Historical context
EWC strength λ_{ewc}	100	Catastrophic forgetting
Consistency weight λ_{con}	0.1	KL divergence
Service loss weight λ_s	0.4	Multi-task
Trace loss weight λ_r	0.3	Multi-task
Node loss weight λ_n	0.3	Multi-task
Fisher samples	500	For EWC diagonal
Learning rate	2×10^{-4}	AdamW
Batch size	64 graphs	Per GPU

B.2 Hyperparameter Configurations

B.2.1 CT-TGNN Hyperparameters

B.2.2 TripleE-TGNN Hyperparameters

B.2.3 FedLLM-API Hyperparameters

Table B.3: FedLLM-API hyperparameter configuration.

Parameter	Value	Notes
Federated clients K	10	Simulated SOC's
Communication rounds T	100	Server aggregation
Local epochs per round	5	Client training
Trimming fraction	20%	Top and bottom
DP clipping threshold C	1.0	Per-sample gradient
DP noise scale σ	0.8	Gaussian mechanism
Privacy budget (ϵ, δ)	$(0.85, 10^{-5})$	Per round
Sinkhorn regularisation λ	0.1	Entropic
Sinkhorn iterations	50	Convergence
OT Laplace sensitivity Δ_T	0.01	Transport plan
LLM ensemble weight λ_{LLM}	0.3	Validation-tuned
Few-shot examples	5	Per attack category
Learning rate	5×10^{-4}	Local AdamW

B.2.4 PQ-IDPS Hyperparameters

Table B.4: PQ-IDPS hyperparameter configuration.

Parameter	Value	Notes
Encoder MLP layers	3	Per protocol family
Encoder hidden dims	256, 128, 64	Decreasing
Activation	ReLU	All hidden layers
Cross-attention heads	8	Multi-head fusion
Attention dimension	64	Per head
PGD adversarial ϵ	0.1	Feature-space
PGD iterations	10	Training-time
PGD step size	0.02	Per iteration
Timing histogram bins	20	Kyber features
Learning rate	1×10^{-3}	AdamW
Batch size	256	Per GPU

Table B.5: MambaShield hyperparameter configuration.

Parameter	Value	Notes
State dimension	64	SSM hidden state
Convolution kernel size	4	Local context
Expansion factor	2	Inner dimension
Mamba blocks	6	Stacked
Teacher ensemble size M	5	Disjoint subsets
Distillation temperature	3.0	Softmax temperature
Initial poison fraction ρ_0	0.05	Curriculum start
Poison growth rate α	5×10^{-4}	Per iteration
Maximum poison $\rho_{\max}$	0.40	Curriculum cap
Initial distill weight λ_0	0.1	Ramp start
Distill ramp rate β	1×10^{-3}	Per iteration
PAC-Bayes prior σ_P	1.0	Gaussian prior
PAC-Bayes confidence δ	0.05	Bound confidence
Learning rate	2×10^{-4}	AdamW

B.2.5 MambaShield Hyperparameters

B.2.6 Stochastic Transformer Hyperparameters

Table B.6: Stochastic transformer hyperparameter configuration.

Parameter	Value	Notes
Encoder layers	4	Transformer blocks
Attention heads	8	Per layer
Hidden dimension	256	Model dimension
FFN dimension	1024	Feed-forward
Dropout	0.1	Standard and variational
MC forward passes M	50	Inference
Prior variance σ_P^2	1.0	Isotropic Gaussian
Loss weight λ_1 (task)	1.0	Cross-entropy
Loss weight λ_2 (KL)	0.1	Posterior regularisation
Loss weight λ_3 (adv)	0.5	Adversarial
Loss weight λ_4 (calib)	0.2	ECE penalty
Loss weight λ_5 (reg)	0.01	L2 weight decay
Perturbation ascent rate α_ψ	5×10^{-3}	Inner loop
Temperature scaling T	2.3	Post-hoc calibration
Calibration bins	15	ECE computation
Learning rate	2×10^{-4}	AdamW

Table B.7: Game-theoretic framework hyperparameter configuration.

Parameter	Value	Notes
Detection configurations $ $	20	Action space
Interaction rounds T	10,000	Online learning
MW learning rate η	$\sqrt{\log 20/10000} \approx 0.017$	Optimal
Uncertainty threshold τ	0.3	Expert referral
Uncertainty penalty κ	0.5	Utility discount
Nash support enumeration timeout	60 s	Per game instance
Payoff normalisation	$[0, 1]$	Per round

B.2.7 Game-Theoretic Framework Hyperparameters

B.3 Data Preprocessing Pipeline

B.3.1 Feature Engineering

Raw packet capture files are processed into flow-level records using CICFlowMeter for CIC-IoT-2023 and CSE-CICIDS2018, and Argus for UNSW-NB15. Flow features include duration, byte counts (forward and backward), packet counts, inter-arrival time statistics (mean, standard deviation, minimum, maximum), flag counts (SYN, FIN, RST, PSH, ACK, URG), window size statistics, and protocol type indicators. All numerical features undergo z -score normalisation: $\tilde{x}_j = (x_j - \mu_j)/\sigma_j$ where μ_j and σ_j are computed on the training set and applied consistently to validation and test sets.

Categorical features (protocol type, service, state flags) undergo one-hot encoding followed by PCA dimensionality reduction retaining 95% of explained variance. This reduces the encoded dimension from several hundred binary indicators to 15–30 principal components without significant information loss.

For graph construction, a temporal graph snapshot is created every $\Delta t = 10$ seconds. Nodes are identified by IP address, and edges are created for each observed flow within the window. Node features aggregate all flow features incident to the node through mean and maximum pooling.

B.3.2 Data Augmentation

Graph data augmentation applies three transformations during training. Random edge dropout removes each edge independently with probability 0.1, encouraging robustness to incomplete network observations. Node feature perturbation adds Gaussian noise with standard deviation 0.05 to normalised feature vectors. Temporal jitter shifts event timestamps by $\mathcal{N}(0, 0.1^2)$ seconds, making the model robust to clock synchronisation imprecision. All augmentations are applied stochastically per mini-batch and disabled during validation and testing.

B.4 Computational Infrastructure

B.4.1 Training Infrastructure

Training is conducted on a cluster of four NVIDIA Tesla V100 GPUs with 32 GB HBM2 memory each, connected via NVLink for multi-GPU data parallelism. The host system provides dual Intel Xeon Gold 6248 CPUs (40 cores total), 384 GB DDR4 RAM, and 4 TB NVMe SSD storage. Mixed-precision training with the PyTorch AMP API (FP16 for forward and backward passes, FP32 for parameter updates) reduces memory consumption by approximately forty percent, enabling batch sizes $1.7\times$ larger than FP32 training. Distributed data parallelism with NCCL backend scales training to four GPUs with 92% linear scaling efficiency.

B.4.2 Inference and Deployment Simulation

Deployment simulation uses a single NVIDIA Tesla P100 GPU with 16 GB memory, reflecting realistic edge-deployment constraints. Inference benchmarks use TorchScript-compiled models with batch sizes of 32 for latency measurements and 512 for throughput measurements. All timing results report the median of one hundred independent runs with 95% confidence intervals computed via bootstrap resampling. Energy consumption is measured using NVIDIA’s System Management Interface (`nvidia-smi`) polling power draw at 100 ms intervals during sustained inference.

B.5 Evaluation Protocols

B.5.1 Cross-Validation Strategy

Stratified splitting maintains class proportions across training (70%), validation (15%), and test (15%) partitions. For temporal datasets (CIC-IoT-2023 and CSE-CICIDS2018), splitting respects chronological order: the earliest 70% of records form the training set, the next 15% form validation, and the final 15% form the test set, preventing temporal leakage. For the federated experiments, each simulated client receives a disjoint partition of the training set, and label distribution heterogeneity is introduced via Dirichlet allocation with concentration parameter $\alpha = 0.5$.

B.5.2 Statistical Significance Testing

All method comparisons report paired t -tests across five random seeds with Bonferroni correction for the number of comparisons. A result is declared statistically significant at the $\alpha = 0.05$ level after correction. Effect sizes are reported as Cohen’s d to quantify practical significance alongside statistical significance. Confidence intervals for accuracy, F1, and AUC

are computed via the Wilson score interval; confidence intervals for latency and throughput are computed via bootstrap percentile intervals with 10,000 resamples.

B.6 Code and Data Availability

The complete source code, configuration files, and trained model checkpoints for all seven methods are available at the following repository: [https://github.com/\[redacted\]/tic-idps](https://github.com/[redacted]/tic-idps). The repository includes automated scripts for data download and preprocessing, training and evaluation pipelines for each method, hyperparameter configuration files in YAML format, and Jupyter notebooks for reproducing all figures and tables in Chapter 3. The CIC-IoT-2023 and CSE-CICIDS2018 datasets are publicly available from the Canadian Institute for Cybersecurity. The UNSW-NB15 dataset is available from the University of New South Wales. The ICS3D collection datasets are available from their respective maintainers under academic licences.

Appendix C

Additional Experimental Results

This appendix presents supplementary experimental results that complement the analyses in Chapter 3. Section C.1 provides per-attack-category performance breakdowns. Section C.2 documents component-wise ablation details and hyperparameter sensitivity studies. Section C.3 reports cross-cloud and cross-domain transfer matrices. Section C.4 analyses the privacy-utility trade-off in detail. Section C.5 presents computational performance benchmarks and scaling characteristics.

C.1 Per-Attack Category Performance

C.1.1 CIC-IoT-2023 Detailed Results

Table C.1 reports per-attack-category precision, recall, F1 score, and AUC-ROC for CT-TGNN on CIC-IoT-2023. All seven major categories exceed 95% recall, with DDoS achieving the highest recall at 99.1% owing to the distinctive volumetric signature, and spoofing attacks showing the lowest recall at 95.2% because protocol impersonation can closely mimic legitimate authentication patterns.

Table C.1: CT-TGNN per-attack-category results on CIC-IoT-2023.

Attack Category	Precision	Recall	F1	AUC-ROC
DDoS	98.7%	99.1%	98.9%	0.998
Reconnaissance	96.2%	96.8%	96.5%	0.991
Web Attacks	97.1%	97.3%	97.2%	0.993
Brute Force	98.4%	98.6%	98.5%	0.997
Spoofing	94.8%	95.2%	95.0%	0.987
Malware	98.6%	98.9%	98.7%	0.998
Mirai Variants	97.1%	97.4%	97.2%	0.994
Benign	98.9%	98.4%	98.7%	0.997

C.1.2 CSE-CICIDS2018 Detailed Results

Table C.2 reports per-attack-category performance for the unified framework on CSE-CICIDS2018. The unified framework achieves the largest improvements over baselines on infiltration (12.7 percentage points above the best baseline), botnet detection (9.3 points), and heartbleed exploitation (8.9 points), categories that involve multi-stage, temporally extended attack sequences where the continuous-time formulation provides the greatest advantage.

Table C.2: Unified framework per-attack-category results on CSE-CICIDS2018.

Attack Category	Precision	Recall	F1	AUC	Δ vs. Baseline
Brute Force (SSH)	95.3%	96.1%	95.7%	0.991	+3.8%
Brute Force (FTP)	96.7%	97.3%	97.0%	0.994	+4.2%
DoS Hulk	97.8%	98.2%	98.0%	0.997	+2.1%
DoS GoldenEye	96.4%	96.9%	96.6%	0.993	+3.4%
DoS Slowloris	95.1%	95.7%	95.4%	0.989	+5.1%
Web Attack (XSS)	93.8%	94.3%	94.1%	0.984	+6.7%
Web Attack (SQL Inj.)	94.2%	94.8%	94.5%	0.986	+5.9%
Infiltration	91.7%	92.4%	92.0%	0.978	+12.7%
Botnet (Ares)	93.1%	93.8%	93.4%	0.982	+9.3%
Heartbleed	92.3%	93.1%	92.7%	0.979	+8.9%
Port Scan	97.6%	98.1%	97.8%	0.996	+2.3%
DDoS LOIC	98.1%	98.4%	98.2%	0.998	+1.7%
Benign	97.4%	96.8%	97.1%	0.995	+2.4%

C.1.3 UNSW-NB15 Detailed Results

Table C.3 reports per-attack-category results on UNSW-NB15. The unified framework achieves the largest improvement on reconnaissance attacks (8.7 percentage points), where the multi-scale temporal modelling captures the slow-and-stealthy scanning patterns that discrete-time baselines systematically miss.

C.1.4 TripleE-TGNN Microservices Attack Breakdown

Table C.4 reports TripleE-TGNN per-attack-type detection accuracy on the Container Security dataset, broken down by the granularity at which each attack is most effectively detected.

C.2 Ablation Study Details

C.2.1 Component-wise Ablation with Confidence Intervals

Table C.5 extends the ablation results of Table 3.8 in the main text with 95% confidence intervals computed from five independent runs with different random seeds. All degradations

Table C.3: Unified framework per-attack-category results on UNSW-NB15.

Attack Category	Precision	Recall	F1	AUC	Δ vs. Baseline
Generic	96.8%	97.2%	97.0%	0.995	+3.2%
Exploits	94.7%	96.3%	95.5%	0.989	+5.4%
Fuzzers	93.2%	94.1%	93.6%	0.984	+4.8%
DoS	96.4%	97.1%	96.7%	0.993	+3.7%
Reconnaissance	92.8%	97.9%	95.3%	0.988	+8.7%
Analysis	97.1%	98.4%	97.7%	0.996	+4.1%
Backdoors	91.9%	93.7%	92.8%	0.981	+6.3%
Shellcode	94.3%	95.8%	95.0%	0.987	+5.1%
Worms	89.7%	91.4%	90.5%	0.973	+7.2%
Normal	97.2%	96.4%	96.8%	0.994	+3.4%

Table C.4: TripleE-TGNN per-attack detection accuracy and dominant granularity.

Attack Type	Accuracy	Dominant Granularity	Single-Gran. Acc.
Container Escape	98.7%	Node + Service	91.2%
Supply Chain	97.3%	Service + Trace	89.8%
Resource Exhaustion	96.9%	Service + Node	90.3%
Privilege Escalation	97.8%	Node	93.1%
API Abuse	95.4%	Service	91.7%
Lateral Movement	96.1%	Trace + Node	88.4%
Data Exfiltration	94.8%	Trace	87.6%
Cryptojacking	97.2%	Node	92.9%
Overall	96.8%	Multi-granularity	88.5%

are statistically significant after Bonferroni correction at the $\alpha = 0.05$ level.

Table C.5: Extended ablation results with 95% confidence intervals (five seeds).

Component Removed	Full System	Without	Cohen’s d
Cont.-time ODE	98.3 $\pm$ 0.2%	91.5 $\pm$ 0.4%	4.82
Multi-scale modelling	98.3 $\pm$ 0.2%	94.4 $\pm$ 0.3%	3.14
Temporal adaptive BN	98.3 $\pm$ 0.2%	93.6 $\pm$ 0.5%	3.67
Cross-gran. attention	96.8 $\pm$ 0.3%	94.7 $\pm$ 0.3%	2.41
Trimmed mean $\rightarrow$ FedAvg	89.7 $\pm$ 0.4%	68.3 $\pm$ 1.2%	7.13
Diff. privacy	87.1 $\pm$ 0.3%	90.2 $\pm$ 0.3%	−3.28
OT alignment	92.7 $\pm$ 0.4%	76.9 $\pm$ 0.8%	5.92
LLM branch (F1)	87.6 $\pm$ 0.5%	34.2 $\pm$ 1.1%	8.47
Progressive distill.	91.4 $\pm$ 0.3%	81.7 $\pm$ 0.6%	4.71
Variational attn. (ECE)	0.078 $\pm$ 0.004	0.192 $\pm$ 0.008	5.38
Game-theoretic equil.	94.7 $\pm$ 0.3%	87.6 $\pm$ 0.4%	4.18

C.2.2 Hyperparameter Sensitivity Analysis

This section presents sensitivity analyses for the most important hyperparameters of each method, varying one parameter at a time while holding all others at their default values.

CT-TGNN: ODE Solver Tolerance

Table C.6 reports the effect of ODE solver tolerance on accuracy, inference latency, and the number of function evaluations (NFE) per forward pass. Tighter tolerances yield marginal accuracy improvements but substantially increase computational cost; the default tolerance of 10^{-5} provides the best accuracy-efficiency trade-off.

Table C.6: CT-TGNN sensitivity to ODE solver relative tolerance.

Rel. Tolerance	Accuracy	Latency (ms)	Avg. NFE
10^{-3}	97.6%	28	14
10^{-4}	98.1%	36	21
10^{-5} (default)	98.3%	47	31
10^{-6}	98.4%	72	48
10^{-7}	98.4%	118	73

FedLLM-API: Privacy Budget and Trimming Fraction

Table C.7 reports the effect of varying the per-round privacy budget ϵ on accuracy, total privacy expenditure over 100 rounds, and membership inference attack success. The default $\epsilon = 0.85$ balances strong privacy with acceptable accuracy degradation.

Table C.7: FedLLM-API sensitivity to differential privacy budget ϵ .

ϵ	Accuracy	Total ϵ_{100}	MI Attack	Δ Acc. vs. No-DP
0.5	84.2%	5.0	51.3%	−7.0%
0.85 (default)	87.1%	8.5	52.7%	−3.1%
1.0	88.4%	10.0	54.1%	−1.8%
1.5	89.7%	15.0	57.8%	−0.5%
2.0	91.8%	20.0	62.4%	+1.6%
∞ (no DP)	90.2%	∞	71.3%	0%

Table C.8 reports accuracy under 30% Byzantine adversaries as the trimming fraction varies. Trimming fractions below 20% fail to remove all Byzantine updates, while fractions above 30% discard too many honest updates.

Table C.8: FedLLM-API sensitivity to trimming fraction under 30% Byzantine adversaries.

Trimming Fraction	Accuracy (30% Byz.)	Convergence Rounds
10%	72.4%	142
15%	81.3%	108
20% (default)	89.7%	78
25%	88.9%	82
30%	86.1%	91
35%	82.7%	104

MambaShield: Distillation Temperature and Ensemble Size

Table C.9 reports the joint effect of distillation temperature and teacher ensemble size on robust accuracy under 25% poisoning. Temperature 3.0 with five teachers yields the best performance; lower temperatures produce sharper distributions that amplify label noise, while larger ensembles provide diminishing returns relative to their computational cost.

Table C.9: MambaShield sensitivity to distillation temperature and ensemble size (25% poisoning).

Temperature	3 Teachers	5 Teachers	7 Teachers	10 Teachers
$T = 1.0$	85.3%	86.7%	87.1%	87.3%
$T = 2.0$	88.4%	89.8%	90.1%	90.2%
$T = 3.0$ (default)	89.7%	91.4%	91.6%	91.7%
$T = 5.0$	88.1%	90.2%	90.5%	90.6%
$T = 10.0$	84.6%	87.3%	87.7%	87.8%

Stochastic Transformer: Monte Carlo Samples

Table C.10 reports the effect of the number of Monte Carlo forward passes on ECE, epistemic uncertainty correlation with prediction error, and inference latency per batch. Early stopping

based on variance convergence reduces the effective number of passes to 18 on average.

Table C.10: Stochastic transformer sensitivity to number of MC forward passes.

MC Passes	ECE	Uncertainty Corr.	Latency (ms/batch)	Eff. Passes
5	0.124	0.43	78	5
10	0.103	0.52	126	10
20	0.089	0.61	204	17
50 (default)	0.078	0.67	340	18
100	0.076	0.68	638	19

C.3 Cross-Cloud and Cross-Domain Transfer

C.3.1 Cross-Dataset Transfer Matrix

Table C.11 reports the accuracy of CT-TGNN when trained on one dataset (rows) and tested on another (columns) without any fine-tuning. The diagonal entries correspond to in-distribution performance. The consistently small off-diagonal degradation, averaging 3.8% across all pairs, confirms the generalisation properties of the continuous-time temporal graph formulation.

Table C.11: CT-TGNN cross-dataset transfer accuracy (%). Rows: training dataset; columns: test dataset.

Train \ Test	CIC-IoT	CICIDS18	UNSW-NB15
CIC-IoT-2023	98.3	92.7	91.4
CSE-CICIDS2018	91.8	96.7	89.3
UNSW-NB15	90.6	88.9	97.1

C.3.2 Cross-Domain Transfer to Speech and Healthcare

Table C.12 reports detailed metrics for the cross-domain transfer experiments described in Section 3.15 of the main text, including the effect of network security pretraining.

C.4 Privacy-Utility Trade-off Analysis

C.4.1 Differential Privacy Impact Across Datasets

Table C.13 reports the accuracy degradation caused by differential privacy at $\epsilon = 0.85$ across all three primary datasets. The degradation is consistent across datasets, ranging from 2.8% to 3.4%, confirming that the privacy-utility trade-off does not vary substantially with data distribution.

Table C.12: Cross-domain transfer results with and without network security pretraining.

Domain	Method	Accuracy	F1	ECE	Pretrained?
Speech (LibriSpeech)	CT-TGNN	90.4%	90.8%	0.091	No
Speech (LibriSpeech)	CT-TGNN	94.2%	94.2%	0.073	Yes
Speech (LibriSpeech)	LSTM baseline	86.9%	86.7%	0.142	No
Healthcare (MIMIC-III)	Stoch. Trans.	86.1%	85.7%	0.098	No
Healthcare (MIMIC-III)	Stoch. Trans.	89.7%	89.3%	0.082	Yes
Healthcare (eICU)	Stoch. Trans.	84.8%	84.2%	0.104	No
Healthcare (eICU)	Stoch. Trans.	88.4%	87.9%	0.087	Yes

Table C.13: Differential privacy impact on accuracy across datasets ($\epsilon = 0.85$, $\delta = 10^{-5}$).

Dataset	Without DP	With DP	Degradation
CIC-IoT-2023	94.7%	91.6%	−3.1%
CSE-CICIDS2018	93.9%	91.1%	−2.8%
UNSW-NB15	93.8%	90.4%	−3.4%

C.4.2 Privacy Accounting Over Communication Rounds

Table C.14 reports the cumulative privacy expenditure as a function of federated communication rounds, computed through the moments accountant with Poisson subsampling amplification. The total budget of $\epsilon_{\text{total}} = 8.5$ after 100 rounds remains within the range typically considered acceptable for sensitive applications.

Table C.14: Cumulative privacy expenditure over federated rounds ($\delta = 10^{-5}$).

Rounds	10	25	50	75	100	150
ϵ_{total}	0.85	2.13	4.25	6.38	8.50	12.75
Accuracy	78.4%	84.1%	87.8%	88.6%	87.1%	86.3%

C.5 Computational Performance Metrics

C.5.1 Inference Throughput and Latency

Table C.15 compares inference throughput, per-sample latency, GPU memory consumption, and energy efficiency across all seven methods and baselines. Measurements are taken on a single NVIDIA Tesla P100 GPU with batch size 32 for latency and batch size 512 for throughput, reporting the median of 100 runs with 95% bootstrap confidence intervals.

C.5.2 Scaling with Input Size

Table C.16 reports how inference latency scales with input sequence length for CT-TGNN, MambaShield, and the standard transformer baseline. CT-TGNN scales linearly through the

Table C.15: Computational performance comparison across methods (P100 GPU).

Method	Throughput (k/s)	Latency (ms)	Memory (GB)	W/1k pred.
CT-TGNN	8,700	47	3.8	34
TripleE-TGNN	4,200	73	4.7	51
FedLLM-API	5,100	62	5.2	47
PQ-IDPS	6,800	38	2.1	28
MambaShield	12,400	18	2.3	19
Stoch. Transformer	2,300	340	8.2	89
Game-Theoretic	7,100	12	0.2	8
LSTM baseline	4,300	38	1.8	42
Std. Transformer	6,200	55	4.1	125
Random Forest	14,200	12	0.8	14
XGBoost	18,700	8	0.6	11

fixed number of ODE solver steps per time constant. MambaShield scales linearly through the selective scan mechanism. The standard transformer scales quadratically through self-attention, becoming impractical for sequences beyond approximately 4096 tokens on 16 GB GPUs.

Table C.16: Inference latency (ms) as a function of sequence length.

Sequence Length	256	512	1024	2048	4096
CT-TGNN	23	41	74	142	278
MambaShield	8	14	26	49	94
Std. Transformer	18	52	178	684	OOM

C.5.3 Training Time and Convergence

Table C.17 reports total training time, number of epochs to convergence, and wall-clock time per epoch for each method on CIC-IoT-2023 using four V100 GPUs with data parallelism.

Table C.17: Training time comparison (four V100 GPUs, CIC-IoT-2023).

Method	Total Time (h)	Epochs	Time/Epoch (min)
CT-TGNN	18.2	142	7.7
TripleE-TGNN	22.7	118	11.5
FedLLM-API	31.4	100 (rounds)	18.8
PQ-IDPS	8.3	87	5.7
MambaShield	14.1	100	8.5
Stoch. Transformer	24.6	96	15.4
Game-Theoretic	6.8	10,000 (rounds)	0.04

C.5.4 Model Size Comparison

Table C.18 reports the total trainable parameter count, compressed model size after INT8 quantisation, and the accuracy retained after quantisation.

Table C.18: Model size and quantisation impact.

Method	Parameters (M)	FP32 Size (MB)	INT8 Size (MB)	INT8 Acc.
CT-TGNN	12.3	47	13	97.9%
TripleE-TGNN	18.7	71	19	96.2%
FedLLM-API	24.1	92	25	86.4%
PQ-IDPS	6.8	26	7	95.8%
MambaShield	8.4	32	9	90.9%
Stoch. Transformer	31.2	119	31	93.4%
Std. Transformer	47.8	182	47	90.8%